THE BASE-β BRANCH OF $3\alpha + 2\beta = \pi$: BOUNDARY WALKS, AND THE STRUCTURE OF A HYPOTHETICAL PRIME TILING

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Abstract. This is a companion note to *A signed-direction invariant for triangle tilings, and the exclusion of primes* $\equiv 3 \pmod 4$. That paper excludes every prime $N \equiv 3 \pmod 4$ exceeding 3 except the base-β candidates $N = 3f^2 - e^2$, which are exactly the primes $\equiv 11 \pmod{12}$. Here we record what is known about the internal structure of a hypothetical tiling in that one open branch: the boundary walk equations and their solutions, the apex-mismatch configuration, and the alignment and far-region theorems. None of this is needed for the main theorem; it is collected here so that the main paper can state the exception cleanly and so that the structural facts are available to anyone attacking the branch.

Two results are worth isolating. First, at $m = 1$ the walk equations are completely solved in a regime: if $f^2 > 2e^2$ then each equal side of the target carries no b-edge, and if $f^2 > 2ef + e^2$ then the base carries exactly e of them, so $R_{\text{base}} \in \{e, 2e\}$. Second, for $e = 1$ this pins the base walk to a single family, reducing that entire subfamily — the candidates $N = 3f^2 - 1$, among them $11, 47, 107, 191, 431, 587, 971$ below 1000 — to the question of whether one explicit walk is realizable.

We also record, as prominently as the positive results, which attacks on this branch are now known to be dead: the whole class of linear direction invariants, the metric bounds derived from them, and the hope of closing $e = 1$ by proving that every side carries at least two c-edges.

1. Scope and conventions

Throughout, the tile is the primitive $3\alpha + 2\beta = \pi$ triangle

$$(a, b, c) = (ef,\ f^2 - e^2,\ f^2), \qquad \gcd(e, f) = 1, \quad 1 \le e < f,$$

with a opposite α, b opposite β, c opposite γ, and $2\gamma = \pi + \alpha$. The base-β target is the isosceles triangle with base angles β,

$$\text{equal sides } S = f^3 m, \qquad \text{base } L = em(3f^2 - e^2), \qquad N = m^2(3f^2 - e^2),$$

and N is prime exactly when $m = 1$. We write $D = 4f^2 - e^2$, so $\sin\alpha = e\sqrt{D}/(2f^2)$ and the tiling's coordinates lie in $\mathbb{Q}(\sqrt{D})$. Numbering of the form "Proposition 4.3" refers to the main paper [3]; results are stated here only when they are not there.

We use freely from [3]: the dissection basics and vertex figures, the corner figures (a base corner carries a single tile presenting β; the apex carries exactly three tiles, each presenting α), the γ-trap (every side of the target carries at least one c-edge), the unsplittability of b, and the corner parallelogram.

2. Boundary walks

A side of the target of length L is exactly partitioned by the tile edges lying on it, so it determines a *walk equation*

$$n_a\, ef + n_b\, (f^2 - e^2) + n_c\, f^2 = L, \qquad n_a, n_b, n_c \ge 0,$$

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together with $n_c \geq 1$ from the γ-trap. The results of this section solve those equations and then constrain the order in which the edges occur.

Theorem 1 (The base-β family at $e = 1$, $f = 2$)**.** *The isosceles triangle with base angles β and sides $(8, 8, 11)$ admits no tiling by $N = 11$ copies of the tile $(2, 3, 4)$. Consequently the base-β branch is closed for $N = 11$, unconditionally.*

Proof. Write $\omega = \alpha/2$. The law of cosines gives $\cos\alpha = (2f^2 - e^2)/(2f^2)$, hence $\sin\omega = e/(2f)$; as $1 \leq e < f$ places this strictly in $(0, \frac{1}{2})$, *Niven's theorem* forces α/π irrational — for every (e, f), with no appeal to [2] (`tile_alpha_irrational`). Irrationality pins the vertex figures: a vertex $x\alpha + y\beta + z\gamma = k\pi$ forces $y + z = 2k$ and $2x + z = 3y$, so a π-vertex is $\{3\alpha, 2\beta\}$ or $\{\alpha, \beta, \gamma\}$, each base corner carries *exactly one* tile (its β-vertex) and the apex *exactly three* (`vertex_pi`, `vertex_beta_corner`, `vertex_apex`); in particular a π-vertex carries at most one γ, since $2\gamma = \pi + \alpha > \pi$. Each side of ABC is partitioned into whole tile edges (a supporting line meets a convex tile in a face).

Two citation-free constraints now cut the base. (a) *Corner parallelogram*: the unique β-corner tile T_B has its b-edge a chord $[P, Q]$, matched (by the mod-f edge arithmetic) by a single tile T' whose α, γ are swapped, so $T_B \cup T'$ is a parallelogram with sides a, c; hence P, Q are $(1, 1, 1)$-vertices whose third tile carries β, and the second base edge from each corner lies in $\{a, c\}$. So a b-edge can occupy only positions $3, \ldots, k - 2$ of a base of k edges. (b) *γ-trap*: no boundary junction carries γ on both sides, and only a c-edge leaves the γ-state, so every side carries at least one c-edge.

At $e = 1$, $m = 1$ the base $Y = 3f^2 - 1$ has $q_{\text{base}} \equiv 1 \pmod{f}$, and the only compositions are $B_0 = \{b, c, c\}$ $(k = 3)$, $B_1 = (a^f, b, c)$ $(k = f + 2)$ and $B_2 = (a^{2f}, b)$ $(k = 2f + 1)$. B_2 dies by (b). B_0 dies by (a) for *every* f: its single b would need a position in $3, \ldots, 1$. At $f = 2$, B_1 also dies by (a): $k = 4$ leaves positions $3, \ldots, 2$. So no base survives, and $N = 11$ admits no tiling. $\square$

Theorem 2 (Boundary walks of the base-β target at $m = 1$)**.** *Let $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$ be the primitive $3\alpha + 2\beta$ tile ($1 \leq e < f$, $\gcd(e, f) = 1$) and let $m = 1$, so that the target is $(f^3, f^3, e(3f^2 - e^2))$ and $N = 3f^2 - e^2$. Write a walk along a side as $P \cdot a + Q \cdot b + R \cdot c$ equal to that side's length. Then both walk sets are cut out by the* same *linear form $\langle e, f, 1\rangle$, the base at level $2e$ and the equal side at level f:*

$$\begin{aligned} &\textit{base:} \quad (P, Q, R) = (je + fp,\ e + fj,\ R), && pe + jf + R = 2e, && j \in \mathbb{Z}_{\geq 0}, \\ &\textit{side:} \quad (P, Q, R) = (qe + fp,\ fq,\ R), && pe + qf + R = f, && q \in \mathbb{Z}_{\geq 0}, \end{aligned}$$

with $p \in \mathbb{Z}$ and $P, R \geq 0$. In the thin regime $f > 2e$ the levels force short lists:

(i) (trichotomy) *the base is exactly one of $\{b^e, c^{2e}\}$, $\{a^f, b^e, c^e\}$, $\{a^{2f}, b^e\}$; in particular it carries* exactly e *b-edges;*

(ii) (dichotomy) *each equal side is $\{a^e, b^f\}$ or $\{a^{fp}, c^{f-pe}\}$ for some $p \geq 0$.*

Feeding in the γ-trap ($R \geq 1$ on every side, Remark 22) removes the thinness hypothesis altogether. For every *(e, f) at $m = 1$:*

(iii) no equal side carries a b-edge — *each is $\{a^{fp}, c^{f-pe}\}$, $p \geq 0$; so* all *boundary b-edges lie on the base;*

(iv) *the base's b-count $Q = e + fj$ obeys $j(f - e) \leq e - 1$.*

In particular $e = 1$ forces $j = 0$ for every f — the base carries exactly one b-edge — *and $f > 2e$ forces $j = 0$, recovering* (i) *minus its $R = 0$ member.*

Proof. Base. Reducing $P \cdot ef + Q(f^2 - e^2) + Rf^2 = e(3f^2 - e^2) \bmod f$ gives $Qe^2 \equiv e^3$, so $f \mid Q - e$ by $\gcd(e, f) = 1$; as $0 \leq Q$ and $0 < e < f$, the least nonnegative residue is e, whence $Q = e + fj$ with $j \geq 0$. Substituting and cancelling one factor f leaves the *halved equation*

$$Pe + j(f^2 - e^2) + Rf = 2ef. \tag{1}$$

Reducing (1) mod f gives $e(P-je)\equiv 0$, so $f \mid P-je$, say $P=je+fp$, $p\in\mathbb{Z}$; substituting into (1) and cancelling f gives $pe+jf+R=2e$. Conversely every such triple solves the original equation. For $f>2e$: from (1), $j(f^2-e^2)\le 2ef$; if $j\ge 2$ then $f^2-e^2\le ef$, i.e. $f^2-ef-e^2\le 0$, whereas $(f-2e)(f+e)=f^2-ef-2e^2>0$ gives $f^2-ef-e^2>e^2>0$ — contradiction. If $j=1$ then $f\mid P-e$ forces $P\ge e$, and (1) gives $e^2+(f^2-e^2)\le 2ef$, i.e. $f^2\le 2ef$, i.e. $f\le 2e$ — contradiction. So $j=0$, whence $f\mid P$, $P=fp$ with $p\ge 0$, and $pe+R=2e$ forces $p\in\{0,1,2\}$ and $R=e(2-p)$, giving the three walks.

Side. The same reduction on $P\cdot ef+Q(f^2-e^2)+Rf^2=f^3$ gives $Qe^2\equiv 0$, so $f\mid Q$, $Q=fq$; cancelling f leaves $Pe+q(f^2-e^2)+Rf=f^2$, whose reduction mod f gives $f\mid P-qe$, say $P=qe+fp$, and cancelling f again gives $pe+qf+R=f$. For $f>2e$: $q\ge 2$ would give $2(f^2-e^2)\le f^2$, i.e. $f^2\le 2e^2<f^2$, absurd. If $q=1$ then $Pe+Rf=e^2$ and $f\mid P-e$ force $P=e$, $R=0$; if $q=0$ then $f\mid P$, $P=fp$ and $pe+R=f$.

(iii) and (iv) are one computation, and need no thinness. Suppose a side walk has $q\ge 1$. From $P=qe+fp\ge 0$ we get $fp\ge -qe>-qf$ (using $e<f$ and $q\ge 1$), so $p>-q$, i.e. $p\ge 1-q$. The level equation then gives

$$R \;=\; f-pe-qf \;\le\; f-(1-q)e-qf \;=\; (1-q)(f-e) \;\le\; 0,$$

contradicting $R\ge 1$. Hence $q=0$, which is (iii). Identically for the base with $P=je+fp\ge 0$: if $j\ge 1$ then $p\ge 1-j$ and

$$R \;=\; 2e-pe-jf \;\le\; 2e-(1-j)e-jf \;=\; e-j(f-e),$$

so $R\ge 1$ forces $j(f-e)\le e-1$, which is (iv) (and it holds trivially at $j=0$ since $e\ge 1$). For $e=1$ this reads $j(f-1)\le 0$, so $j=0$; for $f>2e$ we get $je<j(f-e)\le e-1$, so $j=0$. ☐

Theorem 3 (Apex mismatch: the pierced corner)**.** *Let a base-β target be tiled at any scale m such that both equal sides end (at the apex) with a c-edge — automatic at $m=1$, where the sides are b-free. Then, writing T_2 for the middle of the three apex tiles:*

(i) *the outer apex tiles carry c on the boundary and b inward, and T_2 carries b and c inward, so exactly one inner apex ray carries T_2's c-edge against a neighbour's b-edge: a T-junction at distance b from the apex, whose vertex figure on the neighbour's side is* $(1,1,1)$ *(one α, one β, plus the neighbour's γ);*

(ii) *the remaining segment of T_2's c-edge has length $c-b=e^2$, and $n_a\,ef+n_b(f^2-e^2)+n_cf^2=e^2$ has* no *solution in nonnegative integers, for any $1\le e<f$ with* $\gcd(e,f)=1$*;*

(iii) *hence some far-side edge crosses the far endpoint of T_2's c-edge* straight through*: every such tiling contains a **pierced β-corner** at distance f^2 along an inner apex ray, and the angles on T_2's side of the piercing line beyond the corner form $\{\alpha,\gamma\}$ or $\{3\alpha,\beta\}$.*

Proof. (i) The apex is 3α (three tiles, each α there); the two edges of an α-corner are b and c. The boundary edges at the apex are the sides' last edges, c by hypothesis, so the outer tiles point b inward and T_2 has $\{b,c\}$ inward. Whichever inner ray pairs T_2's c with an outer b, the b-edge ends at distance b: its far endpoint is the γ-corner of its tile (b joins α to γ), and the vertex figure on that side of the straight c-edge sums to π with a γ present, hence is $(1,1,1)$ by the vertex classification. (ii) Any solution has $n_a=n_c=0$ since $ef,f^2>e^2$; then $n_b(f^2-e^2)=e^2$ gives $n_bf^2=(n_b+1)e^2$, and $\gcd(e,f)=1$ forces $f^2\mid n_b+1$, so $n_b\ge f^2-1$ and $e^2=n_b(f^2-e^2)\ge f^2-1\ge e^2$, with equality forcing $f^2-e^2=1$ — impossible, as $f^2-e^2\ge 2e+1\ge 3$. (iii) The far side of the leftover segment is covered by consecutive whole tile edges starting at the T-junction; by (ii) no partial sum lands exactly on the segment's far endpoint, so some edge strictly crosses it. That endpoint is the β-corner of T_2 (c joins α to β). The angles on T_2's side of the crossing line beyond the corner fill $\pi-\beta$; writing them as

$x\alpha + y\beta + z\gamma$ and clearing β, γ, irrationality of α/π forces $y + z = 1$, $2x + z = 3y + 3$, whose only solutions are $(3, 1, 0)$ and $(1, 0, 1)$. ☐

Remark 4 (Validation on the 44-tiling)**.** The theorem can be checked against the genuine 44-tiling ($m = 2$; its equal sides do end with c), where it holds exactly: the apex rays pair one $b \,|\, b$ and one mismatch $c \,|\, b$; the middle tile's c-endpoint $V = (10, 2\sqrt{15})$ is pierced by a straight b-edge, and the sector on the middle tile's side is $\beta + \{\alpha, \gamma\}$ — the first continuation type. Parts (ii) and the vertex figures in (i), (iii) are machine-checked (`apex_leftover_nonrepresentable`, `pierced_corner_types`, `BaseBetaWalks.lean`). The configuration is an interior constraint, uniform in (e, f) and unconditional at $m = 1$: every hypothetical thick-regime tiling, $N = 59$ included, must contain it.

Proposition 5 (Rung two: the pre-piercer chain)**.** *In the configuration of Theorem 3, every full far-side edge strictly between the T-junction and the piercer lies inside $[b, f^2]$, hence has length $\le e^2$; among the tile edges only $b = f^2 - e^2$ can satisfy this, and only in the* super-thick *regime $f^2 \le 2e^2$. So the piercer starts at $(t+1)b$ with $tb < e^2$, where $t = 0$ outside super-thick; and if the piercer is itself a b-edge, its corner at its starting junction is exactly α (the $(1, 1, 1)$ figure offers $\{\alpha, \beta\}$, while a b-edge's endpoints are $\{\alpha, \gamma\}$). In particular for $N = 59$ ($(e, f) = (4, 5)$, super-thick: $25 \le 32$) the piercer starts at distance 9 or 18, and for every non-super-thick member it starts at the T-junction itself.*

Theorem 6 (Alignment of the mismatch ray)**.** *Let $m = 1$, $e \ge 2$, and let r be the mismatch inner apex ray of Theorem 3, of inner length $f \cdot b$ (Proposition 5, $b = f^2 - e^2$). Along r, the near side is $T2$'s single unsplittable c-edge covering $[0, f^2]$, and the far side begins with $T1$'s b-edge. Then:*

(i) no common junction *lies in $(0, f^2)$: for $1 \le k \le f - 1$, $k \cdot b$ is never $n_a\, ef + n_b\, b + n_c f^2$ with $n_c \ge 1$ (so every far-side junction is a genuine T-junction, the near c-edge passing straight through it);*

(ii) *the far side is exactly b^f: the only representation of $f \cdot b$ as $n_a\, ef + n_b\, b + n_c f^2$ with $n_b \ge 1$ is $(0, f, 0)$;*

(iii) *$V = f^2$ is pierced: $b \nmid f^2$, so no far-side junction $k \cdot b$ equals f^2 and a far b-edge straddles $T2$'s β-corner.*

*All three are machine-checked (`far_near_disjoint`, `far_is_bpow`, `b_not_dvd_fsq`); (ii) alone uses γ-vertex induction (*via *the walk arithmetic) to fix the first far edge, the rest is pure number theory.*

Proof. (i) If $kb = n_a ef + n_b b + n_c f^2$ then $(k - n_b)b = f(n_a e + n_c f)$, so $f \mid (k - n_b)e^2$ and $\gcd(e, f) = 1$ give $f \mid k - n_b$; writing $k - n_b = fs$, the equation reads $sb = n_a e + n_c f \ge f > 0$ (as $n_c \ge 1$), so $s \ge 1$ and $k \ge n_b + f \ge f$, against $k \le f - 1$. (ii) $f \mid n_b$ by the same reduction, and $n_b \le f$ by size, so $n_b \in \{0, f\}$; $n_b \ge 1$ forces $n_b = f$, and then $n_a ef + n_c f^2 = 0$ gives $n_a = n_c = 0$. (iii) $b \mid f^2$ would give $b \mid e^2$, hence $b \mid \gcd(f^2, e^2) = 1$, so $b = 1$, impossible as $b \ge 2e + 1 \ge 5$. The far junctions are the multiples kb, none equal to f^2, so the b-edge containing f^2 straddles it. ☐

Theorem 7 (The mismatch ray is completely determined)**.** *Let $m = 1$, $e \ge 2$, and let the family be thick, $f \le 2e$. Then along the mismatch inner apex ray, of inner length $f\,b$:*

$$\textit{far side} = b^{f}\ \ (f\ \textit{edges}), \qquad \textit{near side} = \{a^{f-e},\, c^{f-e}\}\ \ (2(f-e)\ \textit{edges}),$$

and the near side is the unique *possibility. In particular the near side carries equally many a- and c-edges, and no b-edge.*

Proof. The far side is Theorem 6(ii). For the near side, $T2$'s c-edge covers $[0, f^2]$ and the remainder $f\,b - f^2 = f(b-f)$ is covered by whole edges, say n_a of length a, n_b of length b and n_c of length c. Reducing modulo f kills n_b as in Theorem 6(i), and cancelling one factor f leaves

$$n_a e + n_c f + f = b.$$

Now $(n_a, n_c) = (f-e,\ f-e-1)$ solves it, since $(f-e)e+(f-e-1)f = f^2-e^2-f = b-f$. For uniqueness, $\gcd(e, f) = 1$ and the equation give $f \mid n_a+e$, so $n_a+e = ft$; $t = 0$ is impossible and $t \geq 2$ would force $n_a e \geq (2f-e)e$, whence $2ef \leq f^2-f$, i.e. $f \geq 2e+1$, contradicting thickness. Hence $t = 1$, $n_a = f-e$, and then $n_c f = f(f-1-e)$ gives $n_c = f-e-1$. Adding $T2$'s own c-edge produces the multiset $\{a^{f-e}, c^{f-e}\}$, whose total length is $(f-e)(ef+f^2) = f(f-e)(e+f) = f\,b$ as required. □

Theorem 8 (Unique representation of $k\,b$)**.** *For $1 \leq k \leq f-1$ the only way to write $k\,b$ as $n_a\,ef+n_b\,b+n_c f^2$ in nonnegative integers is b^k.* (*Machine-checked,* `kb_unique_rep`*; it generalizes Theorem 6*(ii)*, which needed $n_b \geq 1$.*)

Theorem 9 (The angle $(\pi-\alpha)/2$)**.** *A vertex figure filling $(\pi-\alpha)/2$ is $\alpha+\beta$, and nothing else.* (*Machine-checked,* `halfpi_minus_alpha_unique`*.*)

Theorem 10 (The b-run orientation lemma)**.** *Let a run of b-edges be laid end to end along a ray, seeded at an apex tile. Then every b-edge of the run carries α at its apex end and γ at its far end, and every junction interior to the run is an (α, β, γ) π-vertex. In particular the $3\alpha+2\beta$ figure never occurs along such a run.*

Proof. A b-edge joins its tile's α- and γ-corners, so at a junction the incoming tile presents α or γ, and likewise the outgoing one. The $3\alpha+2\beta$ figure contains no γ, and two γ's cannot share a side of a straight line since $2\gamma = \pi+\alpha > \pi$. Hence if the incoming corner is γ the figure is $(1, 1, 1)$, whose single α must be the outgoing corner, so the next edge again ends in γ. An apex tile has α at the apex, hence γ at the far end of its b-edge, which seeds the induction; it proceeds with no branching. □

Theorem 11 (The far region is a scaled tile)**.** *Suppose the mismatch apex ray is a complete wall, of inner length $f\,b$. Then it cuts off a triangle with angle α at the apex, β at the base corner and hence γ at its third vertex, and with sides $f\,c$, $f\,a$, $f\,b$ — that is,* the tile scaled by f*. It therefore has area f^2 times the tile's and is tiled by the standard f-subdivision into f^2 copies, whose ray side is exactly b^f with the junction chain of Theorem 10.*

Corollary 12 (The far side can never give a contradiction)**.** Under the hypothesis of Theorem 11 *— that the mismatch apex ray is a complete wall — no obstruction to a base-β tiling can be derived from the far side of that ray, at any (e, f): that side admits a genuine tiling. The hypothesis may not be dropped; if the ray is not a complete wall the far region is not a scaled copy of the tile and the conclusion does not follow.*

Remark 13 (Where the obstruction must instead live)**.** Corollary 12 is a negative result and an important one: it explains why the pierced-corner analysis resisted three independent attempts. The configuration it pins down is not contradictory but *realizable*, so no amount of further chasing on that side can close the branch. The check is exact for every family — for $(e, f) = (4, 5)$ the far region has sides $(45, 100, 125) = 5 \cdot (9, 20, 25)$ and contains $25 = f^2$ of the 59 tiles — and it is confirmed on the genuine 44-tiling, whose mismatch ray has *no* straddling tile and whose far region contains $16 = (mf)^2$ tiles with sides $4 \cdot (c, a, b)$.

The live direction is the other ray. For every thick member with $m = 1$, $e \geq 2$, the *matched* inner apex ray is *not* a wall. Indeed if it were, the target would split into $f^2 + f^2 + (f^2 - e^2)$ tiles, the last a middle triangle M with apex α, legs $f\,b$ and base $e\,b$. Since $k\,b$ has the unique representation b^k for $1 \le k \le f-1$ (Theorem 8), both edges of M at its far corner are b-edges; but the angle there is $(\pi-\alpha)/2$, whose only decomposition is $\alpha+\beta$ (Theorem 9), and a b-edge endpoint is an α- or γ-corner, so no tile present can supply the β. This is the precise mechanism behind the failure of the clean-ray corollary retracted in Remark 15, and it is confirmed on the 44-tiling, whose matched ray carries exactly three straddling tiles against the mismatch ray's zero.

Consequently the branch now reduces to a single question: *can the residual region — the target minus the far wall, containing $N-f^2$ tiles and crossed by a necessarily straddled matched ray — be tiled?* Every tool developed here (walk arithmetic, vertex figures, unsplittability) is wall-based and goes silent once a tile straddles. What is needed is a bound on where a tile may straddle a ray whose two ends are pinned by b-edges.

Remark 14 (Tiles do straddle: the no-straddle hypothesis is false)**.** Remark 13 asks for a bound on where a tile may straddle a ray whose two ends are pinned by b-edges. The strongest form one might hope for is that no such straddle occurs at all:

(DL) if a segment carries a whole b-edge at each end, no tile straddles it.

(DL) would make the walk arithmetic of §5.5 applicable to interior segments, and it is the form the funnel arguments implicitly assume. It is false.

Testing it against the three tilings of base-β targets exhibited in this note and the main paper — all with tile $(2,3,4)$, all verified exactly — for every b-edge we take the line through it and count tiles with interior on both sides whose centroid projects near the segment:

tiling	b-edges	lines with a straddling tile	worst
44-tiling `N44B` of $(16,16,22)$, $m=2$	44	19	3
44-tiling `N44C` of $(16,16,22)$, $m=2$	44	30	4
99-tiling of $(24,24,33)$, $m=3$	99	64	4

Straddling near b-edges is common rather than exceptional. Consequently the missing ingredient is not a prohibition but a *quantitative* bound: a description of the set in which a straddling tile's crossing points must lie. Since every vertex of a tiling lies in a finitely generated $\mathbb{Z}$-module over $\mathbb{Q}(\sqrt{D})$, and b-pinning fixes the segment's endpoints inside that module, such a bound is the natural object to seek; we have not found it. The computation is `rust/walls` (binary `straddle`), and it is recorded here because a negative result of this shape saves the reader the attempt, and because it explains why the interior argument of §5.5 has never been written out.

Remark 15 (Two corrections this forces)**.** Theorem 10 is uniform in (e,f) and in m, and is confirmed on the genuine 44-tiling, where all four far b-edges of the mismatch ray and both sides of the clean ray read $\alpha \to \gamma$. It strictly generalizes Proposition 5, which established only the first step. But it also refutes two statements of an earlier version, which we withdraw.

First, the assertion that the *clean* apex ray is b^f against b^f. Its justification — "by the symmetry of the isosceles target both inner rays have the same inner length $f\,b$" — conflates two objects. The symmetry does apply to the geometric rays; it does not apply to the tiling's maximal edge *segment* along a ray, which may terminate early at a vertex where the edges turn away. In fact the clean ray's segment can never reach the base: if it did, both final b-edges would contribute γ at an interior point of the base, where the tiles above sum to π, contradicting $2\gamma > \pi$. On the 44-tiling the two segments have lengths 12 and 9 respectively, and the clean ray terminates at an interior $\beta+3\gamma$ vertex — which is precisely the M_4 vertex the census forces.

Second, and more seriously, the same objection applies to the hypothesis under which Theorems 6 and 7 were stated: that the *mismatch* ray's maximal segment attains the full length $f\,b$. That too was justified by symmetry, and it too now requires a genuine proof. The arithmetic of those theorems is unaffected — given a chain of total length $f\,b$ containing a b-edge, the conclusions stand and remain machine-checked — but their geometric hypothesis must be stated explicitly. Establishing that the mismatch segment reaches the base is the single highest-value open step in this branch: it would restore Theorems 6 and 7 unconditionally for every thick member.

Remark 16 (Scope and consequences)**.** Thickness is used exactly once, and sharply: the next solution would be $n_a = 2f - e$, which needs $f \geq 2e + 1$ — the thin regime. So in the regime that is open, the entire mismatch ray is now pinned on *both* sides, with no freedom beyond the order of the near-side edges. Explicitly: $N = 23$ has far 5^3, near $\{6, 9\}$; $N = 59$ far 9^5, near $\{20, 25\}$; $N = 66$ far 16^5, near $\{15^2, 25^2\}$; $N = 83$ far 11^6, near $\{30, 36\}$; $N = 167$ far 39^8, near $\{40^3, 64^3\}$. The count of interior junctions is therefore also determined — $f - 1$ on the far side and $2(f - e) - 1$ on the near side, none shared (Theorem 6(i)) — so a hypothetical thick tiling carries exactly $3f - 2e - 2$ T-junctions along this ray, each a π-vertex. The arithmetic is machine-checked (`near_side_unique`, `BaseBetaWalks.lean`).

Remark 17 (Scope, and why $m = 1$ is the rigid case)**.** Theorem 2(i) generalizes the $e = 1$ base analysis used for Theorem 1 to the infinite family $f > 2e$ — $e = 1$, $f \geq 3$; $e = 2$, $f \geq 5$; $e = 4$, $f \geq 9$; $\ldots$ — covering the primes $N = 47, 71, 107, 191, 227, 239, 359, \ldots$, and its parts (iii)–(iv), which need no thinness, deliver the $e \geq 2$ analogue of the $e = 1$ structure theorem of Remark 14 for *every* (e, f): the equal sides are b-free and the base's b-count is bounded by $j(f - e) \leq e - 1$. The step that fails for $m \geq 2$ is the very first one: there $Q \equiv em \pmod f$, so j may be *negative* and the level rises to $2em$. This is not an artifact. The genuine 44-tiling ($e{=}1, f{=}2, m{=}2$) has base walk *aaaaccca*, i.e. $(P, Q, R) = (5, 0, 3)$, which is the parameter $j = -1$, $p = 3$; and the 99-tiling ($e{=}1, f{=}2, m{=}3$) has base walk *aabbbbbbbcc*, i.e. $(2, 7, 2)$, the parameter $j = 2$, $p = 0$. Both satisfy $R \geq 1$, as the γ-trap requires.

Theorem 18 (Walk structure at $m = 1$)**.** *Let $m = 1$, $\gcd(e, f) = 1$, $1 \leq e < f$, and consider a tiling of the base-β target.*

- (i) *If $f^2 > 2e^2$, then each equal side carries* no *b-edge. Its edges are a's and c's with $n_a = fk$ and $n_c = f - ke \geq 1$, and it ends with a c-edge at the apex.*
- (ii) *If $f^2 > 2ef + e^2$ — equivalently $f/e > 1 + \sqrt{2}$ — then the base carries* exactly *e b-edges, and $n_a = f\ell$, $n_c = e(2 - \ell)$ with $\ell \in \{0, 1\}$. In particular*

$$R_{\text{base}} \in \{e,\, 2e\}.$$

Proof. A side of length L is exactly partitioned by the tile edges on it, giving $n_a\,ef + n_b\,b + n_c\,f^2 = L$ with $n_a, n_b, n_c \geq 0$, and $n_c \geq 1$ by Proposition 25.

(i) Here $L = f^3$. Rearranging with $b = f^2 - e^2$ gives $n_b e^2 = f\,(n_a e + n_b f + n_c f - f^2)$, so $f \mid n_b e^2$ and hence $f \mid n_b$ by coprimality; write $n_b = ft$. If $t \geq 2$ the b-edges alone cost at least $2fb = 2f^3 - 2fe^2$, which together with $n_c f^2 \geq f^2$ exceeds f^3 unless $f^2 \leq 2e^2$. If $t = 1$, cancelling f leaves $n_a e + n_c f = e^2$, so $n_a \leq e$ and $f \mid e(e - n_a)$; coprimality with $0 \leq e - n_a \leq e < f$ forces $n_a = e$ and then $n_c = 0$, contradicting $n_c \geq 1$. So $t = 0$. With $n_b = 0$ the equation is $n_a e + n_c f = f^2$, whence $f \mid n_a$; writing $n_a = fk$ gives $n_c = f - ke$. Finally, the apex tile on that side presents α (Proposition 24), whose incident edges are b and c; as the side has no b-edge, it ends with a c-edge.

(ii) Here $L = e(3f^2 - e^2)$, and the same rearrangement gives $(n_b - e)e^2 = f\,(n_a e + n_b f + n_c f - 3ef)$, so $f \mid n_b - e$; write $n_b = e + ft$. Negative t makes $n_b < 0$. For $t \geq 1$ the b-edges alone cost at least

$$(e + f)b = (f + e)^2(f - e) = e(3f^2 - e^2) + f\,(f^2 - 2ef - e^2),$$

which already exceeds L under the hypothesis, before the mandatory c-edge is placed. So $n_b = e$; subtracting and cancelling f leaves $n_a e + n_c f = 2ef$, whence $f \mid n_a$, and writing $n_a = f\ell$ gives $n_c = e(2-\ell)$. Then $n_c \geq 1$ forces $\ell \leq 1$.

Machine-checked (`equal_side_no_b`, `equal_side_shape`, `base_b_count` in `lean/BaseBetaWalkArith.lean`), and cross-checked by enumerating all walk solutions for the 198 coprime pairs with $f \leq 25$: no discrepancies. □

Remark 19. Both hypotheses are sharp: $n_b = 2f$ on an equal side is admissible exactly when $f^2 \leq 2e^2$, and $n_b = e + f$ on the base exactly when $f^2 \leq 2ef + e^2$, by the displayed identity. Of the 42 base-β prime candidates below 1000, 28 satisfy (i) — including the thick values 23, 131 and 167, since (i) needs only $f/e > \sqrt{2}$ — and 17 satisfy (ii). The theorem structures those cases; it does not by itself exclude any of them.

Theorem 20 (The $e = 1$ subfamily reduces to one walk)**.** *Let $e = 1$, $m = 1$ and $f \geq 3$, so the target is $(f^3, f^3, 3f^2 - 1)$, the tile is $(f, f^2 - 1, f^2)$ and $N = 3f^2 - 1$. Both hypotheses of Theorem 18 hold ($f^2 > 2$ and $f^2 > 2f + 1$). In any tiling:*

(i) *neither equal side carries a b-edge; each equal side begins and ends with a c-edge, and therefore carries at least two;*
(ii) *the base carries exactly one b-edge and exactly one c-edge — so $R_{\text{base}} = 1$ — and its first and last edges are a-edges.*

Thus the base walk is a permutation of (a^f, b, c) beginning and ending with a, with the b in position $3, \ldots, f$.

Proof. (i) An equal side has length f^3 and its edges lie in $\{f,\ f^2 - 1,\ f^2\}$. Reducing $n_a f + n_b(f^2 - 1) + n_c f^2 = f^3$ modulo f gives $f \mid n_b$. If $n_b \geq 2f$ then $n_b(f^2 - 1) \geq 2f^3 - 2f > f^3$ for $f \geq 2$. If $n_b = f$ then $f(f^2 - 1) = f^3 - f$ leaves f, forcing $n_c = 0$ and $n_a = 1$, hence no c-edge on that side — contradicting Proposition 25. So $n_b = 0$, and then $n_a + n_c f = f^2$, so $f \mid n_a$; write $n_a = fk$, $n_c = f - k$.

By Proposition 24 the apex tile on that side presents α, whose incident edges are b and c; since the side carries no b-edge, it ends with a c-edge. Its other end is handled in *(ii)*: there we show the base begins and ends with a-edges, so the base-corner tile's c-edge lies on the equal side, which therefore begins with a c-edge. Were $n_c = 1$ the single c-edge would be both the first and the last, forcing the side to consist of one edge and $f^3 = f^2$; so $n_c \geq 2$.

(ii) The base has length $3f^2 - 1$. Reducing modulo f gives $n_b \equiv 1$. If $n_b \geq f + 1$ then $n_b(f^2 - 1) \geq f^3 + f^2 - f - 1 > 3f^2 - 1$ for $f \geq 3$, so $n_b = 1$. Subtracting, $2f^2 = n_a f + n_c f^2$, so $n_a = f(2 - n_c)$ with $n_c \in \{0, 1, 2\}$, and $n_c \geq 1$ by Proposition 25.

If $n_c = 2$ then $n_a = 0$ and the base is three edges, a permutation of (b, c, c). By Proposition 28 the first two edges lie in $\{a, c\}$ and so do the last two; with no a-edge available this forces all three to be c-edges, contradicting $n_b = 1$. Hence $n_c = 1$ and $n_a = f$, so the base is a permutation of (a^f, b, c) with $n = f + 2$ edges, and Proposition 28 places the b away from the first two and last two positions.

It remains to exclude $E_1 = c$. Suppose it. Then the unique c-edge is E_1, so every one of $E_2, \ldots, E_n$ is an a- or b-edge and places a γ at one of its endpoints; there are $f + 1$ of them and $f + 1$ interior junctions $J_1, \ldots, J_{n-1}$. The corner tile T_A has its c-edge on the base, so its α vertex is J_1 and its γ vertex lies on the equal side; by Proposition 28 its partner T_3 presents γ at J_1. So J_1 already carries a γ, and since a π-vertex carries at most one (Proposition 23), E_2 must place its γ at J_2; inductively E_i places at J_i, and E_n places at J_n, a corner — contradicting Proposition 24. Hence E_1 is an a-edge, and the same argument at B gives E_n an a-edge. This is what *(i)* used. □

Remark 21. Theorem 20 reduces the whole $e = 1$, $m = 1$ subfamily — which contains the base-β prime candidates $N = 3f^2 - 1$, among them 11, 47, 107, 191, 431, 587 and 971 below 1000 — to the single question of whether that one base walk is realizable. It also corrects a plausible-looking plan: one might hope to close the subfamily by proving $R \geq 2$ on every side. That is false for the base. The walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$; on the base, "$R \geq 2$" is not a lemma reducing the problem but a restatement of it. On the two equal sides $R \geq 2$ does hold, and is proved above.

Remark 22 (The $e \geq 2$ residue)**.** The $e \geq 2$ residue ($N = 23, 59, 71, 83, 131, 167, \ldots$; the members 23 and 71 are now settled by search, Table 1) resists all three natural invariant methods, and in two cases *provably*:

- **Linear/colouring invariants** — dead by Remark 20.
- **b-edge census.** On a maximal segment of length L the two sides carry $q_1 \equiv q_2 \pmod f$ (both $\equiv -e^{-2}L$), and the entire census — in *every* modulus, and in its per-direction refinement — is a formal consequence of the single parity identity $N(a+b+c) \equiv 2X+Y \pmod 2$, valid for all (e, f, m). So no congruence of this type can bite.
- **Euler / vertex counting.** For any such tiling $V - E + F = 2$ is *identically* the sum of the three corner identities $\#\alpha = \#\beta = \#\gamma = N$; the vertex-type system collapses to $Q_{013} = 1 + P_{320} + 2Q_{640} + Q_{431} + W_{320}$ and $3P_{320} + P_{111} + 6Q_{640} + 4Q_{431} + 2Q_{222} + 3W_{320} + W_{111} = N - 3$, which is feasible for every $e \geq 2$, $m = 1$ target. So Euler carries no information here. Here W_{320} and W_{111} count the interior vertices at which some tile is met in the *relative interior* of one of its edges: such a tile contributes a straight angle, so the remaining tile angles sum to π rather than 2π and the figure is $(3, 2, 0)$ or $(1, 1, 1)$ (Proposition 23). These types are not optional — the dissections here are not edge-to-edge (Remark 18) — and an earlier version of this census omitted them. Their presence only enlarges the feasible set, so the conclusion is unaffected.

What *is* gained is citation-free structure, valid for all (e, f, m): the γ-trap (Proposition 25) and the corner parallelogram (Proposition 28), both proved above from the corner figures and the unsplittability of b, and both used by the exact search as prunes. They were checked against the two genuine certificates before being proved: the first-two-edges rule of Proposition 28 holds on all six sides of the 44- and 99-tilings, and at each of the four base corners exactly one of the two sides begins with a c-edge.

This isolates the residue to one clean statement. Since $q_{\text{base}} \equiv em \pmod f$ while $q_{\text{equal}} \equiv 0$, we have:

if the base carries no b-edge, then $f \mid em$, hence $f \mid m$, hence no tiling with $m = 1$ — for every e at once.

The 44-tiling ($m = 2$) indeed carries *zero* b-edges on its entire boundary, consistent with this. But Remark 21 now shows the converse direction is a dead end: the 99-tiling has $f \nmid m$ and its base carries seven b-edges, so "the base never carries a b-edge" is *false* in general, and the displayed implication — whose hypothesis is equivalent to $f \mid m$ — cannot be used to exclude $m = 1$ without already knowing $f \mid m$, which is false. A fourth method also fails, and provably: *boundary counting*. Each side S of ABC, cut into k_S whole edges, forces at least $2k_S - 1$ tiles to meet it (the k_S inner tiles, plus a distinct "middle" tile at each of the $k_S - 1$ junctions, since a triangle has no two collinear edges); this is exactly tight on the 44-tiling (base $k = 8$, $15 = 2 \cdot 8 - 1$ tiles). But at $m = 1$ the boundary is *minimal* — for $e = 1$ the least feasible total is $\sum k_S = 3f + 2$ — so counting sees only $O(f)$ of the $N = \Theta(f^2)$ tiles, and $\sum(2k_S - 1) = 6f + 1$ exceeds $N = 3f^2 - 1$ only at $f = 2$, the case already closed. The ratio $(6f + 1)/(3f^2 - 1) \to 0$: $m = 1$ is the case *least* constrained by counting, not the most.

So no *invariant* route survives: linear invariants, the census, Euler and counting are all provably vacuous, and the one arithmetic statement that would settle it ($f \mid m$) is now known to be false. What does survive is *structure*. Theorem 2 classifies the boundary walks for every (e, f) at $m = 1$, and in the thin regime $f > 2e$ reduces the whole branch to two possible bases and b-free equal sides — the $e \geq 2$ analogue of Remark 14, and the first statement in this branch that is uniform in e. Parts (iii)–(iv) need no thinness at all: at $m = 1$ *no equal side carries a b-edge*, for every (e, f), so all boundary b-edges lie on the base; and the base's b-count $Q = e + fj$ obeys $j(f - e) \leq e - 1$. It does not close the branch, but it does locate the difficulty exactly. The bound $j \leq (e - 1)/(f - e)$ is uniform, and it degrades precisely as f falls towards e: at $e = 1$ it pins $j = 0$ for every f (one b-edge on the base, the $e = 1$ rigidity of Remark 14, now a corollary); at $f > 2e$ it still pins $j = 0$; but in the *thick* regime $f \leq 2e$ it leaves $\lfloor (e - 1)/(f - e) \rfloor + 1$ values of j and the trichotomy dissolves. That is precisely where the open members sit — $N = 59$ is $(e, f) = (4, 5)$, so $j \leq 3$; $N = 66$ is $(3, 5)$, so $j \leq 1$; likewise $N = 23, 83, 131, 167$. We record the branch as open and settle its members one at a time by exhaustive search. Table 1 lists every member below 110; of the eleven, eight are settled and 59 and 66 are now excluded by exhaustive search (apex-down with the walk prune; 1 838 175 and 7 232 464 nodes), leaving 83 under search. ($N = 83$, $(e, f) = (5, 6)$, is the smallest base-β prime not yet settled.) The walk classification also feeds back into the search: the γ-trap and the corner parallelogram have the multiset corollaries $R \geq 1$ on every side, $Q_{\text{base}} \leq k - 4$ and $Q_{\text{side}} \leq k - 2$ (k the number of edges of the walk), and the engine now prunes any partial boundary walk that fits no surviving multiset (prune P5). The rule was validated positionally on all six sides of the two genuine tilings before use — the 44-tiling's base *aaaaccca* and the 99-tiling's base *aabbbbbbbcc*, which is exactly tight for $Q \leq k - 4$ — the strengthened engine re-finds the 44-tiling from scratch (852,093 nodes; the found tiling passes the independent verifier: congruence, exact area, containment, pairwise disjointness) — and the corner rule is applied only after checking that the tile's b-edge is unsplittable ($n_a a + n_c c = b$ has no solution), which is what forces the parallelogram. For $N = 59$ this cuts the admissible bases from seven to four, for $N = 66$ from four to three.

3. The $e = 1$ family

Theorem 23 (The $e = 1$ base-β family)**.** *Let $f \geq 2$, and consider the base-β instance $(e, f) = (1, f)$ at $m = 1$: the tile $(f,\ f^2 - 1,\ f^2)$ on the isosceles target with base angles β, base $Y = 3f^2 - 1$ and equal sides $X = f^3$, with tile count $N = 3f^2 - 1$. No such tiling exists. In particular no prime $p = 3f^2 - 1$ whose only base-β representation is $(1, f)$ is a number of congruent triangles into which a triangle can be cut; this covers $p = 11, 47, 107, 191, 431$.*

The proof is a finite chain of forcing lemmas; we state the architecture and prove the three arithmetic pivots, deferring the full strip-and-column exposition to the extended version of this note. The chain: the walk trichotomy (base walks $(2f, 1, 0)$, $(f, 1, 1)$, $(0, 1, 2)$, the last dead by Theorem 20); the corner and partner forcing; for each corner, either the quadratic growth completes ($k = f$) or a first break occurs, on the base (then the walk letter is forced into $\{a, c\}$, and two base-breaks are incompatible with the single c of $(f, 1, 1)$) or on a side. A side-break at level j forces a horizontal column of f tiles whose junction fillers are exactly the triangles on consecutive apexes (Lemma 24); the structure recurses one level per step and reaches the base, where the apexes must be walk junctions (no vertex of a tile can lie interior to a boundary edge), pinning the orientations (Lemma 25) and forcing the letters a^f on $[0, f^2]$ (Lemma 27). In $(f, 1, 1)$ the walk then ends in b or c, contradicting the corner forcing; in $(2f, 1, 0)$ the letter after f^2 is the b, while the terminal vertex at $(f^2, 0)$ carries exactly $\gamma + \alpha$ from the filler and column, leaving a wedge of exactly β, whose tile presents a or c — never b. Every case terminates in one of these contradictions.

Lemma 24 (Filler identity)**.** *With J_i the i-th column junction and ap_i the i-th apex, the triangle $(J_i, \mathrm{ap}_i, \mathrm{ap}_{i+1})$ has sides exactly c, b, a: $\left(\frac{3f^2-1}{2f}\right)^2 + c^2\sin^2\beta = f^4$ and $\left(\frac{f^2-1}{2f}\right)^2 + c^2\sin^2\beta = (f^2-1)^2$, both identities in f (clear denominators and expand; machine-checked in `lean/W2Core.lean`, axiom-free). Hence the column and its fillers tile the full inter-level strip and lay a run of a-edges one level down, whose far sides decompose only as single a-edges.*

Lemma 25 (Offset congruence)**.** *The terminal column's base apex lies at $\left[j(3f^2-1)+\sum_i \varepsilon_i\right]/(2f)+kf$ with $\varepsilon_i \in \{-(f^2-1), 3f^2-1\}$ per level. Modulo f — for every f, of either parity, since $f^2 \equiv 0$ — these are $+1$ and -1. Let p be the number of levels taking $+1$ and $q = j-p$ the rest, so $\sum \varepsilon_i \equiv p-q$ and*

$$\sum \varepsilon_i - j = -2q.$$

Integrality of a base vertex gives $f \mid 2q$ when f is odd, and $2f \mid 2q$ when f is even; either way $f \mid q$, and $q \le j < f$ forces $q = 0$. Hence $\sum \varepsilon_i = j$, every level is (γ,β)-oriented, and the apexes lie at consecutive multiples of f.

Remark 26 (The evenness hypothesis is removable)**.** Earlier versions of Lemma 25 worked modulo $2f$, where the two offsets are $+1$ and -1 only when f is even, since that congruence needs $f^2 \equiv 0 \pmod{2f}$; for odd f the residues are $f+1$ and $f-1$ and the computation does not transfer. That is why Theorem 23 was previously stated for even f only, leaving the members $N = 26, 74, 146, \ldots$ claimed by Theorem 29 but uncovered.

Both defects disappear at modulus f. There the residues are ± 1 with no parity hypothesis, and parametrising by the numbers p, q of $+1$ and -1 levels makes $\sum \varepsilon_i - j = -2q$ manifestly even, so the parity check that the odd case appeared to require never has to be made: it is absorbed into the parametrisation. The forcing is `lean/OffsetForcing.lean` (`forcing_odd`, `forcing_even`, `forcing`), general in f and axiom-clean, superseding the per-member even-f checks of `lean/W2Core.lean`.

Lemma 27 (Interior multiples)**.** *A base edge of length f^2-1 or f^2 placed with its left end below f^2 contains a multiple of f in its open interior, which would be a column apex interior to a boundary edge — impossible. Hence the letters covering $[0, f^2)$ are exactly f copies of a.*

The corner cases $(1,2)$, $(1,3)$, $(1,4)$ are additionally settled by exhaustive certificates (the seeded refutations of the side-break configurations; 267, 1809 and 1728 nodes, all branches dead), and the members $11, 26, 47$ by the full engine searches recorded in the main paper. The arithmetic layers of the chain are machine-checked in `BAdjacency.lean`, `Rigidity.lean`, `W2Core.lean`, `MidTriangle.lean` and `SurplusLattice.lean` (all kernel-only, axiom-clean).

4. The general base-β architecture

The $e=1$ machinery of Section 3 extends to every coprime pair. Four pillars are identities in (e,f): the column reach $b\sin\gamma = c\sin\beta$ (law of sines); the filler-mesh identity (the column's junction fillers are exactly the triangles on consecutive apexes, sides c, b, a; `lean/GeneralPillars.lean`); the sector inequalities $\gamma-(\alpha+\beta)=\alpha$ and $\cos\gamma = -e/2f < 0$; and the interior-multiple bounds. Two master lemmas complete the frame (`lean/MasterLemmas.lean`): a base segment of length a between forced junctions decomposes only as a single a (two applications of $\gcd(e,f)=1$), and the two corner-block types — f a-feet, or dually e c-feet — have identical footprints ef^2 (the relation $ec = fa$), so in every walk of the trichotomy the two corner structures consume exactly the non-b letters and the middle is exactly the e b-letters. Walks containing an a-block die at the middle's $\alpha+\beta$ corner with (b,b) rays; the walk $(0, e, 2e)$ is forced to $c^e b^e c^e$, its two dual blocks complete, and the resulting pentagon is refuted (certificate at $(2,3)$: 85 nodes, all branches dead; the general argument runs through the dual-junction overhangs). The orientation congruences force (γ, β) at every level for both parities of f. The assembled exposition of the general recursion is in preparation; every arithmetic component cited above is kernel-checked, and the certified refutations cover $(1, 3)$, $(1, 4)$, $(2, 3)$.

5. The base-β branch: no instance at $m = 1$

This section assembles the complete argument. Throughout, (e, f) is coprime with $1 \le e < f$, the tile is $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$ with angles (α, β, γ), $\gamma = 2\alpha + \beta$, $3\alpha + 2\beta = \pi$, α/π irrational; the target Δ is isosceles with base angles β, base $Y = e(3f^2 - e^2)$, equal sides $X = f^3$, and $N = 3f^2 - e^2$.

The argument below is *conditional*, on one geometric statement about the base corners that we isolate here and do not prove. We state it first, so that nothing downstream is read as unconditional.

Hypothesis 28 (Complete corner walls)**.** Let $\mathcal{T}$ be a tiling of the base-β target at $m = 1$. Then neither base-corner structure is starved or broken: each is the complete block — f a-feet at the west corner, e c-feet at the east — whose hypotenuse is a straight segment, of length $f\,b$ resp. $e\,b$, partitioned into whole b-edges.

Theorem 29 (The base-β branch, conditional on Hypothesis 28)**.** *Assume Hypothesis 28. Then Δ admits no tiling by N congruent copies of the tile. Consequently no prime $N \equiv 11 \pmod{12}$ is a number of congruent triangles into which a triangle can be cut, and with the results of the main paper the folklore conjecture would follow: no prime $N \equiv 3 \pmod 4$, $N > 3$, is such a number.*

Remark 30 (Why the hypothesis is needed, and where it bites)**.** Hypothesis 28 is exactly the step Remark 15 already identifies as "the single highest-value open step in this branch"; isolating it here makes the dependence explicit rather than leaving it distributed over the funnels. Two places consume it.

First, the funnels of §5.5 dispose of a walk by completing both corner blocks; if a corner is *starved or broken* the text routes the case to "the walk arithmetic … exactly as in the $e = 1$ case", and the $e = 1$ case is Theorem 23, whose proof defers the strip-and-column exposition. At $e = 1$ this is not a corner case but the whole problem: Theorem 20 forces the base walk to begin and end with a-edges, which is incompatible with the complete-block reading of the base, so *every* hypothetical $e = 1$ tiling falls into the starved-or-broken clause.

Second, Lemma 39 is stated "after both dual blocks complete", and its stubs are region-boundary segments only because the opposite side is a wall of whole b-edges. In a dissection that is not edge-to-edge — which the main paper explicitly permits — a far-side edge may begin inside a stub and overhang it, so no such segment need be created. The arithmetic of the lemma is unaffected and is machine-checked general in (e, f) (Lemma 40, `lean/PentagonLemma.lean`); what is conditional is that a stub arises at all.

The unconditional consequences of this section are therefore the arithmetic ones, together with the members settled by certified exhaustive search independently of any of it: $N = 11, 23, 26, 47, 59, 66, 71, 107$. The member $(1, 2)$, i.e. $N = 11$, is moreover closed unconditionally by Theorem 1, whose argument is boundary-only and needs no corner-block hypothesis.

Remark 31 (A corner block can break: the hypothesis needs $m = 1$ essentially)**.** It is worth recording what the corner blocks are, and that Hypothesis 28 is genuinely a statement about $m = 1$ and not a local fact.

First the identification. At a base corner the complete block with f a-feet has footprint $fa = ef^2$ and, by the law of cosines against the base angle β, its far side has length exactly

$f^3 = fc$; so *the block is the tile scaled by* f, and it contains f^2 tiles. Dually the c-feet block is the tile scaled by e, with footprint $ec = ef^2$, far side $e^2 f = ea$, and e^2 tiles. At $m = 1$ the f-block's far side is the whole equal side, so its hypotenuse is the cevian from the apex. The two counts and the middle add up,

$$N \;=\; f^2 \;+\; e^2 \;+\; 2\,(f^2 - e^2),$$

which is $3f^2 - e^2$, as it must be.

Now the point. The local ingredients of the strip argument — the corner decomposition $\{\beta\}$ with flanks $\{a, c\}$, the single-b partner, the parallelogram mate, the kite kills, the surplus lattice — are all statements about the tile and about angles. None of them mentions m. If they sufficed to force a complete block they would force one at every m. They do not: among the three genuine tilings of base-β targets that we possess, all with tile $(2, 3, 4)$ and all verified exactly, the 44-tiling `N44B` of $(16, 16, 22)$ has base walk a^4c^3a, and at its right-hand corner *neither* candidate block is complete — the f-block (4 straddling tiles) nor the e-block (1 straddler) — because the last base letter there is a single a where a block would need the footprint $ef^2 = 4$ to be cleanly composed. The other two, `N44C` and the 99-tiling of $(24, 24, 33)$, do have both corners complete.

So a broken corner is not merely permitted in principle; it occurs in a tiling we have exhibited. Any proof of Hypothesis 28 must therefore use $m = 1$ in an essential way — the natural candidate being the base b-count, which is e at $m = 1$ but is 0 for both 44-tilings and 7 for the 99-tiling — and no argument phrased purely in the local toolkit above can succeed. We record this because it is the sharpest constraint we have on how the gap can be closed, and because it disqualifies the shortest-looking route.

5.1. **The angle calculus.** Every corner argument below is an instance of one finite computation, which we isolate once.

Since α/π is irrational (Sec. 1), so is α/β: from $\alpha/\beta = p/q$ and $3\alpha + 2\beta = \pi$ one gets $\beta = q\pi/(3p + 2q)$, making β and hence α rational multiples of π. Consequently an angle has *at most one* representation $x\alpha + y\beta$ with $x, y \geq 0$: two would give $(x - x')\alpha = (y' - y)\beta$. Encoding

$$\alpha \mapsto (1, 0), \qquad \beta \mapsto (0, 1), \qquad \gamma = 2\alpha + \beta \mapsto (2, 1), \qquad \pi = 3\alpha + 2\beta \mapsto (3, 2),$$

a vertex figure at an angle (X, Y) with n_α, n_β, n_γ tile corners is exactly a solution of $n_\alpha + 2n_\gamma = X$, $n_\beta + n_\gamma = Y$ in nonnegative integers. All of the following are that system, solved; each is machine-checked in `lean/AngleArithmetic.lean` (core toolchain, no `sorry`).

Lemma 32 (Angle calculus)**.** *For every member* (e, f)*:*

1. (No right angle.) $\pi/2$ *has no representation: it would force* $2x = 3$*. Hence no piece of any dissection of a base-*β *region has a right angle. In particular the target cannot be split along its altitude, and no perpendicular cut is available anywhere in the branch.*
2. (Apex.) *At the apex,* $(X, Y) = (3, 0)$*, so* $n_\beta = n_\gamma = 0$ *and* $n_\alpha = 3$*: the apex carries exactly three* α*-corners.*
3. (Base corner.) *At* $(0, 1)$ *the only figure is a single* β*-corner, with flanks in* $\{a, c\}$*.*
4. (γ-trap.) *At a straight angle,* $(3, 2)$*, the only figures are* $\{\alpha, \alpha, \alpha, \beta, \beta\}$ *and* $\{\gamma, \alpha, \beta\}$*; in particular at most one* γ *meets any interior boundary point.*
5. (Wedge $\alpha + \beta$.) *At* $(1, 1)$ *the only figure is one* α *and one* β*.*

Remark 33 (Where β sits)**.** $3\alpha > \beta$ iff $\beta < \pi/3$ iff $\cos\beta > \frac{1}{2}$ iff $e(3f^2 - e^2) > f^3$. For $e = 1$ this is $f^3 - 3f^2 + 1 < 0$, which holds only at $f = 2$; in general it is $f/e < 2.879\ldots$, the real root of $t^3 - 3t^2 + 1$. So $(1, 2)$ — the one member whose instance spectrum is completely determined (Theorem 165) — is the unique member of its family with $\beta < \pi/3$, and should not be treated as typical of $e = 1$.

5.2. **The γ-grading of directions.** Before the forcing arguments, one structural fact about the whole branch, which does not seem to have been noticed and which costs two lines.

Proposition 34 (γ-grading)**.** *Let $\mathcal{T}$ be any tiling of any target by a base-β tile. Then every edge direction occurring in $\mathcal{T}$ is an integer multiple of γ modulo π. Writing a direction as $x\alpha + y\beta$, the integer is $L = 2x - 3y$; it is well defined because $3\alpha + 2\beta = \pi$.*

Proof. From $\gamma = 2\alpha + \beta$ and $3\alpha + 2\beta = \pi$:

$$2\gamma = 4\alpha+2\beta = 4\alpha+(\pi-3\alpha) = \alpha+\pi \equiv \alpha, \qquad -3\gamma-\beta = -6\alpha-4\beta = -2(3\alpha+2\beta) = -2\pi \equiv 0,$$

so $\alpha \equiv 2\gamma$ and $\beta \equiv -3\gamma$ modulo π. Hence $\langle\alpha, \beta\rangle = \langle 2\gamma, 3\gamma\rangle = \langle\gamma\rangle$, as $\gcd(2, 3) = 1$.

It remains to see that all directions lie in $\theta_0 + \langle\alpha, \beta\rangle$ for one reference θ_0. Read around a tile, the direction turns by the exterior angle, $-$(interior) modulo π, so a tile's three directions are θ, $\theta-\alpha$, $\theta-\alpha-\beta$: a translate of a fixed three-element set S. If two tiles share a boundary segment of positive length they share a direction, so their offsets differ by an element of $S - S \subseteq \langle\alpha, \beta\rangle$. The tile adjacency graph is connected, the target being connected and the tiles finitely many closed sets with disjoint interiors covering it; propagating along a chain from a reference tile gives the claim. $\square$

Two consequences are used below. Round a tile the label shifts are $-1, -2, +3$ at the γ-, α- and β-vertices, so a tile's three edge labels are $\{L - 2, L, L + 1\}$, or $\{L - 1, L, L + 2\}$ in the mirrored chirality: *span exactly three*, with the b-edge carrying L in either case. And the target's own sides are graded 0 (base) and ∓ 3 (the equal sides).

Remark 35. The point of Proposition 34 is that α/π is irrational, so the group generated by α and β is dense modulo 2π; one expects no finite structure on directions at all, and indeed $\langle\gamma\rangle$ *is* dense in $\mathbb{R}/\pi\mathbb{Z}$. What the grading says is not that the direction set is discrete but that it is *cyclic*: the directions are the multiples of one angle, so they carry a canonical integer label, and a given tiling uses only a narrow window of those labels. This is worth isolating: the invariant no-go of the main paper is proved for functionals on a dense direction set, and does not automatically apply to functionals graded by L. The arithmetic is machine-checked in `lean/LabelCalculus.lean`, and the grading was tested against every tiling in this work — the 44-tilings, the 99-tiling, the 28-tiling on a cevian target and the scalene four-component 77-tiling — with no direction failing.

Proposition 36 (The direction group of a branch)**.** *Let T be a triangle whose angles satisfy a single primitive relation, i.e. the lattice $\Lambda = \{(x, y) \in \mathbb{Z}^2 : \ x\alpha + y\beta \equiv 0 \pmod{\pi}\}$ is generated by one vector (p, q). Then in any tiling by congruent copies of T all edge directions lie in one coset of $\langle\alpha, \beta\rangle \subset \mathbb{R}/\pi\mathbb{Z}$, and*

$$\langle\alpha, \beta\rangle \ \cong\ \mathbb{Z}^2/\langle(p, q)\rangle \ \cong\ \mathbb{Z} \oplus \mathbb{Z}/d, \qquad d = \gcd(p, q).$$

Writing $(p, q) = d\,(p', q')$, the free coordinate is the label $L(x, y) = q'x - p'y$, which is surjective onto $\mathbb{Z}$ with kernel exactly $\langle(p', q')\rangle$.

Proof. The coset statement is the propagation argument of Proposition 34, which used no property of the angles. For the quotient, $\gcd(p', q') = 1$ gives u, v with $up' + vq' = 1$; then $L(vn, -un) = n$, so L is onto, and $L(x, y) = 0$ means $q'x = p'y$, whence $p' \mid x$ and (x, y) is a multiple of (p', q'). The remaining generator contributes $\mathbb{Z}/d$. $\square$

Remark 37 (A torsion dichotomy across the classification)**.** Solving for Λ in each branch of the Laczkovich classification:

branch	relation	(p,q)	direction group
base-β	$3\alpha+2\beta=\pi$	$(3,2)$	$\mathbb{Z}$
$\gamma=2\alpha$	$3\alpha+\beta=\pi$	$(3,1)$	$\mathbb{Z}$
$\gamma=2\pi/3$	$\alpha+\beta=\pi/3$	$(3,3)$	$\mathbb{Z}\oplus\mathbb{Z}/3$
$\gamma=\pi/3$	$\alpha+\beta=2\pi/3$	$(3,3)$	$\mathbb{Z}\oplus\mathbb{Z}/3$
right angle	$\alpha+\beta=\pi/2$	$(2,2)$	$\mathbb{Z}\oplus\mathbb{Z}/2$

The two branches in which α/π is irrational — base-β and $\gamma=2\alpha$ — are exactly those with $d=1$, that is, with torsion-free direction group. Every special-angle branch carries torsion. We record the coincidence without asserting a mechanism.

The phenomenon that tiles related by rotation through a single irrational angle generate a cyclic orientation group is not itself new; it is familiar from substitution tilings, where the pinwheel tiling's orientations are the multiples of $2\arctan(1/2)$ and are dense in the circle. What is recorded here is the computation of Λ for each branch, the resulting torsion dichotomy, and, in the base-β case, the identification of the free generator as γ itself. The arithmetic is machine-checked in `lean/DirectionGroup.lean`.

5.3. **The forcing toolkit.** *Corners.* A vertex of angle θ between two boundary rays admits only the tile-angle decompositions of θ, and the extreme tiles must present flanks compatible with the rays' first edges; for $\theta=\beta$ the unique decomposition is $\{\beta\}$ and the flanks are $\{a,c\}$, one per ray (the two orientations). *Partners.* The corner tile's b-edge is a boundary-anchored chord; both sides partition into whole edges of total b, and $x\,a+y\,b+z\,c=b$ forces $(x,y,z)=(0,1,0)$ (reduce mod f; sizes), so the far side is a single b-edge; of the two congruent mates the reflected one repeats the corner angle and overflows ($2\gamma=\pi+\alpha$), leaving the parallelogram partner. *Strips.* Inductively, the k-th chord ($k<f$ resp. $k<e$ for the dual) carries k aligned b-edges; the surplus lattice (Sec. 4) forbids $0<|dQ|<f$, so the far side matches b-for-b; kites die bottom-up (γ-repeat at the base, then $2\gamma>\pi$ at each junction), partners are forced, and each junction wedge is exactly β resp. the dual angles. All statements are symmetric under the a-feet/c-feet duality; footprints agree by $ec=fa$.

5.4. **Runs, columns, fillers, recursion.** Along any covered run of an anchored line both sides partition into whole edges of equal total, and the surplus lies in $\Lambda=\mathbb{Z}(f,0,-e)\oplus\mathbb{Z}(f-e,-f,f-e)$. A break tile's c-edge admits no whole-edge completion of $c-a=f(f-e)$ (mod f and gcd), so the run closes minimally as the generalized column: e c-edges against f a-edges ($ec=fa$). The column's junction fillers are exactly the triangles on consecutive apexes with sides c,b,a (the filler identity, an identity in (e,f)), so column and fillers tile the inter-level strip completely and lay an a-run one level down; an interior a-edge's far side decomposes only as a single a (Master Lemma 1: $\gcd(f^2-e^2,ef)=1$), so the run forces the next level's column. The recursion reaches the base; a tile vertex on the base is a walk junction (no one-sided T-junction), and the accumulated orientation offsets satisfy, modulo $2f$, the congruence that forces every level to the (γ,β) orientation — for both parities of f — placing the terminal apexes at consecutive multiples of a. Between consecutive apexes the walk letters decompose only as single a's (Master Lemma 1 again), so the letters spanning the apex range are forced.

5.5. **The funnels.** By Master Lemma 2 the two corner structures — f a-feet or e c-feet — have equal footprints ef^2, so in each walk of the trichotomy they consume exactly the non-b letters and the middle is the e b-letters: $(2f,e,0)\mapsto(a,a)$, $(f,e,e)\mapsto(a,c)$, $(0,e,2e)\mapsto(c,c)$. (The step just used — that a stub is a genuine region-boundary segment, so that whole-edge partition applies — assumes no tile straddles the ray. That assumption is false in general; see Remark 14, and Remark 30 for how it is absorbed into Hypothesis 28.) If a corner structure is starved or broken, the recursion of the previous subsection forces the a-prefix regardless (the apex lattice), and the β-wedge at a column terminal (filler γ + column α leave exactly β, whose tile presents a or c) forbids a b-letter there; the walk arithmetic then kills exactly as in the $e = 1$ case.

Lemma 38 (Terminal wedge)**.** *Let a base-reaching column terminate at base vertex $V = (x_0 + fa, 0)$, x_0 its anchor offset. Then the angle at V decomposes as γ (the last filler) + α (the last column tile) + β, and the base edge east of V is an a- or a c-letter.*

Proof. The filler $(J_{f-1}, \mathrm{ap}_{f-1}, \mathrm{ap}_f)$ has its a-edge on the base ending at V with the a–b corner there, i.e. γ; the column tile ends at V with its apex angle α. The remaining budget is $\pi - \gamma - \alpha = \beta$, whose unique decomposition is a single β-tile; its flanks are a and c, one of which lies along the base east of V. A b-letter there would have to be that flank, and $b \notin \{a, c\}$. □

Lemma 39 (Pentagon)**.** *In the walk $(0, e, 2e)$, after both dual blocks complete, the middle region — base b^e, dual hypotenuses b^e, side segments of length fb, apex 3α, containing $3(f^2 - e^2)$ tiles — admits no tiling.*

Proof. Consider the west base corner $W_0 = (ef^2, 0)$, of interior angle $\pi - \alpha$ (the dual block's α sits west of it). Its decompositions are $\{\gamma, \beta\}$ and $\{2\alpha, 2\beta\}$; in each, the tile adjacent to the dual hypotenuse presents along it a flank from $\{a, c\}$ — β-tiles carry (a, c), γ-tiles (a, b) with the b unavailable (it would repeat the hypotenuse edge inside the wedge), α-tiles (b, c) likewise contribute c. The hypotenuse is partitioned into whole b-edges of the dual strips, so a flank of length a or c never terminates at a hypotenuse junction: each placement leaves a stub of length $s_1 \in \{c - b, |a - b|\} = \{e^2, |ef - f^2 + e^2|\}$ against the next b-edge. Such a stub is a region boundary segment and must be partitioned into whole tile edges. Neither can be: the first lies in $(0, \min(a, b))$, which contains no element of the numerical semigroup $\langle a, b, c\rangle$, and the second is a gap of that semigroup by Lemma 40. So no partition exists and the region dies. The certified refutation at $(2, 3)$ (85 nodes, every branch dead, all kills of run or wedge type) instantiates the argument; the stub arithmetic above is member-independent. □

For the middles: any walk containing an a-block dies at the middle's a-side corner (Lemma 38 and the (b, b) row of the vertex table), of angle $\alpha + \beta$ with both rays presenting b — the T_{mid} kill of the main paper. The walk $(0, e, 2e)$ is forced to $c^e b^e c^e$ (the dual break flanks are $\{a, c\}$ and no a exists); both duals complete and the middle is the pentagon: base b^e, two dual hypotenuses b^e, two side segments of length fb, apex 3α. Its west wedge is $\pi - \alpha$; every decomposition places against the dual hypotenuse a flank from $\{a, c\}$, never matching b: each placement leaves a stub, and the stubs walk in steps of e^2 (through-vertex overhangs) or $|a - b|$; each such stub is a gap of the numerical semigroup $\langle a, b, c\rangle$ — the first, $e^2 \bmod b$, because it lies in $(0, \min(a, b))$, and the second, $|a - b|$, by Lemma 40 — so no whole-edge partition exists and the region dies. No iteration occurs: both stubs are gaps at the first step, for every member. (Certificate at $(2, 3)$: 85 nodes, all dead, kills exclusively of run/wedge type.)

Lemma 40 (The second stub is a gap)**.** *Let $\gcd(e, f) = 1$ with $1 \le e < f$, and let $(a, b, c) = (ef,\, f^2 - e^2,\, f^2)$. Then $|a - b|$ is a gap of the numerical semigroup $\langle a, b, c\rangle$: it is nonzero and admits no representation $xa + yb + zc$ with $x, y, z \ge 0$.*

Proof. First $\gcd(a, b) = 1$: a common divisor of e and b divides $b + e^2 = f^2$, hence divides $\gcd(e, f^2) = 1$; a common divisor of f and b divides $f^2 - b = e^2$, hence divides $\gcd(f, e^2) = 1$; so $\gcd(ef, b) = 1$. Next both generators exceed 1: $a = ef \ge 2$, and $b = (f - e)(f + e) \ge 3$ since $f - e \ge 1$ and $f + e \ge 3$. Hence $a \ne b$ — equality with $\gcd(a, b) = 1$ would force $a = b = 1$ — so $t := |a - b| > 0$.

Now $t < c$: if $a > b$ then $t = ef + e^2 - f^2 < f^2$ because $ef < f^2$ and $e^2 < f^2$; if $b > a$ then $t = f^2 - e^2 - ef < f^2$. So $z = 0$ in any representation. Also $t < \max(a, b)$ trivially, so the larger of a, b has coefficient 0 as well. What remains is $t = k\cdot\min(a,b)$ for some $k \geq 1$, i.e. $\min(a,b)$ divides $|a-b|$ and therefore divides $\max(a,b)$ too; with $\gcd(a,b)=1$ this forces $\min(a,b)=1$, contradicting $a \geq 2$, $b \geq 3$. □

Remark 41. The lemma is stronger than the pentagon argument needs and removes its only iterative step. The earlier treatment sent the stub sequence walking by e^2 modulo b until it entered $(0, \min(a,b))$, a gap whose reachability had to be argued geometrically; the statement above shows the second stub is *already* a gap, for every member, so no walk occurs. This is not merely a shorter route to the same place: an exhaustive check over every member with $f \leq 200$ (12 231 pairs, `rust/stubgap`) finds that *both* initial stubs are gaps at the first step in every single case, so the walk the earlier argument invoked never executes even one step. The mechanism can be deleted from the lemma rather than justified. It is formalized in `lean/PentagonLemma.lean` (`pentagon_lemma`, general in (e,f), standard axioms only), together with $\gcd(a,b)=1$, and cross-checked by exhaustive semigroup search over all coprime pairs with $f \leq 400$ (`rust/stubgap`). Note that $(0, \min(a,b))$ and $\{|a-b|\}$ are genuinely different gaps: at $(1,3)$ the tile is $(3,8,9)$, $\min(a,b)=3$, and $|a-b|=5$ lies outside $(0,3)$ while still being a gap of $\langle 3,8,9\rangle$.

Lemma 42 (Two c-edges are forced; $p \leq f-2$)**.** *At $e=1$, $m=1$, the first and last edges of each equal side are c-edges: the first is the base-corner tile's side flank (its base flank is an a-edge by Theorem 20), and the last belongs to the apex figure $\{3\alpha\}$, whose side edge is b or c, with b excluded by Lemma 43. Hence $f-p \geq 2$, i.e. $p \leq f-2$. In particular $p=0$ for $f=2$: the member $(1,2)$ satisfies Hypothesis 28 outright, by boundary analysis alone, and the closure of $N=11$ in Theorem 1 is recovered through Remark 45. For $f=3$ the only surviving value is $p=1$, whose side word is forced to $c\,a\,a\,a\,c$; by the run rigidity its chirality word is $(-4)^j(-2)^{3-j}$ for some $0 \leq j \leq 3$, so the entire boundary configuration space at $(1,3)$ consists of four explicit placements.*

Lemma 43 (The equal sides carry no b-edge)**.** *Let $m=1$ and let (e,f) be any member. Then every side walk $P'a + Q'b + R'c = f^3$ obeying the γ-trap $R' \geq 1$ has $Q'=0$.*

Proof. Reducing the walk modulo f kills $P'a$ and $R'c$ and leaves $f \mid Q'e^2$; as $\gcd(e,f)=1$, $f \mid Q'$. Write $Q' = qf$ and suppose $q \geq 1$. Dividing the walk by f,

$$P'e + q\,b + R'f = f^2, \qquad b = f^2 - e^2,$$

so in particular $q\,b + R'f \leq f^2$. Reducing *this* modulo e gives $f(qf+R') \equiv f^2$, and f is invertible mod e, so $R' \equiv f(1-q) \pmod{e}$.

Case $q=1$. Then $R' \equiv 0 \pmod{e}$, and $R' \geq 1$ forces $R' \geq e$, so $R'f \geq ef$. But $q\,b + R'f \leq f^2$ gives $R'f \leq f^2 - b = e^2$, whence $ef \leq e^2$ and $f \leq e$, contradicting $e < f$.

Case $q \geq 2$. From $q\,b + R'f \leq f^2$ with $q \geq 2$, $R' \geq 1$ we get $2(f^2-e^2) + f \leq f^2$, i.e. $f^2 + f \leq 2e^2$; hence $f < e\sqrt{2} < 2e$, so $u := f - e$ satisfies $1 \leq u \leq e-1$. Put $t := q-1 \geq 1$. Since $f \equiv u \pmod{e}$, the congruence above reads $R' \equiv -ut \pmod{e}$. Write $ut = v + we$ with $0 \leq v \leq e-1$ and $w \geq 0$. Substituting $f = e+u$ and $b = u(2e+u)$, the constraint is

$$(t+1)\,u\,(2e+u) + R'(e+u) \;\leq\; (e+u)^2.$$

If $v=0$ then $e \mid R'$ and $R' \geq e$, so the display gives $(t+1)u(2e+u) \leq (e+u)^2 - e(e+u) = u(e+u)$, i.e. $(t+1)(2e+u) \leq e+u$, impossible as $t+1 \geq 2$. If $v \geq 1$ then $R' \geq e-v$, so $(t+1)u(2e+u) \leq (e+u)(u+v)$. But $(t+1)u = ut + u = v + we + u$, so

$$(t+1)u(2e+u) - (e+u)(u+v) = (v+u)\big[(2e+u)-(e+u)\big] + we(2e+u) = e\big[(v+u) + w(2e+u)\big] > 0,$$

a contradiction. Hence $q=0$, i.e. $Q'=0$. □

Remark 44. The special case $e^2 < f$ — which already covers every $e = 1$ member — has the one-line proof "$Q' \neq 0 \Rightarrow Q' \geq f \Rightarrow Q'b + R'c \geq f^3 - fe^2 + f^2 > f^3$", and it is that case which is machine-checked in `lean/SideNoB.lean` as `side_no_b`, with `side_no_b_e_one` the $e = 1$ specialisation. The unconditional statement is `side_no_b_uncond` in the same file, formalised by the identity $(\star)$ $t\,b + R'f + P'e = e^2$ used in the proof above; both depend only on the standard axioms. Cross-checked by exhaustive search over every member with $f \leq 200$.

Remark 45 (At $e = 1$, Hypothesis 28 is exactly "the sides carry no a-edge")**.** Lemma 43 is one of the few statements in this development genuinely special to $m = 1$ — at $m = 2$ it is false, by the identity $e(ef) + f(f^2 - e^2) + f \cdot f^2 = 2f^3$ — and by Remark 31 any proof of Hypothesis 28 must use $m = 1$ essentially. It is therefore worth following where it leads at $e = 1$, where it reduces the hypothesis to a single integer.

With $Q' = 0$ the side equation $P'f + R'f^2 = f^3$ gives $P' + R'f = f^2$, so $f \mid P'$; writing $P' = fp$,

$$(P', R') = (fp,\ f - p), \qquad 0 \leq p \leq f - 1,$$

the γ-trap being $R' \geq 1$. The complete west block is the tile scaled by f, so its far side — which at $m = 1$ is the whole equal side — is subdivided into exactly f c-edges, i.e. $p = 0$. Hence, at $e = 1$,

$$\text{Hypothesis 28 at the west corner} \iff p = 0 \iff \text{the equal side carries no } a\text{-edge.}$$

Two corner facts pin the ends of that walk. At the base corner the unique figure is a single β-tile with flanks $\{a, c\}$, and Theorem 20 puts an a-edge at the base end, so the side's first edge is a c. At the apex the figure is exactly three α-corners (Lemma 32(2)), and the two edges at a tile's α-vertex are b and c, so with $Q' = 0$ the side's last edge is a c as well. Every equal side therefore reads $c\,(\cdots)\,c$ with fp interior a-edges.

So the open step at $e = 1$ is the single statement "$p = 0$", a direct analogue of Lemma 43 with a in place of b, and it is not arithmetic: the walk equation admits every $p \leq f - 1$. Finally, note that $p = 0$ closes the subfamily outright, without any further funnel analysis: complete blocks put f a-feet at one base corner and $e = 1$ c-foot at the other, so the base reads $a^f\,b\,c$ and ends in a c-edge, whereas Theorem 20 forces a-edges at both ends. The contradiction is immediate. Hypothesis 28 is thus not a step towards the $e = 1$ case; at $e = 1$ it *is* the case.

Lemma 46 (First-run orientation)**.** *In any tiling of the* $(1, f)$ *target at* $m = 1$*, every tile of the first* a*-run of an equal side is mirrored: its* γ *sits at the upper junction and its label is* -2*. In particular at* $(1, 3)$*, where Lemma 42 forces the side word* $c\,a\,a\,a\,c$*, the boundary configuration is* unique*: both* c*-tiles direct with label* -1*, all three* a*-tiles mirrored with label* -2*.*

Proof. The corner tile's b-edge is a chord with both endpoints on the target boundary: its upper end is the first side junction J, its lower end the base point $(f, 0)$, and the ray beyond either end leaves the target (past J it crosses the side, the two directions being non-parallel; past $(f, 0)$ it descends below the base). So no covering edge overshoots, the far side partitions into whole edges of total b, and by the partner relation $xa + yb + zc = b \Rightarrow (x, y, z) = (0, 1, 0)$ (machine-checked, general in (e, f): `PentagonLemma.partner_unique`) the far side is a single b-edge: a partner tile P shares the whole edge. The ends of a b-edge host α and γ, so P contributes α or γ at J; the corner tile contributes α there. If the first a-tile were direct it would contribute γ at J, and the straight-angle figure at J would contain one α, one γ, and a further corner from $\{\alpha, \gamma\}$; neither admissible figure does (machine-checked: `OrderForcing.first_run_kill`). So the first a-tile is mirrored, and the run rigidity (`gamma_far_absorbing`: mirrored is absorbing along the run) makes the whole run mirrored. $\square$

Lemma 47 (Top junction; $p \le f-3$ for $f \ge 4$)**.** *Let an a-run end at the last side junction J (final c-block of length 1) with the run's last tile mirrored. The junction figure is $\{\gamma,\alpha,\beta\}$, its interior α-filler has its edges along the two junction rays, and the edge along the top tile's a-ray is $c=f^2$ (mirrored filler) or $b=f^2-1$ (direct filler), with endpoint $J+(f^2,0)$ resp. $J+(f^2-1,0)$. Testing these against the right side of the target reduces, exactly, to the signs of*

$$f^3-3f^2+1 \quad (\textit{mirrored filler outside} \iff >0 \iff f\ge 3), \qquad f^3-3f^2-f+1 \quad (\textit{direct filler outside} \iff >0$$

Hence for $f\ge 4$ no such junction exists; for $f=3$ the filler is forced direct with label 0. Consequently: $p=f-2$ gives c-count 2, forcing the single-run word $c\,a^{f(f-2)}\,c$, which is all-mirrored by Lemma 46 and ends at such a junction — dead for $f\ge 4$. So

$$p \;\le\; f-3 \qquad (f\ge 4),$$

and at $(1,4)$ in particular $p\le 1$. The first polynomial is the thin/thick discriminant of Remark 33: the member $(1,2)$ is on its other side, which is again why $f=2$ is the degenerate case. The sign reductions are verified exactly (rational cross products over $\mathbb{Q}(\sqrt{D})$) and the two positivity facts are machine-checked (`OrderForcing.mirrored_filler_outside`, `direct_filler_outside`), as are the single-run combinatorics and the count bound (`two_c_single_run`, `p_le_f_sub_three`).

Lemma 48 (c-chord dichotomy)**.** *Let a chord of length c be blocked at both ends, so that its far side partitions into whole tile edges of total c. Then $xa+yb+zc=c$ forces $(x,y,z)=(0,0,1)$, or, exactly when $e=1$, $(x,y,z)=(f,0,0)$: the far side is a single c-edge or (at $e=1$ only) a run of f a-edges. Machine-checked general in (e,f) (`CChord.c_chord_dichotomy`); the modular step is the identity $(y-1)f^2=e(ye-xf)$.*

Remark 49 (Tile-interior blocking)**.** Two ends of a chord can be blocked by the target boundary; they can also be blocked by the *interior of an already-forced tile*, since a covering edge extended past the chord's end would enter it. Forced tiles thereby create new fully-blocked chords, and the forcing propagates.

Theorem 50 (Hypothesis 28 holds at $(1,3)$)**.** *No tiling of the $(1,3)$ target at $m=1$ carries an a-edge on an equal side. Hence, by Remark 45, no tiling of the $(1,3)$ target at $m=1$ exists at all.*

Proof. Suppose an equal side carries an a-edge. By Lemma 42, $p=1$ and the side word is $c\,a\,a\,a\,c$; by Lemma 46 the boundary configuration is unique, and the corner tile T_1, its b-partner P, and the three mirrored a-tiles are all pinned. Write the base corner at the origin, J for the first side junction, and $Q=J+(f,0)$ for the end of P's horizontal a-edge, so that P's c-edge runs from $(f,0)$ to Q parallel to the side. That edge is blocked at $(f,0)$ by the base and at Q by the interior of the second side tile (Remark 49; the directions differ, so the crossing is transversal), and Lemma 48 applies: its far side is a single c-edge or a run of three a-edges.

The vertex figure at $(f,0)$ already contains T_1's γ and P's α, leaving exactly one β (the representation of $\pi-\gamma-\alpha$ is unique). The tile carrying it presents, from its β-corner with flanks $\{a,c\}$, the base's second edge.

Single-c branch. The c-tile C has β at $(f,0)$ and its a-edge on the base: the base word is $a\,a\,b\,c\,a$, so the third base edge is the b-edge $[2f,\,2f+b]$. Label consistency forces C direct, and $C=T_1+(f,0)$: the corner block grows by one cell. Now C's b-edge is blocked at $(2f,0)$ by the base and at Q by the same tile interior, so its far side is a single b-edge (Lemma 43's companion fact, `partner_unique`), carried by a tile Z whose corners at the two ends are α and γ in one of two orientations. The reflected orientation puts γ at $(2f,0)$, where C's γ already sits: the straight-angle figure admits at most one γ. So $Z=P+(f,0)$, with α at $(2f,0)$ and γ at Q, and the figure at Q closes as $\{\beta, \alpha, \gamma\}$. At $(2f, 0)$ the figure now contains C's γ and Z's α, leaving exactly one β, whose tile presents the base edge from $(2f, 0)$ out of flanks $\{a, c\}$. But that edge is the b-edge of the forced base word. Contradiction.

Three-a branch. The first a-tile A_1 has its corner at $(f, 0)$ among the a-edge's end angles $\{\beta, \gamma\}$; the single open slot is β, so A_1 has β at $(f, 0)$ and its other flank, the c-edge, on the base: the base word is $a\,c\,b\,a\,a$, with the b-edge at $[f + c,\ f + c + b] = [12, 20]$. Label consistency forces A_1 mirrored. A_1's b-edge runs from the first trisection point of P's c-edge down to $(f + c, 0) = (12, 0)$; it is blocked at $(12, 0)$ by the base and at the trisection point by P's interior, so its far side is again a single b-edge, carried by Z'. The reflected orientation repeats γ at the trisection point, where A_1's γ already sits — impossible. So Z' is the parallelogram mate with γ at $(12, 0)$. The figure at $(12, 0)$ then contains A_1's α and Z''s γ, leaving exactly one β — but the base edge from $(12, 0)$ is the b-edge of the forced word, whose ends host α or γ. Contradiction.

Both branches of Lemma 48 are impossible, so no such tiling exists; $p = 0$. □

Remark 51. Three remarks. First, the terminal mechanism is the same in both branches: the cascade advances one cell from the corner and the base word, fixed at the fork, collides with it one base point later. Second, the proof is search-free; it re-derives the exclusion of the $(1, 3)$ instance ($N = 26$) established by the certified exhaustion (2 025 nodes), and the two agree. Third, the arithmetic components are machine-checked (`partner_unique`, `c_chord_dichotomy`, `first_run_kill`, `straight_junction_cases`, the slot classifications `beta_slot_unique`, `alpha_slot_unique`, `wedge_gamma_cases`, and the polynomial signs of Lemma 47); the geometric steps are exact coordinate computations over $\mathbb{Q}(\sqrt{35})$. Hypothesis 28 is thereby proved for the members $(1, 2)$ (Lemma 42) and $(1, 3)$; it remains open for $f \geq 4$, where $p \leq f - 3$, and for $e \geq 2$.

Remark 52 (The cascade at $(1, 4)$, and the general mechanism)**.** Running the same cascade at $(1, 4)$ ($p \leq 1$ by Lemma 47) kills five of the *six* admissible base words (an earlier count of five missed $a\,c\,b\,a\,a\,a$, which dies at once: its b-edge sits at $[f + c,\ f + c + b]$, exactly the four-a branch's first β-slot). For every side word with initial c-block 1, the corner partner and the c-chord dichotomy force, at each base point $(kf, 0)$ reached, a β-slot whose tile presents the next base edge from flanks $\{a, c\}$: the words with b in position 3 die at $(2f, 0)$; the word $a\,a\,a\,b\,c\,a$ dies at $(3f, 0)$; the word $a\,a\,c\,b\,a\,a$ dies at $(f + c, 0) = (24, 0)$ through the a-run partner, exactly as at $(1, 3)$. Two configurations survive the present tools, and the obstruction is located precisely in each: for the base word $a\,c\,a\,b\,a\,a$ the deviating a-run's new c-chord $[(20, 0),\ (20, 0) + cu]$ has its upper endpoint beyond the second side tile's c-edge ($25.875 > 21.875$ exactly), so tile-interior blocking fails and the dichotomy does not apply; and for the side word $c\,c\,a^4\,c$ the first run does not abut the corner tile, so Lemma 46 gives no chirality forcing. Hypothesis 28 at $(1, 4)$ is thereby reduced to these two configurations; the first is now eliminated, and the second is narrowed, as follows. *The double-c side word dies.* With the second c-tile forced direct and the run all-direct, the β-slot tile B at the run's first junction has exactly two label-consistent forms, and both are impossible. The junction ray toward the second c-tile's b-edge extends, exactly, to the base point $(2f, 0)$: the points $2c\,u$, Q and $(2f, 0)$ are collinear with $|2c\,u \to (2f, 0)| = 2b$. If B is direct, its a-edge covers the first 4 units of that line's far side, and the remaining stretch to Q has length $11 \notin \langle 4, 15, 16\rangle$; some covering edge must therefore cross Q, a through-edge forces the west α-slot tile $X = T_1 + (f, 0)$ (the mirrored X dies by base overshoot), so the whole segment $[Q, (2f, 0)]$ is an edge and the far side must partition $26 \notin \langle 4, 15, 16\rangle$ with no crossing at the γ-end. If B is mirrored, its c-edge itself crosses Q by $e^2 = 1$, forcing X the same way, and the remaining stretch is $14 \notin \langle 4, 15, 16\rangle$ with no crossing at the α-end. Both die; the side word $c\,c\,a^4\,c$ admits no completion for any base word. The crossing exclusions are instances of the general principle that a through-edge occupies the whole half-turn on its side (`through_edge_exclusive`); the three gaps are machine-checked (`gaps_4_15_16`). Each remaining configuration is narrowed as follows. In the double-c configuration $c\,c\,a^4\,c$, the second c-tile is forced direct: mirrored would place its α at the first junction beside the corner tile's α and the partner's γ, and $\pi-2\alpha-\gamma$ has no representation (`first_run_kill`); moreover the final c-block has length 1, so by Lemma 47 the run must be entirely direct, and the configuration now hinges on the single unforced β at the run's first junction. In the base word $a\,c\,a\,b\,a\,a$, the β-slot at $(20,0)$ forces the tile $T_1+(20,0)$, whose new c-chord is covered on its far side, in the mirrored-filler sub-case, by the mate's a-edge, two further a-edges, and the filler's own a-edge, whose endpoint is exactly the escape point $E=Q+(c,0)=(20,0)+cu$; the direct-filler sub-case leaves E unblocked and remains open.

Proposition 53 (The double-c kill at any initial block length)**.** *Let $e=1$, $m=1$, $f\geq 3$, and let an equal side carry an a-edge and have initial c-block of length $j\geq 2$. Write u for the unit vector along that side and place the base corner at the origin. Then:*

- (i) *$j\leq f-1$, and the chord from $j\,c\,u$ to the base point $(jf,0)$ has length exactly jb, divided into j segments of length b by $R_i=i\,c\,u+(j-i)(f,0)$ (so R_1 is the point called Q in the $(1,3)$ argument; the Q_j of Lemma 67 are a different family, spaced c apart along the interior wall);*
- (ii) *that chord admits exactly one partition into tile edges, namely j b-edges;*
- (iii) *the β-slot tile at $j\,c\,u$ presents, from its β-corner, a flank of $\{a,c\}$, and each choice leaves a run outside $\langle a,b,c\rangle$;*
- (iv) *the α-slot tile at R_{j-1} cannot lay a c-edge along the chord.*

Proof. (i) The side begins and ends with a c-edge (Theorem 20(i)), so an initial block of length ≥ 2 leaves a further c at the far end and $n_c\geq 3$; with $n_c=f-k$ and $k\geq 1$ (the side carries an a-edge) this gives $j\leq n_c=f-k\leq f-1$. For the chord, $\cos\beta$ is $(3f^2-1)/(2f^3)$ by the law of cosines on (f,f^2-1,f^2), which is the isosceles ratio of the target, so with $c\,u=(f^2\cos\beta,f^2\sin\beta)$,

$$|j\,c\,u-(jf,0)|^2=j^2\big(f^2-1\big)^2=(jb)^2,$$

and R_i is the affine combination dividing the segment in ratio $i:(j-i)$, so consecutive R_i are b apart (`chord_jb`, `chord_jb_segment`).

(ii) Reducing $xa+yb+zc=jb$ modulo f forces $y\equiv j$, while $yb\leq jb$ forces $y\leq j$; as $0\leq j\leq f-1$ this gives $y=j$, hence $xa+zc=0$ and $x=z=0$ (`partition_jb_gen`, which is general in (e,f); `partition_30` is its case $f=4$, $j=2$).

(iii) The β-slot exists: at the side junction $j\,c\,u$ where the block ends, Lemma 67 gives C_j's α and M_j's γ, which force the next side tile's corner to β, and that lemma is unconditional — it uses no base-letter hypothesis and no line closure, so it applies at every block length. Now β is opposite b, so its incident edges are a and c: the tile in the β-slot lays an a-edge (direct) or a c-edge (mirrored), never a b-edge. Both runs left on the chord are gaps, by the uniform statement that $jb-t\notin\langle a,b,c\rangle$ whenever $1\leq j\leq f-1$ and $f\mid t$, $t>0$ (`gap_jb_minus_multiple`): indeed $a=f$ and $c=f^2$ are both multiples of f. Concretely, if the tile is direct its a-edge does not reach R_{j-1} and the remainder $b-a$ is a gap, so no edge ends at R_{j-1} and some edge crosses it; if it is mirrored its c-edge overshoots R_{j-1} by $c-b=e^2=1$ and crosses it outright. Either way a through-edge occupies the whole half-turn on its side (`through_edge_exclusive`), the segment to $(jf,0)$ is a single edge, and the far side must partition $jb-a$ resp. $jb-c$ — gaps.

(iv) The last segment $[R_{j-1},(jf,0)]$ of the chord has length exactly b by *(i)*, and $c-b=e^2>0$, so a c-edge laid from R_{j-1} along the chord passes $(jf,0)$ by e^2. The chord descends: $j\,c\,u$ has ordinate $jf^2\sin\beta>0$ and $(jf,0)$ lies on the base, so the continuation beyond $(jf,0)$ has negative ordinate and leaves the target. This is the directional overshoot test of Remark 66: a c laid downward onto a base point exceeds the boundary and dies, whereas the same edge laid upward from the base would merely pass an interior point. Hence the α-slot flank along the chord is

a whole b-edge, and the reflected placement is excluded. Nothing here depends on f, nor on e beyond $e^2 > 0$. □

Remark 54. Proposition 53 is the $f = 4$ argument of Remark 52 with every constant replaced by its general form: the three runs $11, 26, 14$ against $\langle 4, 15, 16\rangle$ are $b - a$, $2b - a$, $2b - c$, i.e. the cases $(j, t) = (1, a), (2, a), (2, c)$, and the $2b$ chord is $j = 2$. The bound $j \leq f - 1$ of *(i)* is sharp for *(iii)*: at $j = f$ both $jb - a = f(f^2 - 2)$ and $jb - c = f(f^2 - f - 1)$ are multiples of f and lie in the semigroup.

Part *(iv)* supplies what the $f = 4$ argument records as "the mirrored X dies by base overshoot": there $X = T_1 + (f, 0)$ is the triangle $(f, 0), (2f, 0), R_1$, and the exclusion is the case $j = 2$. The general reason is that the last segment of the chord has length exactly b while the competing edge has length $c = b + e^2$, and the chord's far end is a base point, so the excess leaves the target. Consequently escape route 2 of Conjecture 57 is closed at every initial block length, for every $f \geq 3$. Route 1 — a deviating a-run whose new c-chord ends beyond the forced region — is what now stands between this and Hypothesis 28 at $e = 1$, and the next remark says exactly how it behaves in f.

Remark 55 (Route 1 escapes by exactly a, uniformly in f)**.** The escape recorded at $(1, 4)$ as "$25.875 > 21.875$ exactly" is not a numerical accident of $f = 4$, and it does not improve with f. Write $A = c\,u$ for the upper end of the side's first c-edge and $B = (c + a)u$ for the upper end of the run's first a-edge. The second side tile carries its a-edge AB on the side and, being mirrored (Lemma 46), presents β at A, so its c-edge runs from A; its third vertex is

$$V = c\,u + (c, 0), \qquad |V - A| = c, \quad |V - B| = b,$$

so that c-edge is *horizontal*, at height $c \sin\beta$. The deviating a-run's new c-chord starts at the base junction $(f + c, 0)$ and has upper endpoint $E = (f + c, 0) + c\,u$, at the same height. Hence

$$x(E) - x(V) = f$$

identically: the chord's endpoint overshoots the blocking edge by exactly one a-length, for every f. At $f = 4$ this is $25.875 - 21.875 = 4$.

Two consequences. The escape condition $x(E) > x(V)$ reads $f > 0$, so it holds in every member: there is no analogue here of the sign conditions $f^3 - 3f^2 + 1$ and $f^3 - 3f^2 - f + 1$ by which Lemma 47 disposes of its case, and no threshold in f beyond which route 1 closes itself. Any proof must therefore recover the blocking by a different mechanism rather than by showing the overshoot absent.

Second, the configuration on that horizontal line is completely rigid, and says what the mechanism has to be about. Writing $R_1 = c\,u + (f, 0)$ for the point called Q in the $(1, 3)$ argument (the midpoint of the $2b$ chord of Proposition 53), all four relevant points lie at the common height $c \sin\beta$, in the order

$$A \overset{a}{—} R_1 \overset{c-a}{—} V \overset{a}{—} E, \qquad |AV| = |R_1E| = c,$$

so the line carries *two overlapping c-segments offset by exactly* a: the second side tile's c-edge $[A, V]$, which is a genuine tile edge, and $[R_1, E]$, whose right end is the chord's unblocked endpoint. In particular the gap $[V, E]$ between the blocking edge's end and the chord's endpoint has length exactly a — the length of an a-edge. What route 1 needs is therefore a statement about what covers $[V, E]$: if that segment is forced to be a whole a-edge then E is a junction and the vertex figure there is constrained, which is the substitute for the tile-interior blocking that fails. We do not prove that here, and record only that the geometry leaves no other target: every length in the display is exact and independent of f.

Remark 56 (The blocking hypothesis cannot be weakened to "both ends are vertices")**.** It is worth recording what the step left open in Remark 55 may *not* appeal to. In Lemma 48 the hypothesis that the chord is blocked at both ends is what converts it into a union of whole tile edges; one cannot replace it by the weaker hypothesis that both ends are vertices of the tiling, because the tilings in this family are not edge-to-edge. The certified 99-tiling of member $(1,2)$ settles this concretely. Its 99 tiles span 216 distinct edges, and 52 of those carry a vertex of the tiling in their relative interior — each exactly one. Moreover its 89 vertices contain 19 pairs at distance exactly a, 23 at distance exactly b and 45 at distance exactly c that span no tile edge at all. (These counts are read off the coordinates of the machine-checked certificate; in that embedding every x is rational and every y a rational multiple of $\sqrt{15}$, so the chart $(x, y/\sqrt{15})$ is an affine rescaling of the plane and decides collinearity and betweenness faithfully.)

Hence knowing that a segment has both endpoints at tiling vertices and has the length of a tile side carries *no* information about how it meets the tiling. Since V and E are vertices at distance exactly a, this is precisely the shape of the hypothesis one is tempted to use, and it is not available. Whatever forces $[V,E]$ to be a whole a-edge must consume the cascade's tile data — which tiles have been forced and where — and not merely its vertex set. The same remark explains why the tile-interior blocking of Remark 49, and not a vertex-incidence argument, is the mechanism the cascade actually runs on.

Conjecture 57 (Advance-and-collide)**.** Let $e=1$, $m=1$, $f\geq 3$, and suppose an equal side carries an a-edge. Then the corner cascade — the b-partner of the corner tile, the c-chord dichotomy on successive both-ends-blocked chords, and the β-slot at each base point reached — terminates in every branch with a collision: a β-slot, whose tile presents the next base edge from flanks $\{a,c\}$, at a base position where the word of Theorem 20 carries its b-edge, or a repeated γ. Consequently $p=0$: Hypothesis 28 holds for every $(1,f)$.

The conjecture is proved here for $f=2$ (Lemma 42), $f=3$ (Theorem 50) and $f=4$ (Theorem 58, which exhausts all six admissible base words). The two ways a branch escapes the collision in the general argument are exactly: a deviating a-run whose new c-chord ends beyond the forced region (so that tile-interior blocking fails), and an initial side c-block of length at least 2 (so that the first-run orientation is not forced). Any proof of the general case must close these two gaps and no others.

The second gap is now general except for its enumeration. Its ingredients carry no restriction on f: `first_run_kill` and `through_edge_exclusive` are statements in $(n_\alpha, n_\beta, n_\gamma)$ alone; Lemma 47 reduces to the signs of f^3-3f^2+1 and f^3-3f^2-f+1; the chord carrying the three runs has length exactly $2b$ with Q its midpoint for every f, since with the base corner at the origin and u the unit vector along the equal side,

$$|2c\,u-(2f,0)|^2 = 4(f^2-1)^2 = (2b)^2, \qquad |2c\,u-Q|^2 = |Q-(2f,0)|^2 = (f^2-1)^2 = b^2$$

(`chord_two_b`, `chord_two_b_half`); and the three runs $b-a=f^2-f-1$, $2b-a=2f^2-f-2$, $2b-c=f^2-2$ are gaps of $\langle f,\ f^2-1,\ f^2\rangle$ for every $f\geq 3$ (`double_c_kill`), the values $11, 26, 14$ against $\langle 4, 15, 16\rangle$ being the case $f=4$. At $f=2$ the last two runs lie in the semigroup, which is where the hypothesis $f\geq 3$ of Theorem 20 is used. What is not yet general is the enumeration: a side with initial c-block at least 2 begins and ends with a c-edge, so $n_c\geq 3$ and the double-c walks are $(fk, 0, f-k)$ with $1\leq k\leq f-3$; at $f=4$ that is the single word $c\,c\,a^4\,c$ treated in Remark 52, while for $f\geq 5$ there are $f-3$ walks and the argument as written covers an initial block of exactly 2 followed by an a-run.

Theorem 58 (Hypothesis 28 holds at $(1,4)$)**.** *No tiling of the $(1,4)$ target at $m=1$ carries an a-edge on an equal side; hence, by Remark 45, no such tiling exists, re-deriving the exclusion of the $(1,4)$ instance ($N=47$) established by certified exhaustion (12 440 nodes).*

Proof. Suppose a side (named the left) carries an a-edge; then $p = 1$ there (Lemma 47). Both corner partners are forced with γ at their first junctions, unconditionally: the reflected orientation repeats the corner tile's γ at the base end and overflows.

Left kills. For left side words with initial c-block 1, Lemma 48 runs on P's c-chord as at $(1, 3)$: base words with second letter a die at $(2f, 0)$ or $(3f, 0)$ (β-slot against the forced b-edge); $a\,a\,c\,b\,a\,a$ dies at $(f + c, 0)$ through the a-run partner; $a\,c\,b\,a\,a\,a$ dies at $(f + c, 0) = (20, 0)$, where its b-edge occupies the four-a branch's first β-slot. Five of the six words die, leaving $a\,c\,a\,b\,a\,a$. The double-c left word dies for every base word by the line-total kill of Remark 52.

Right kill of $a\,c\,a\,b\,a\,a$. The right cascade is the left cascade of the reversed word $a\,a\,b\,a\,c\,a$. The right corner tile, its partner P', and the β-slot at $(Y - f, 0) = (43, 0)$ force $C' = T_1' - (f, 0)$, whose b-edge is collinear with the second right tile's b-line and meets the base at $(Y - 2f, 0) = (39, 0)$ — an endpoint of the base word's b-edge $[24, 39]$. If the right side is all c's, the second right tile is forced direct (`first_run_kill` with P''s γ), the doubled b-line is edged on its east side, and its west side partitions $2b = 30$, whose unique decomposition is $b + b$ (`partition_30`): the split falls at the mid-junction, producing the two b-partners. If the right side carries an a-run with initial block 1, the second right tile's horizontal c-edge blocks the mid-junction and the partner of C' is forced with γ there by the through-edge figure. Either way, γ at $(39, 0)$ would double C''s γ, so the partner's α sits at $(39, 0)$; the figure there closes with exactly one β adjacent to the base-left ray, and that tile must present the b-edge $[24, 39]$ from flanks $\{a, c\}$ — impossible. The double-c right word dies by the mirrored line-total kill. Every combination of side and base words is exhausted; $p = 0$. □

Remark 59 (The reversal principle)**.** The mechanism is symmetric: the corner cascade runs from both base corners, and the right cascade of a base word is the left cascade of its reversal. A base word survives only if both it and its reversal survive the left cascade; at $(1, 4)$ the reversal involution pairs the six words so that none does. This is the intended proof shape for Conjecture 57: show the left cascade kills every word with second letter a and every word whose b-edge occupies the four-a branch's first slot, and that the survivors' reversals always lie in the killed class.

Theorem 60 (The cascade closes every initial-block-1 configuration, for all f)**.** *Let $f \geq 3$, $m = 1$, and suppose an equal side of the $(1, f)$ target carries an a-edge whose first run is preceded by exactly one c-edge, and that the opposite side either has no a-edge or also has initial c-block 1. Then no tiling exists. Consequently Hypothesis 28 fails to hold at $(1, f)$ only if some hypothetical tiling has a side whose first a-run sits behind at least two c-edges; for $f \leq 4$ no such configuration survives either (Theorems 50, 58), so Hypothesis 28 holds outright there.*

Proof. Three lemmas. *(L1: the single-c chain.)* With initial block 1 the second side tile is the first a-tile, mirrored (Lemma 46), and its horizontal c-edge spans offsets $[0, c]$ above the corner. The cascade produces cells $C_k = T_1 + (kf, 0)$ and partners $Z_k = P + (kf, 0)$ by induction: the k-th chord's upper end is $cu + (kf, 0)$, interior to that horizontal edge exactly when $kf < f^2$; a base word whose b sits at position $q \leq f$ is reached at step $q - 2 \leq f - 2$, so every needed chord is blocked (`cascade_reaches`). At each base point the figure closes with exactly one β whose flanks are $\{a, c\}$; reflected partners die by repeating γ at the base end. The letter b therefore collides: every word with its b before its c dies. *(L2: the f-a branch.)* If the c immediately precedes the b, the f-a branch of Lemma 48 runs at the c's position, its first a-tile's b-partner is forced (the reflected orientation repeats γ at the chord's division point), its γ lands at the next base point, and the β-slot there collides with the b-edge. *(L3: reversal.)* Every admissible word has its b before its c, or its c immediately before its b, or its c before its b with a gap; in the third case the reversed word's b-position is $f + 3 - q \in [3, f]$ and precedes its c (`reversal_covers`), so the mirrored cascade from the opposite corner kills it — the opposite corner's chain is available because the opposite side has no a-run (then the doubled b-line's far side partitions $2b$ uniquely as $b+b$, `partition_2b`, forcing the partners) or has initial block 1 (then its horizontal c-edge blocks the mid-junction and the through-edge figure forces the partner). In every case the collision closes the configuration. □

Lemma 61 (The jb-line partition)**.** *For $1 \le j < f$ the only decomposition of $j\,b$ as $xa+yb+zc$ is $x=z=0$, $y=j$. Hence a line of length jb that is covered by whole edges on a side splits there into exactly j whole b-edges. (Whether a given line is so covered is a separate question: see Remark 14.)*

Proof. Reducing $xf+y(f^2-1)+zf^2 = j(f^2-1)$ modulo f gives $y \equiv j$; the size bound gives $y \le j$, and $j<f$ then forces $y=j$, hence $x=z=0$. Machine-checked (`partition_jb`). □

Remark 62 (The block cascade)**.** In a residual configuration (initial c-block of length $\ell \ge 2$), the line through $j\,cu$ in the b-direction reaches the base at $(jf,0)$ with total length jb: each step is the vector $cu-(f,0)$, of length b, by the translation structure alone. Three consequences, all side-word-independent, and all conditional on the relevant segment being uncrossed (Remark 14). (i) The $(f,0)$-slot closes with exactly one β whose flanks are $\{a,c\}$, so the second base letter is a or c, with its tile forced: $X_1 = T_1+(f,0)$ or the run-start A_1. (ii) When the second letter is a, the $2b$-line $[2cu, Q_1, (2f,0)]$ is fully west-edged by C_2's and X_1's b-edges, and at Q_1 the west side closes exactly (C_2's $\gamma+P$'s $\beta+X_1$'s $\alpha=\pi$); Lemma 61 forces the far side to split at Q_1 into both b-partners, the reflected lower partner repeats X_1's γ at the base point $(2f,0)$ and dies, and the $(2f,0)$-slot closes with exactly one β: *the third letter is never b.* Every residual configuration whose base word has second letter a and b in position 3 is dead. (iii) At the $2cu$ junction the figure C_2's α + upper mate's γ + corner $=\pi$ forces C_3 direct when $\ell \ge 3$: the block's chirality forcing propagates upward with the line, one junction per depth, and Lemma 61 supplies the partition at every depth $j<f$. The remaining freedom is the row-2 chord $[Q_1, Q_1+cu]$, whose east side forks single-c/f-a exactly as the base did: the corner block grows as a two-dimensional array of words, one chord fork per row. Closing the residual class for all f means closing this array; that is the precise remaining content of Conjecture 57 at $e=1$.

Lemma 63 (The east fan at the fork is forced)**.** *Let $f \ge 3$, $\ell \ge 2$, and let the second base letter be a, so that the brick X_1 and the mate M are in place. At the fork point $Q_1 = cu+(f,0)$ the tiles east of the chord form the figure $\{\gamma,\alpha,\beta\}$ in that angular order from the base side: the mate M_1 of X_1, then a β-tile, then the direct partner D_2 of C_2. The β-tile has flanks $\{a,c\}$, one along the floor and one against D_2's c-edge, so exactly two configurations occur: a on the floor (matching M_1's a-edge), or c on the floor — the tile A_2.*

Proof. The chord through Q_1 joins the side vertex $2cu$ to the base vertex $(2f,0)$, so both ends lie on the boundary and no edge extends past them; its east side is a union of whole edges of total length $2b$, which is $b+b$ for $f \ge 3$ (Lemma 61), split exactly at Q_1. Hence Q_1 is a vertex of that cover and each east tile meets it at a vertex. The lower half is X_1's b-edge; the single whole b covering it from the east belongs to a partner of X_1, and the reflected partner would repeat X_1's γ at the base junction $(2f,0)$, giving $2\gamma>\pi$ there since $\cos\gamma = -1/(2f) < 0$ (`straight_junction_gamma_bound`). So it is the direct partner M_1, which presents γ at Q_1 — its edges there being b and a — and lays an a-edge east along the floor. A straight-junction figure containing a γ is $\{\gamma,\alpha,\beta\}$ (`straight_junction_cases`), so exactly two further tiles meet Q_1. The one adjacent to the chord shares C_2's whole upper b, so its angle at Q_1 is an endpoint angle of a b-edge, namely α or γ; γ is taken, so it is α, and the reflected alternative would again double C_2's γ at Q_1. That tile is therefore D_2. The remaining tile carries β, whose flanks are $\{a,c\}$. □

Theorem 64 (The row fork kill)**.** *Let $f \geq 3$. In a residual configuration (initial c-block $\ell \geq 2$) whose base word has second letter a, the row-2 fork resolves single-c: the β-tile lays a along the floor.*

Proof. Lemma 63 leaves exactly two configurations, and Proposition 65 excludes the second. □

Proposition 65 (The A_2 branch dies)**.** *In a residual configuration (initial c-block $\ell \geq 2$) whose base word has second letter a, suppose the tile east of the row-2 fork is the mirrored $L = -2$ tile A_2, laying a horizontal c-edge along the floor line east of the fork point. Then no tiling exists.*

Proof. A_2's c-edge is an edge of the tiling, so no tile crosses the floor east of the fork point, and the south side of that edge is a union of whole edges with vertices on it. At a junction J of that cover the fan below the floor is the previous element's β, the row-1 brick's α, and a remaining wedge of exactly γ. Since $\gamma < \pi$, no tile can meet J through the interior of one of its edges (that contributes exactly π), so the wedge is filled by tiles having a vertex at J: the fill is $\{\gamma\}$ or $\{2\alpha, \beta\}$ (`straight_junction_cases`). The wedge is bounded by the floor and by the brick's b-edge running *down* to the base. A filler laying c along that ray leaves the target (it exceeds the ray by $c - b = 1$ past the base); a filler laying a along it makes that a the first element of the b-edge's east cover, leaving $f^2 - 1 - f$, which admits no completion (`east_cover_gap`); a filler laying a whole b along it is the reflected partner of the brick and repeats γ at the base junction (`straight_junction_gamma_bound`). Hence every fill is the brick's mate, the cover is f whole a-edges, their feet force base junctions at $(kf, 0)$ for $k = 2, \ldots, f+1$, and the first $f+1$ letters are all a — contradicting the b at position at most f. (The non-brick a-up placements are excluded by `alpha_vertex_gap` and `anti_brick_side` as before.) □

Remark 66 (Why $f \geq 3$, and what the $(1, 2)$ tilings show)**.** The hypothesis $f \geq 3$ in Lemma 63 is doing real work, and the $(1, 2)$ member shows exactly what it buys. At $f = 2$ one has $2b = 6 = b + b = a + c = 3a$: the chord's partition is not unique, the split need not fall at Q_1, and a tile may straddle there. The kernel-checked 99-tiling does precisely this — its tile with vertices $(4, 0)$, $\left(\frac{23}{4}, \frac{\sqrt{15}}{4}\right)$, $(5, \sqrt{15})$ lays a c-edge from the base along the chord which covers the row-1 brick's entire b-edge and overshoots Q_1 by exactly $c - b = 1$. For $f \geq 3$ that escape costs $2b - c = f^2 - 2$, which lies below b and is not divisible by f: a gap (`gap_2b_minus_c`). So the phenomenon is real, is realised, and is confined to $f = 2$.

Two methodological points are worth recording. First, the overshoot test is *directional*: a c laid downward from a floor junction exceeds the boundary and dies, while a c laid upward from the base overshoots an interior point and is legal. Second, an argument of the form "a segment pinned at both ends is covered by whole edges" is sound only where the ends are genuinely pinned — on the boundary, or against a tile edge that blocks continuation — and should say which. Both uses in Theorem 60 are of that kind: its chords are blocked above by the first a-tile's horizontal c-edge, and its doubled b-line runs boundary to boundary.

Lemma 67 (The wall climb)**.** *For every $\ell \geq 2$ and $2 \leq j \leq \ell$, the block tile C_j is direct and its mate M_j is forced, with M_j's c-edge $[Q_{j-1}, Q_j]$ continuing the interior wall; the first run tile above the block is mirrored. This is unconditional: no base-letter hypothesis and no line closure is used.*

Proof. Induction on j. C_2 is direct by the cu-junction figure (T_1's $\alpha + P$'s $\gamma + \beta$). Given C_j direct, the west fan at Q_{j-1} consists of β (P's for $j = 2$, else M_{j-1}'s), C_j's γ, and a wedge between C_j's b-ray and the u-line of exactly $\pi - \gamma - \beta = \alpha$: it is filled by a single α-tile (α is the minimal angle). Its flank along C_j's b-ray cannot be a c-edge — it would overshoot the boundary point jcu — so it is a whole b-edge shared with C_j, and the reflected partner is excluded by the wedge's α: the filler is the mate M_j, its c-edge running up the u-line. At the side junction

jcu, C_j's $\alpha + M_j$'s γ force the next side tile's corner to β: C_{j+1} is direct while the block lasts ($\alpha+\gamma+\alpha$ and $\alpha+\gamma+\gamma$ are impossible straight figures), and the first run tile is mirrored when it ends. □

Remark 68 (The pincer)**.** The assembly frame for the residual class: each corner runs its own chain, and a chain needs only its *own* second letter to be a. If $cp > bp$ the left chain meets the b first (second letter a since $cp > bp \geq 3$); if $cp < bp$ the reversed word has b before c and the right corner — whose side is all-c, block-1, or again residual — runs the mirrored chain (reversed second letter a since $bp \leq f$). Both chains advance through slots, lines and row forks; with the three leaks excluded the forks resolve to bricks, the lines close at every depth (Lemma 61), and the colliding b kills the configuration. Hypothesis hyp:walls at $e = 1$ is therefore reduced to the leak exclusion.

Theorem 69 (L2 at every reached slot)**.** *Let $f \geq 3$ and suppose the chain from a corner has closed slots $1, \ldots, k$. If the base letter presented at slot k is c and the next letter is b, no tiling exists. This holds for every block length of that corner's side.*

Proof. The slot-k β-tile lays c on the base and its a up the u-ray from $(kf, 0)$, which is the Z_{k-1}-mate's c-edge (for $k = 1$: P's c-edge) — an edge of the chain. The branch tile's b runs from the fu-division point of that edge to the base point $(kf + c, 0)$; its upper end abuts the edge's interior, so no covering edge crosses it, and the lower end is on the base: the far side is a single whole b (`partner_unique`). The reflected partner repeats γ at the division point, an interior straight junction, and $2\gamma > \pi$; the mate's γ lands at $(kf + c, 0)$, the figure there closes with exactly one β whose flanks are $\{a, c\}$, and it must present the b-edge: impossible. □

Theorem 70 (The block-two chain runs to arbitrary depth)**.** *Let $f \geq 3$ and let a side carry its first a-run behind exactly two c-edges. Then the chain from that corner closes every slot it needs: the base letters are presented one by one, a b collides at its slot, and a c followed by b collides by Theorem 69. In particular a base word whose b precedes its c dies from this corner.*

Proof. By Lemma 67, T_{a1} is mirrored and lays its horizontal c-edge at height $2cu_y$: the ceiling. The column-k line through $((k+1)f, 0)$ meets the ceiling line after two pieces, at a point interior to T_{a1}'s c (offset $kf < c$), so its west-edged portion is X_k's b plus the row-2 brick's b, pinned above by the ceiling edge and below by the base: `partition_2b` splits the east side at the joint, the reflected partners die by repeating γ (at the base, resp. at the joint), and both mates are forced: the next slot closes. The row-2 bricks advance by the fork at each column: the fork line is pinned the same way, the fan at the fork point is $\gamma \,|\, \alpha \,|\, \beta$ exactly as in Lemma 63 with the ceiling in place of the wall, and the A-branch dies by Proposition 65, whose south feet lie on the base and whose leak inventory is closed by `east_cover_gap` and the directional overshoot (both valid one strip above the base). Induction on the column. □

Theorem 71 (Reach three behind a thick block, and the pincer window)**.** *Let $f \geq 3$. From a corner whose side has initial block $\ell \geq 3$ (including the all-c side), slots $1, 2, 3$ close: the lines for columns $k \leq 2$ terminate on the wall (Lemma 67) with $k+1$ pieces and split by Lemma 61; the single row-2 brick needed is forced by Theorem 64. Consequently every residual configuration dies unless its base word escapes all four kills: direct-b from the b-first corner at depth ≤ 4, and L2 (Theorem 69) from the c-first corner when the c sits at depth ≤ 4 with the b adjacent. For $f \leq 5$ no admissible (bp, cp) escapes (`pincer_window`).*

Corollary 72. *Hypothesis 28 holds at $(1, 5)$: together with blocks ≤ 2 (Theorems 60, 70) the window covers every configuration. This is the first member closed with no engine certificate.*

Lemma 73 (The apex)**.** *The target's apex angle is $\pi - 2\beta = 3\alpha$, and the only vertex figure filling it is three α-corners (`apex_figure`). Since the sides carry no b-edge (`side_no_b_uncond`),* *the two outer apex tiles lay c-edges on the sides — the top letters of both side words are c — and the three wedges' rays sit at labels $-3, -1, +1, +3$. The apex therefore admits exactly two configurations, distinguished by the middle tile's chirality: its b shares the -1-ray with the left tile's b (both γ-ends meeting below), or its c covers that b with a T-junction at depth b. The side conclusion is specific to $m = 1$: the kernel-checked tilings at $m \geq 2$ carry b-edges on their sides, and their apex regions, while all realizing the forced three-α figure, diverge from one another immediately below it. Any forcing chain from the apex must therefore anchor on the $m = 1$ side structure; no m-independent reach exists.*

Remark 74 (The straddler: the unique leak at rows ≥ 3)**.** At every strip junction of a row-j A-branch the wedge is exactly α, and its filler admits exactly two label assignments: the mirrored $L = -2$ mate (whole b shared with the run tile; the strip structure stays rigid) or the direct $L = 0$ *straddler* — b horizontal under the next c-line, c laid along the run tile's b-ray, covering that whole b and overshooting the run tile's α by exactly $c - b = 1$, poking below the floor line just past the horizontal c-edge's east end. At row 2 the analogous fill dies against the boundary ($2b - c = f^2 - 2$ is a gap, `gap_2b_minus_c`); at rows ≥ 3 the poke is interior and no local figure excludes it. The straddler does not cascade: it couples strips only upward, so no descent-to-the-base argument applies. Killing this one shape at rows ≥ 3 is equivalent to the full chain reach, hence to Conjecture 57 at $e = 1$.

Lemma 75 (Descent identities and the ladder)**.** *Let $f \geq 3$. (i) A line in the b-direction (label -1) crosses the horizontal strip lines at intervals of exactly b along itself and advances exactly c horizontally per crossing: $\sin\beta = b\sqrt{D}/(2f^3)$ with $D = 4f^2 - 1$, and $fu_x + b\,|\cos 2\gamma| = c$ (`descent_ident`, `sinb_ident`). (ii) A straddler's c-edge therefore covers the run tile's b and pokes one unit past its α-vertex T-junction; at every T-junction on the line the fan-adjacent tile lays a whole further edge along it, so the straddler forces a doubly edged ladder: east breaks in $b+\langle a, b, c\rangle$, west breaks in $c+\langle a, b, c\rangle$, staggered by $c-b = 1$. (iii) The ladder cannot terminate on the base: the termination abscissa is $\big(f^3(m+1)-K(f^4-3f^2+1)\big)/f^2$ with $K = (j-1)f+i$, and $f^4-3f^2+1 \equiv 1 \pmod{f^2}$ forces $f^2 \mid K$, impossible since $K \equiv i \not\equiv 0 \pmod{f}$ (`ladder_no_base`). (iv) The ladder can terminate only at an interior mutual vertex, where the cumulative east and west sums agree; the stagger makes this $s_1 = s_2 + 1$ with $s_1, s_2 \in \langle a, b, c\rangle$, realized earliest by $(s_2, s_1) = (b, c)$: every straddler forces at least $b + c = 2f^2 - 1$ of doubly edged ladder, descending two full strips and gliding exactly c east of the strip ends.*

Lemma 76 (Termination is one condition, boundary or interior)**.** *A ladder terminates only where* both *of its staggered covers end: at an interior mutual vertex by definition, and at a boundary point because the boundary ends both covers at once. So side exit is not a separate escape route; every termination satisfies the same condition. With $c = b+1$ that condition reads: $T - b$ and $T - c$ are consecutive members of $S = \langle a, b, c\rangle$. Consecutive members of S begin only at f^2 (`consecutive_gap`), so*

$$T \;\geq\; b + c \;=\; 2f^2 - 1.$$

Proof. Write $m + 1 = T - b$ and $m = T - c$, both in S. Reducing $xf + y(f^2 - 1) + zf^2$ modulo f shows an element of S congruent to $-y$; if $m + 1 < f^2$ then both representations have $z = 0$ and $y \leq 1$, and the four cases give $f \mid 1$, $f \mid 2$, or a direct size contradiction. Hence $m + 1 \geq f^2$ and $T = b + (m + 1) \geq b + c$. □

Theorem 77 (Straddlers need three strips, so rows $1, 2, 3$ are clean)**.** *Let $f \geq 3$. The target is exactly f strips tall ($f^3 \sin\beta$ over $c \sin\beta$), and a ladder descends exactly b along itself per strip (Lemma 75). By Lemma 76 its length satisfies $T \geq b + c > 2b$, so it crosses strictly more than two strips; since it cannot terminate on the base (Lemma 75(iii)), a straddler whose* *ladder starts at the floor of row j requires $j-1>2$. Hence no straddler occurs at rows 1, 2 or 3, the row-3 fork admits only the two branches of Lemma 63. The A-branch's exclusion at row 3 invokes Proposition 65 whose south-cover feet land on the row-2 floor, edged only as far as the staircase has reached: that step is incomplete as written, and the structural claim of reach 4 is open pending its repair. The $(1,6)$ conclusion itself is unaffected: it is certified by the independent exhaustion* `w107`.

Corollary 78. *Hypothesis 28 holds at $(1,6)$, hence for every $f\le 6$: with reach 4 the four kills cover depth ≤ 5 and no admissible (bp,cp) escapes (*`pincer_window_four`*). The $(1,5)$ and $(1,6)$ cases are additionally certified by independent exhaustions of the full $m=1$ targets (*132,519 *and* 1,251,382 *nodes, instances* `w74`, `w107`*); the $(1,7)$ exhaustion is running.*

Lemma 79 (Corner lines are column lines)**.** *Let $f\ge 3$, $m=1$. Every b-direction line through two vertices of the corner lattice $\mathbb{Z}(a,0)+\mathbb{Z}\,cu$ meets the base at $(kf,0)$ and the left side at $k\,cu$ for an integer k, and its horizontal distance to the side at row r is exactly $(k-r)a$. Its length is kb and both ends lie on the boundary, so for $k<f$ Lemma 61 splits both of its sides into exactly k whole b-edges, breaking at every strip junction on both sides at once: the line is* unstaggered. *Since $k\,c\le f^3$ forces $k\le f$, the only b-direction line of the left corner that escapes is $k=f$ — the line from $(f^2,0)$ to $f\,cu$, which is the apex. A ladder is staggered by $c-b=1$, so no straddler sits on a corner line with $k<f$.*

Lemma 80 (The chain never needs the apex line)**.** *The base b-letter sits at a position $bp\in[3,f]$. With the preceding letters forced to a it begins at $\big((bp-1)f,0\big)$, and the collision figure there consumes only the brick and mate of column $bp-2$, hence only the lines L_k with $k\le bp-1\le f-1$. Every line the chain requires therefore has $k<f$, where Lemma 79 applies (*`chain_needs_small_lines`*); the apex line L_f, the one line escaping Lemma 61, is never required.*

Remark 81 (The residue, as one statement, and why it must be $m=1$)**.** Combining Lemmas 79 and 80: no straddler can sit on any line the chain needs, *provided no tile crosses that line.* Since L_k runs boundary to boundary and every needed k satisfies $k<f$, this crossing statement is the only remaining hypothesis at $e=1$; with it, every fork closes, the chain reaches the b, and Hypothesis 28 follows at $e=1$ for every f.

The statement is false without the hypothesis $m=1$. Tested against the five kernel-checked tilings, all of which have $m\ge 2$, tiles cross boundary-to-boundary b-lines freely: 42, 15, 24, 10 and 98 crossing incidences respectively. This is consistent with Remark 31 — at $m=2$ the corner block can break — and it settles the shape of any proof: the crossing statement admits no m-independent argument, and must consume the $m=1$ hypothesis exactly where the corner block is rigid. That input is not supplied here.

Lemma 82 (The vertex census)**.** *In every tiling of the base-β target, write n_1,n_2 for the numbers of straight figures $\{\gamma,\alpha,\beta\}$ and $\{3\alpha,2\beta\}$, and $v_1,\dots,v_4$ for the interior figures $\{\beta,3\gamma\}$, $\{2\alpha,2\beta,2\gamma\}$, $\{4\alpha,3\beta,\gamma\}$, $\{6\alpha,4\beta\}$. The base corners fill uniquely as $\{\beta\}$ and the apex as $\{3\alpha\}$, and balancing the three corner types across the N tiles gives*

$$v_1 \;=\; 1+n_2+v_3+2v_4 \qquad (\texttt{vertex\_census}).$$

In particular $v_1\ge 1$: a $\{\beta,3\gamma\}$ vertex is mandatory, and each γ-poor figure demands one more. The identity is verified exactly on all five kernel-checked tilings. It is also the entire *content of corner counting: the three balance equations admit exactly one relation, and it does not involve the side structure — corner counting alone cannot see p, consistent with the census-based routes logged as dead.*

Definition 83 (The frontier invariant)**.** A *frontier* is a finite family of disjoint uncovered polygons inside the target together with, for each target side, the multiset of tile-edge types already laid on it. Say a frontier is *admissible* if all five of the following hold.

(P1) Each polygon's area is a positive integer multiple of the tile's area, and the multiples sum to the number of tiles remaining.
(P2) Every maximal straight run of a polygon's boundary joining two convex corners has integral length lying in $\langle a, b, c\rangle$.
(P4) Every convex corner of every polygon has angle at least α.
(P5) The per-side edge-type multiset is dominated componentwise by an admissible walk profile.
(P6) An edge lying in a target side and meeting a corner of that side has the type forced at that corner.

Admissibility is decidable, and it is preserved by no tiling step: a tiling of the target is exactly a sequence of placements through admissible frontiers ending at the empty frontier.

Lemma 84 (The angle threshold)**.** *For the tile $(a, b, c) = (f, f^2 - 1, f^2)$ one has $\cos\alpha = (2f^2 - 1)/(2f^2)$ (`cos_alpha_closed`), alongside $\cos\gamma = -1/(2f)$ and $\cos\beta = (3f^2 - 1)/(2f^3)$. Condition (P4) is therefore the statement that the uncovered region never acquires a corner sharper than the tile's own sharpest angle.*

Remark 85 (The exhaustion invariant, and the shape of a uniform proof)**.** Definition 83 is the invariant maintained by the exhaustive search whose verdicts certify the small members. Its content relative to the hand arguments of this paper is that (P2) generalises the semigroup gap lemmas used throughout, while (P4) is a constraint on the uncovered region rather than on completed vertices, and is not used anywhere in the arguments above. In the certified cases the searches reach a contradiction long before the tiling is complete. For $f = 2, 3, 5, 6, 7$ the maximum refutation depth is $6, 13, 24, 29, 35$ against $N = 3f^2 - 1$ placements, so the fraction of the tiling ever built is 0.55, 0.50, 0.32, 0.27, 0.24 and falling, while the share of (P4) violations rises from 50% to 98%. The vanishing fraction is the robust part of this observation. The linear form is not: the ratio depth$/f$ reads 3.00, 4.33, 4.80, 4.83, 5.00 and is still increasing at $f = 7$, so five instances do not separate a bound Cf from a slowly superlinear law. A proof of Hypothesis 28 for all f would follow from a uniform bound of the form: from a corner-anchored start with $p \geq 1$, no admissible frontier survives past depth Cf. That statement is open and the figures quoted are measurements, not evidence for the constant.

Lemma 86 (The thick-member monochotomy)**.** *Let $e \geq 2$, $e < f$, $\gcd(e, f) = 1$, tile $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$. The only decomposition of c as $xa + yb + zc$ is the single c: writing $xef + y(f^2 - e^2) = f^2$, coprimality forces $e \mid y - 1$, and both $y = 1$ (then $xf = e < f$) and $y \geq e + 1$ (then the sum exceeds f^2 since $f^2 > e^2 + e$) are impossible (`c_chord_unique_thick`). Consequently the f-a branch of Lemma 48 is an $e = 1$ phenomenon: at thick members every c-chord fork of the corner chain is forced, and the chain machinery, where it applies, is stiffer than at $e = 1$.*

Lemma 87 (Thick-side quantization)**.** *At any member (e, f), $e \geq 1$, a side's edge counts satisfy $xe + zf = f^2$ with no b (`side_no_b_uncond`), so $f \mid x$ by coprimality (`side_quantized`): the number of a-edges on a side is a multiple of f. For $e \geq 2$ this quantizes the walls violation: any bound $p < f$ forces $p = 0$, the side condition of Hypothesis 28.*

Lemma 88 (The thick base trichotomy)**.** *For a separated member ($f^2 > 2ef + e^2$) the base admits exactly three edge decompositions: $(x, y, z) \in \{(0, e, 2e),\ (f, e, e),\ (2f, e, 0)\}$ (`base_trichotomy`); the walls form is the middle one, f a-feet, e b-edges, e c-feet. Close pairs ($f^2 \leq 2ef + e^2$) admit in addition finitely many decompositions with $y > e$, computable per member.*

Lemma 89 (The shadow at a c-corner). *Suppose a base corner tile lays c on the base (then it is mirrored with its a-edge first on the side), and the first side run has length at least 2. The run is γ-upper by absorption, so its second tile lays a horizontal c-edge at height $a\sin\beta$ — exactly one tile-height (`strips_tall` family identities). Every tile in the strip below spans it and carries its c on the floor or the ceiling; a tile with b on the floor and c on the ceiling would have two parallel sides. Hence every base edge meeting the shadow $(0, a\cos\beta + c)$ is a c-edge. Consequences: at such a corner either the first side run has length 1, or the second base edge is a c and no b-edge meets the shadow. At $e = 1$ the two-sided shadow window is always smaller than the b-edge (`shadow_footage_e1`: the comparison reduces to $3f^2 > 1$), an independent exclusion of c-corners; at $e \geq 2$ the comparison reverses, and the shadow is a constraint rather than a kill.*

Lemma 90 (The thick base dichotomy). *The γ-trap (Proposition 25, every side of a base-β target carries a c-edge, general in (e, f)) forces $z \geq 1$, deleting the third column of Lemma 88. At a separated thick member only two base decompositions survive: $(0, e, 2e)$ and the walls form (f, e, e) (`base_dichotomy_thick`).*

Lemma 91 (The c-corner is rigid). *Suppose a base corner tile lays c on the base. Since the corner angle is β, with flanks $\{a, c\}$, that tile lays a on the side. Its b-edge is a chord joining the base point $(c, 0)$ to the side point $a\,u$, so both of its ends lie on the boundary and its far side carries a single whole b (`partner_unique`, `lean/PentagonLemma.lean`). The corner tile presents α at the base end of that chord and γ at the side end; the reflected partner would repeat γ at the side end, where the junction is straight and $2\gamma > \pi$ because $\cos\gamma = -e/2f < 0$. Hence the direct partner is forced, contributing γ at the base end and α at the side end, and each end closes with exactly one β whose flanks are $\{a, c\}$: the second base edge and the second side edge are each an a or a c.*

In the base column $(0, e, 2e)$ of Lemma 90 the base carries no a-edge, so both corner tiles lay c on the base and the hypothesis holds at both corners: both sides begin with an a-edge, and by Lemma 87 each side then carries at least f of them. The γ-trap bounds the side profiles to a-count fj with $je \leq f - 1$.

This is the unconditional entry point that Lemma 39 lacks: that lemma disposes of the walk $(0, e, 2e)$ only after both dual blocks are complete, and its stubs are region-boundary segments only under that hypothesis, whereas the forcing here uses nothing but the boundary position of the corner chord and $2\gamma > \pi$. What remains for the column is to run the chain from the rigid c-corner far enough to create the stubs of Lemma 39 unconditionally; the stub arithmetic itself is already general and machine-checked (Lemma 40).

Theorem 92 (The column $(0, e, 2e)$ admits no tiling). *Let (e, f) be a separated member with $e \geq 1$, $\gcd(e, f) = 1$, $f \geq 2$, and suppose the base walk is $(0, e, 2e)$: no a-edge on the base. Then no tiling exists.*

Proof. Write $\gamma = 2\alpha + \beta$; a wedge $X\alpha + Y\beta$ filled by corner counts (x, y, z) satisfies $x + 2z = X$, $y + z = Y$, so the fills of β, of α and of $\alpha + \beta$ are $\{\beta\}$, $\{\alpha\}$ and $\{\alpha, \beta\}$ respectively (`fill_beta`, `fill_alpha`, `fill_alpha_beta`).

The corner angle is β, so a single tile T_0 occupies the corner; its flanks are $\{a, c\}$ and the base carries no a-edge, so T_0 lays c on the base and a on the side, and its b-edge joins $(c, 0)$ to $a\,u$. At the side vertex $a\,u$ the junction is straight and T_0 presents γ, leaving $\alpha + \beta$, so exactly two further tiles meet there, one with α and one with β. Their order is forced. The edge T_0 presents along the chord is an edge of a tile, so no tile crosses it, and both of its ends lie on the boundary, so no edge overruns them; hence the far side of that chord is a union of whole edges of total length b, which by `partner_unique` is a single b. The tile carrying it meets $a\,u$ at an endpoint of a b-edge, so presents α or γ there, and γ is already taken. It is therefore the α-tile, and it is the partner P_0, forced with no hypothesis on corner blocks.

Inductively, suppose T_k (a translate of T_0 by $(kc, 0)$) and P_{k-1} are in place. At the base vertex $((k+1)c, 0)$, T_k presents α and P_k will present γ; at the vertex $V_k = (kc, 0) + a\,u$, T_k presents γ and P_{k-1} presents β, leaving α, whose unique filler has flanks $\{b, c\}$; a c laid along T_k's b-ray would run from V_k past $((k+1)c, 0)$, a point of the base, and so leave the target, so the filler lays a whole b there and is the partner P_k. At the base vertex $(kc, 0)$, T_{k-1}'s α and P_{k-1}'s γ leave exactly β, whose filler has flanks $\{a, c\}$: the next base edge is an a or a c, hence a c, and its tile is T_k. The induction therefore forces every base edge to be a c-edge.

The base has length $e(3f^2 - e^2)$, so this requires $f^2 \mid e(3f^2 - e^2)$, hence $f^2 \mid e^3$, hence $f = 1$ by coprimality (`base_not_all_c`), contradicting $f \geq 2$. $\square$

Corollary 93. *At a separated thick member the base walk is the walls form (f, e, e): Lemma 88 gives three columns, the γ-trap deletes $(2f, e, 0)$ (Lemma 90), and Theorem 92 deletes $(0, e, 2e)$.*

Separation enters that chain only through Lemma 88, to rule out columns with $y > e$. That can be done under the weaker hypothesis $f > 2e$, so the trichotomy and the dichotomy both extend. Proposition 146 is not available for this purpose, because it assumes $x, z \geq 1$ while Lemma 88 allows $x = 0$ and $z = 0$; the argument below is run with $x, z \geq 0$ throughout.

Corollary 94 (The base trichotomy and dichotomy without separation)**.** *Let $\gcd(e, f) = 1$ and $f > 2e$. Then the base admits exactly the three decompositions $(0, e, 2e)$, (f, e, e), $(2f, e, 0)$ of Lemma 88, and after the γ-trap exactly the two of Lemma 90, namely $(0, e, 2e)$ and the walls form (f, e, e).*

Proof. Reducing $xa + yb + zc = e(3f^2 - e^2)$ modulo f kills $a = ef$ and $c = f^2$ and leaves $-ye^2 \equiv -e^3$, so $f \mid e^2(y - e)$ and hence $f \mid y - e$ by coprimality. Write $y = e + kf$; since $0 \leq y$ and $e < f$, $k \geq 0$.

Suppose $k \geq 1$. The base equation is then $xe + zf = 2ef - kb$ (Proposition 145(i), whose derivation uses no positivity), and $f \mid x - ke$, so $x = ke - mf$ for an integer m; from $x \geq 0$, $mf \leq ke$. The identity

$$(ke - mf)e + k(f^2 - e^2) - 2ef \;=\; f\big(kf - (m+2)e\big)$$

together with $z \geq 0$, i.e. $xe + kb - 2ef \leq 0$, gives $kf \leq (m+2)e$ after dividing by $f > 0$. Now $2m < k$: for $m \leq 0$ this is clear from $k \geq 1$, and for $m \geq 1$ it follows from $2me < mf \leq ke$. Finally $f > 2e$ and $k \geq 1$ give $2ke < kf \leq (m+2)e$, hence $2k < m + 2$, and with $2m \leq k - 1$ this yields $4k < k + 3$, i.e. $k < 1$, a contradiction.

So $y = e$. Then $f \mid x$, say $x = jf$, and the base equation becomes $je + z = 2e$, i.e. $z = e(2 - j)$, so $z \geq 0$ gives $j \in \{0, 1, 2\}$ and the three columns are exactly those listed. The γ-trap forces $z \geq 1$, deleting $j = 2$. We cite the γ-trap (Proposition 25) rather than Lemma 90: the latter's formal counterpart `base_dichotomy_thick` carries a separation hypothesis, whereas the γ-trap is proved for an arbitrary side of a base-β target and its corner input (Proposition 24) puts no γ at a base corner or at the apex, so it applies here unchanged. (`no_extra_column_of_f_gt_two_e`.) $\square$

Remark 95 ($f > 2e$ is sharp, and $(1, 2)$ is the unique exception)**.** The inequality cannot be weakened to $f \geq 2e$. Coprimality makes $f = 2e$ possible only at $e = 1$, $f = 2$, and that member does carry a fourth decomposition: $1 \cdot a + 3 \cdot b + 0 \cdot c = 2 + 9 = 11 = e(3f^2 - e^2)$, i.e. $(x, y, z) = (1, 3, 0)$ with $y = 3 > e$. It escapes the argument above exactly at $z = 0$. Checked by structured enumeration of the base equation on all 165038 members with $f > 2e$, $e \leq 300$: no decomposition outside the three, and the same enumerator does find $(1, 3, 0)$ at $(1, 2)$.

Remark 96 (The band this opens, and what still blocks the walls form there)**.** Separation is $f > (1 + \sqrt{2})e$, so Corollary 94 is strictly weaker in hypothesis and covers the band $2e < f \leq$

$(1+\sqrt{2})e$ that Lemma 88 left to a per-member computation: 5064 of the 17296 close pairs with $e \le 200$ lie in it. For those members the base dichotomy now holds for the same reason it holds at a separated member.

To reach the walls form one further needs Theorem 92. Its separation hypothesis is indeed inert — no step of its proof invokes $f^2 > 2ef + e^2$, and its arithmetic inputs (`partner_unique`, $c > b$, `base_not_all_c`) were checked on all 36896 coprime members with $e \le 200$, $e < f < 4e+2$, of which 17296 are close pairs, with no failure. But deleting the hypothesis does not deliver the conclusion, because the proof of Theorem 92 has a gap at $e \ge 2$, recorded next.

Remark 97 (A gap in the induction of Theorem 92)**.** The induction step of Theorem 92 argues that at $V_k = (kc, 0) + a\,u$ the tiles T_k and P_{k-1} present γ and β, "leaving α". That is the straight-junction identity $\pi - \gamma - \beta = \alpha$, which follows from $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta$. It is legitimate at $k = 0$, where $V_0 = a\,u$ lies on an equal side and the junction is genuinely straight. For $k \ge 1$ the point V_k is *strictly interior*: the angle around it is 2π, and

$$2\pi - \gamma - \beta \;=\; 4\alpha + 2\beta \;=\; \pi + \alpha \;\ne\; \alpha.$$

The discrepancy is exactly π. For instance at $(e, f) = (2, 5)$ one has $V_1 = (30.68, 8.23)$, interior to the target $(0, 0), (142, 0), f^3u$; the residue there is $203.07°$, not the $\alpha = 23.07°$ the step requires. The same holds at every member we computed, $(2, 3)$, $(3, 8)$, $(1, 4)$ and $(3, 7)$ among them. With the residue $\pi + \alpha$ the partner P_k is not forced, and the chain stops after T_0, P_0, T_1; that much is exactly Proposition 28.

At $e = 1$ the theorem is nevertheless true, by a different and much shorter route: the column $(0, e, 2e) = (0, 1, 2)$ has a base word of only three letters, and Proposition 28 places the first two and the last two base edges in $\{a, c\}$, which for a three-letter word is all of them; the word then carries no b-edge, contradicting $y = 1$. That argument uses no induction and no separation.

At $e \ge 2$ the word has $3e \ge 6$ letters, the four pinned positions leave $3e - 4 \ge e$ interior slots, and the b-edges fit inside them: at $(2, 5)$ the word $c\,c\,b\,b\,c\,c$ satisfies $25+25+21+21+25+25 = 142 = e(3f^2 - e^2)$ and survives every filter we have. So Theorem 92 is established at $e = 1$ only, and its conclusion at $e \ge 2$ — on which Corollary 93 depends — must be regarded as **open**. We have not refuted the conclusion; we refute only the proof. The missing ingredient is sharp: force the β-tile at V_k, whose flanks are $\{a, c\}$, to lay its c rather than its a along $[V_k, V_{k+1}]$. This is an instance of the crossing question of Remark 14, which Remark 154 already identifies as the obstruction, and Remark 66 already warns that a pinning argument must say against what the ends are pinned. Here the would-be blocking edge at the upper end is P_k's own c-edge, the very thing being derived.

Remark 98 (The gap is exactly one crossing, and there is no cheaper repair)**.** The step admits a sharp reduction. Let Q be the tile across T_k's b-edge $[V_k,\ ((k+1)c, 0)]$, whose base end lies on ∂ABC and whose end V_k is interior. Measured from the anchored base end, Q lays a, b or c along that edge:

- b: an exact match, so Q is the partner P_k — the case the induction wants;
- c: overruns V_k by $c - b = e^2$;
- a: falls short of V_k by $b - a$, leaving a run of that length still to be covered.

The third case does not close either, because $b - a$ is representable in $\langle a, b, c\rangle$ for no member whatever. Indeed if $xa + yb + zc = b - a$ then $y = 0$, since $y \ge 1$ would give $yb \ge b > b - a$; reducing $x\,ef + z\,f^2 = f^2 - e^2 - ef$ modulo f then gives $f \mid e^2$, hence $f = 1$ by coprimality. (`gap_b_sub_a`; checked on all 29435 coprime members with $e \le 200$, $e < f < 4e + 3$, with no representation in any of them.)

So both surviving branches force a tile edge past V_k, and the induction step of Theorem 92 is *equivalent* to the assertion that nothing overruns V_k. That is the one-end-anchored pinning question in its purest form: the base end cannot be overrun, because the continuation would leave the target, while V_k is interior and, as computed in Remark 97, is not blocked by any tile already forced. Repairing Theorem 92 at $e \geq 2$ is therefore neither more nor less than settling that question, and the same question is what stops the corner cascade of Remark 55 and the filler dichotomy of Section 5.6. Three routes, one wall.

Remark 99 (The overrun configuration at V_k, determined)**.** The crossing that has to be excluded is more rigid than it looks. By Remark 56's accounting, a tile whose edge passes through V_k contributes exactly π there, so an overrun forces the tiles with a *vertex* at V_k to form a π-figure. The forced ones are T_k, presenting γ, and P_{k-1}, presenting β, and the only π-figure containing $\{\gamma, \beta\}$ is $\{\gamma, \alpha, \beta\}$. So an overrun admits *exactly one* further tile at V_k, and it presents α.

That tile has room. The chord leaves the base at angle α, so its extension at V_k lies at $\pi - \alpha$, while P_{k-1}'s c-edge lies along π; the wedge between them is exactly α. The configuration is therefore angularly consistent and cannot be refuted by counting angles, which is why it has survived.

Its two edges at V_k are b and c, one along the chord's extension and one along P_{k-1}'s c-edge. Along the latter, of length c:

- laying c matches exactly;
- laying b falls short of V_{k-1} by $c - b = e^2$, and e^2 lies in $\langle a, b, c\rangle$ for no member (`gap_e_squared`), so that residue admits no whole-edge closure either.

The overrun therefore forces either a second crossing further along the same horizontal, or the α-tile to share P_{k-1}'s c-edge exactly and lay its b along the chord's extension. We record this because it reduces the open question to a single configuration, not because it settles it: the second branch remains open, and a descent on the first is blocked by the fact that the e^2-residue may be closed by an edge overrunning its *other*, interior end rather than V_{k-1}.

Where that surviving branch lands is worth recording, because it is not an accident. The chord's direction vector is $a\,u - (c, 0)$, which is T_0's own b-edge, so

$$W \;=\; V_k + \big(a\,u - (c, 0)\big) \;=\; \big((k-1)c,\, 0\big) + 2a\,u.$$

At $k = 1$ this is $W = 2a\,u$, a point of the equal side, and R then lays an a-edge on that side from $a\,u$ to $2a\,u$. Far from being absurd, that is exactly the configuration Lemma 91 predicts for this column. So the overrun is geometrically coherent, not merely angularly so, and it will not be refuted locally.

Remark 100 (Theorem 92 at $e \geq 2$, repaired conditionally)**.** The previous remark also supplies the honest repair. Lemma 91 already observes that in the column $(0, e, 2e)$ the base carries no a-edge, so both corner tiles lay c on the base, both equal sides begin with an a-edge, and by Lemma 87 each carries at least f of them. In the notation of §5.6 that is $p \geq 1$ on both sides. Hence

Proposition 101. *Assume Hypothesis 28 at the member (e, f), i.e. $p = 0$. Then no tiling has base walk $(0, e, 2e)$, for every $e \geq 1$.*

Proof. By Lemma 91 the column $(0, e, 2e)$ forces $p \geq 1$ on each equal side, contradicting $p = 0$. □

This is two lines and uses no induction, no chord pinning, and no crossing question. It is weaker than Theorem 92 as printed, being conditional; but Hypothesis 28 is assumed throughout this programme anyway, so for every downstream use the conditional form is as good. Corollary 93 and Corollary 108 may accordingly be read as holding under Hypothesis 28 at every member, and unconditionally at $e = 1$, where Remark 97's three-letter argument applies.

The moral is worth stating. Theorem 92's induction was attempting to prove, unconditionally and by a long chord cascade, a statement that follows from Hypothesis 28 immediately. Its gap is therefore not an accident of that particular proof: excluding the overrun at V_1 means exactly bounding the side's run of a-edges, which is Hypothesis 28 again. The two are one problem, and the cascade was the long way round.

Remark 102 (Counting cannot decide the crossing question)**.** Since an overrun at Y is equivalent to the tiles with a *vertex* at Y forming a π-figure, it is natural to attack it with the vertex census of Lemma 82, which counts exactly those figures. That cannot work, for a structural reason worth recording so it is not attempted again.

Write n_1 for the number of $\{\gamma, \alpha, \beta\}$ figures and n_2 for the number of $\{3\alpha, 2\beta\}$ figures, and $v_1, \dots, v_4$ for the four interior types. The census consists of one equation per angle, and its identity is obtained by subtracting the β equation from the α equation:

$$v_1 \ = \ 1 + n_2 + v_3 + 2v_4,$$

in which n_1 *cancels*. The third equation then pins $n_1 = N - 3v_1 - 2v_2 - v_3$, but supplies no bound of its own. So the census constrains $\{3\alpha, 2\beta\}$ and says nothing bounding about $\{\gamma, \alpha, \beta\}$.

That is precisely the wrong way round for us: at V_k the forced tiles are T_k, presenting γ, and P_{k-1}, presenting β, and $\{\gamma, \alpha, \beta\}$ is the only π-figure containing both. The census is therefore blind to exactly the configuration in question, and no counting argument built on it can decide the crossing.

The kernel-checked 99-tiling illustrates both points. Its vertex figures are: on the boundary, 22 of type $\{\gamma, \alpha, \beta\}$, with the corners contributing $\{\beta\}$ twice and $\{3\alpha\}$ once; in the interior, 50 of type $\{\gamma, \alpha, \beta\}$, 2 of type $\{3\alpha, 2\beta\}$, and $(v_1, v_2, v_3, v_4) = (4, 7, 1, 0)$. The identity reads $4 = 1 + 2 + 1 + 0$, exactly. And the $50 + 2 = 52$ interior π-figures agree exactly with the 52 edges carrying an interior vertex found in Remark 56 by an unrelated computation — a useful cross-check on both. Note the ratio: 50 of the 52 crossings are of the type the census cannot see.

Remark 103 (Anchoring one end is not enough, and the data says which edges get crossed)**.** It is tempting to hope for a lemma of the shape "a tile edge with one end on ∂ABC is not crossed". No such lemma exists. Across the four kernel-checked tilings, the split edges with exactly one endpoint on the boundary number 19, 8, 6 and 5 respectively, so crossings of that shape are routine.

Sorting them by edge type is more informative. Writing crossed/total for edges with exactly one end on the boundary:

	a	b	c
99-tiling	0/27	4/12	15/22
77-tiling	0/9	1/5	7/13
44-tiling	0/10	1/13	5/9
parallelogram 52	0/8	3/14	2/13

Two things follow. First, b-edges with one anchored end *are* crossed, in every one of the four. The chord at V_k in Theorem 92 is exactly such an edge, so the crossing there cannot be excluded by any statement about anchoring alone: the exclusion has to consume the cascade's forced tiles, and that is precisely what has resisted.

Second, a-edges with one anchored end are crossed in none of them. That one is not a coincidence, and the next lemma explains it with a sharp constant.

Lemma 104 (Blocked-end quantization)**.** *Let $[X, Y]$ be an edge of a tile of the tiling whose extension beyond X is blocked — either because $X \in \partial ABC$, or because the continuation enters* *the interior of another tile (Remark 49). Then every vertex of the tiling lying in the relative interior of* $[X, Y]$ *is at distance from* X *belonging to the numerical semigroup* $\langle a, b, c\rangle$. *Consequently* $[X, Y]$ *carries no interior vertex whenever*

$$\langle a, b, c\rangle \cap \big(0,\ |XY|\big) = \varnothing,$$

and in particular whenever $|XY| = \min(a, b, c)$.

Proof. Let T be the tile having $[X, Y]$ as an edge. On T's side of the segment nothing can contribute an interior vertex, since $[X, Y]$ is a single edge of the triangle T. So any vertex in the relative interior of $[X, Y]$ comes from the opposite side.

That side is covered by tiles abutting $[X, Y]$, each contributing a subsegment of one of its own edges, and consecutive contributions meet at points of $[X, Y]$; a vertex in the relative interior is exactly such a meeting point. By hypothesis no covering edge overruns X: its continuation would leave ABC, or would enter the interior of a tile. Hence the first covering edge begins exactly at X, and the covering proceeds by whole tile edges laid end to end. The meeting points are therefore the partial sums of a sequence of tile side lengths, so each lies at distance from X in $\langle a, b, c\rangle$. (The last covering edge may overrun Y; that affects no meeting point interior to $[X, Y]$.) The two consequences are immediate, the second because $\langle a, b, c\rangle$ has no element strictly between 0 and its smallest generator. □

Corollary 105 (One-end-blocked chord dichotomy)**.** *Let* $[X, Y]$ *be a tile edge with the extension beyond* X *blocked. Then the far side of* $[X, Y]$ *is covered by whole tile edges laid end to end from* X, *and exactly one of the following holds:*

(i) *they partition* $[X, Y]$ *exactly, so* $|XY|$ *is a sum of tile sides and the arithmetic of Lemma 48 applies as it stands;*
(ii) *the last one overruns* Y, *and those before it sum to some* $s \in \langle a, b, c\rangle \cap \big(0, |XY|\big)$, *with* $s = 0$ *permitted.*

In particular, if $|XY| = \min(a, b, c)$ *then that intersection is empty, so* $s = 0$ *and the far side of* $[X, Y]$ *is a* single *tile edge, either matching* $[X, Y]$ *exactly or overrunning* Y.

Proof. Immediate from Lemma 104: the covering begins at X and proceeds by whole edges, so the partial sums before the final edge lie in $\langle a, b, c\rangle$, and the final edge either ends at Y or passes it. □

This is the common form of three arguments that were being made separately. Lemma 48 is the case where *both* ends are blocked, which deletes branch (ii) outright. The chord at V_k in Remark 98 is the case $|XY| = b$ with the base end blocked, where $\langle a, b, c\rangle \cap (0, b)$ consists of the multiples of a below b — which is exactly the "j a-edges, then an overrun" description obtained there by hand. And the filler dichotomy of §5.6 is the same statement at a side junction. Branch (ii) is the crossing that Remark 103 shows cannot be removed by anchoring alone; the corollary does not close it, it states it once instead of three times.

Remark 106 (Branch (ii) occurs, so no general lemma will delete it)**.** It is worth being explicit that branch (ii) is not a formal possibility awaiting exclusion: it is realised in a kernel-checked tiling. In the 99-tiling, whose tile scaled by 16 is $(a, b, c) = (32, 48, 64)$, the b-edge from $(88, 24)$ — a boundary point — to the interior point $(136, 24)$ carries one interior vertex, at distance 32 from the blocked end. The remaining tail is $48 - 32 = 16$, and $16 \notin \langle 32, 48, 64\rangle$, so the covering edge that meets Y must overrun it. Four b-edges of that shape occur in that tiling alone.

Together with Remark 103 this settles the shape of the obstruction. Three successive generalisations have now been refuted by explicit certified witnesses: that an edge with one end on the boundary is uncrossed (false, $19 + 8 + 6 + 5$ counterexamples); that anchoring plus a length condition suffices (false, Remark 56); and that branch (ii) can be deleted (false, above). What survives is Corollary 105 itself, which is sharp, and the conclusion that the crossing at a given chord can only be excluded using the tiles the cascade has already forced there. No lemma quantified over chords will do it. That is a constraint on how the remaining work must be organised, and it is the reason the open steps in this note are stated as facts about particular configurations rather than as general pinning principles.

Remark 107 (The quantization is exactly tight)**.** The lemma is sharp in the strongest sense: not only does every crossing sit at a semigroup position, but *every* semigroup position below the edge length is realised. Across the eight kernel-checked tilings, all 52 crossings on edges with exactly one anchored end lie at a distance in $\langle a, b, c\rangle$ from that end, with no exception; and edges of minimal length are crossed 0 times out of 74. Comparing permitted with observed positions, by edge length:

44-, 77-, 99-tilings	$(a,b,c) = (16,24,32)$ or $(32,48,64)$	
length a	permitted $\varnothing$	observed $\varnothing$
length b	permitted $\{a\}$	observed $\{a\}$
length c	permitted $\{a,b\}$	observed $\{a,b\}$
parallelogram 52	$(a,b,c) = (18,48,54)$	
length a	permitted $\varnothing$	observed $\varnothing$
length b	permitted $\{18,36\}$	observed $\{18,36\}$
length c	permitted $\{18,36,48\}$	observed $\{18,36,48\}$

So no strengthening of the conclusion is possible: the permitted set is attained.

Which side is minimal depends on the member: $a < b$ exactly when $f > \varphi e$ with φ the golden ratio, since $a < b$ reads $f^2 - ef - e^2 > 0$. At the members carrying the certified tilings — $(1,2)$ and $(1,3)$ — one has $a = m$, which is why the observation appeared as a statement about a-edges.

The lemma is elementary and geometric, and its arithmetic content is only that a tile side is at least the least tile side, so we do not formalise it; it carries the label PROVED. It does not touch the chord at V_k, whose length is b and which is crossed in practice (Remark 103). It is recorded because it is the first general statement we have that excludes crossings outright, and because it fixes the exact radius within which the boundary still controls the interior.

Corollary 108 (The walls form at $e=1$, $f \geq 3$)**.** *Let $e = 1$ and $f \geq 3$. Then the base walk is the walls form $(f,1,1)$.*

Proof. Corollary 94 applies, since $f \geq 3 > 2 = 2e$, and leaves $(0,1,2)$ and $(f,1,1)$; the first is excluded by the three-letter argument of Remark 97. ☐

5.6. **The side parameter, and where its range is bounded.** Once the base walk is pinned, what remains is the side. The side question has a single integer parameter, and on the family of Corollary 108 that parameter takes only two open values, uniformly in the member.

The parametrisation itself is already in hand and is not new here. By Lemma 87 an equal side carries no b-edge at any member and its a-count is a multiple of f; writing that count as pf, the side equation $pf \cdot ef + n_c f^2 = f^3$ gives

$$n_c + pe = f,$$

and the γ-trap's $n_c \geq 1$ gives $0 \leq p \leq (f-1)/e$. Hypothesis 28 is exactly the assertion $p = 0$. Lemma 91 already records this profile in the form "a-count fj with $je \leq f - 1$", and the Lean statements are `SideNoB.side_no_b_uncond`, `SideNoB.side_quantized` and `SideNoB.c_corner_forces_side_a`. What follows is the one consequence we did not find recorded:

on the close pairs with $f > 2e$ the range of p is not merely bounded but equal to $\{0, 1, 2\}$, uniformly in the member.

Corollary 109 (A uniform two-case side problem)**.** *In the setting above, suppose in addition that $2e < f$ and that (e, f) is a close pair, $f^2 \leq 2ef + e^2$. Then $\lfloor (f-1)/e \rfloor = 2$. Hence on that family Hypothesis 28 reduces to excluding $p \in \{1, 2\}$ — the same two cases for every member.* (`side_p_le_two`.)

Proof. From $f \geq 2e + 1$ we get $(f - 1)/e \geq 2$. Conversely $p \geq 3$ would give $3e \leq pe \leq f - 1$, so $f \geq 3e + 1$ and then $f^2 - 2ef - e^2 \geq (3e + 1)(e + 1) - e^2 = 2e^2 + 4e + 1 > 0$, contradicting the close-pair inequality. □

Remark 110 (What this buys, and what it does not)**.** Corollary 109 is independent of Theorem 92 and of the base walk: it constrains the side at any member. On the close pairs with $f > 2e$ — 203 of them with $e \leq 40$, the smallest being $(3, 7)$ with $N = 138$, then $(4, 9)$ with $N = 227$ and $(5, 11)$ with $N = 338$ — the side parameter is bounded by 2. Checked on all 126003 close pairs with $f > 2e$ and $e \leq 1000$: $\lfloor (f - 1)/e \rfloor = 2$ in every one.

On that family Corollary 94 also reduces the base to two columns. It does not reduce it to one: that step needs Theorem 92 at $e \geq 2$, which Remark 97 leaves open. So for these members the current state is two base columns and two side parameters, not one of each.

The contrast with $e = 1$ is the point. There the range is $1 \leq p \leq f - 1$ and grows without bound with f, which is why the corner cascade of Conjecture 57 has to be run as an induction over block lengths. Here the range is $\{1, 2\}$ for every member of an infinite family, so the side question is a fixed finite case analysis rather than an induction. This is the first sub-family in which that happens.

At $(3, 7)$, concretely: $p = 1$ gives a side of 7 a-edges and 4 c-edges, and $p = 2$ gives 14 a-edges and 1 c-edge. Excluding those two remains open; the reduction bounds the problem, it does not solve it.

5.7. **A parity of the straight figures.** Lemma 82 records three counting identities whose published consequence is $v_1 = 1 + n_2 + v_3 + 2v_4$. The α-identity also carries a parity statement, which we record together with the reason it does *not* exclude any member.

Lemma 111 (Straight-figure parity)**.** *Write $S = n_1 + n_2$ for the total number of straight (π) figures. Then $S \equiv N + 1 \pmod 2$.*

Proof. The α-identity is $3 + n_1 + 3n_2 + 2v_2 + 4v_3 + 6v_4 = N$; substituting $n_1 = S - n_2$ gives $S + 2(n_2 + v_2 + 2v_3 + 3v_4) = N - 3$. (`census_parity`.) □

The three identities have rank two and this is the only parity they contain: substituting the v_1 relation into the β-identity returns the α-identity exactly.

Remark 112 (Why this does not exclude a member, and the trap it sets)**.** It is tempting to combine Lemma 111 with the boundary walks. On a member all of whose walks have odd edge-counts — $(3, 7)$ is one, with $k = 4p + 7$ on a side and 9 or 13 on the base — the boundary contributes $\sum k_i - 3$ straight figures, an even number, against the odd value $N + 1 = 139$ that Lemma 111 demands. That looks like an exclusion. It is not.

S counts *every* straight figure, and a straight figure need not lie on the boundary: at an interior T-junction one tile has the vertex in the relative interior of a side, contributing a straight angle and no corner, while the tiles on the other side have corners summing to π. Such a vertex is a $\{\gamma, \alpha, \beta\}$ or $\{3\alpha, 2\beta\}$ figure and is counted in $n_1 + n_2$. Writing T for their number, $S = \sum k_i - 3 + T$, so the parity constrains only T: at $(3, 7)$ every tiling has an *odd* number of interior T-junctions. Since non-edge-to-edge incidences are exactly what this problem must allow, nothing is excluded.

This is the failure mode Lemma 82 already warns of — corner counting cannot see the side structure — and the boundary walks are side structure. It is recorded here because the temptation is sharp and the arithmetic is otherwise correct.

5.8. **Apex rigidity, and a bound on the side parameter.** The apex of a base-β target is completely rigid, and the rigidity is an identity of the whole family rather than a coincidence at one member. It forces the covering of the first chord exactly, and from that it bounds p.

Throughout, A is the apex, the equal sides have length f^3, the apex angle is 3α, and $\sin(\alpha/2) = e/(2f)$, whence

$$\cos\tfrac{\alpha}{2} = \frac{\sqrt{4f^2-e^2}}{2f}, \qquad \cos\tfrac{3\alpha}{2} = \frac{(f^2-e^2)\sqrt{4f^2-e^2}}{2f^3}, \qquad \sin\tfrac{3\alpha}{2} = \frac{e(3f^2-e^2)}{2f^3}.$$

By Lemma 43 an equal side carries no b-edge, and the apex figure is $\{3\alpha\}$, so the tile at A presents α and its two edges there are b and c; the one on the side is therefore a c. Hence the first junction J of the side sits at distance $c = f^2$ from A. Write C for the chord through J parallel to the base. It cuts off a triangle similar to the target with ratio $f^2/f^3 = 1/f$, so the area above C is N/f^2 tile areas and $|C| = e(3f^2-e^2)/f$.

Lemma 113 (The apex identities)**.** *For every (e, f),*

$$b\cos\tfrac{\alpha}{2} = c\cos\tfrac{3\alpha}{2}, \qquad c\sin\tfrac{3\alpha}{2} - b\sin\tfrac{\alpha}{2} = a.$$

Proof. Both sides of the first are $(f^2-e^2)\sqrt{4f^2-e^2}/(2f)$. For the second, $c\sin\frac{3\alpha}{2} = e(3f^2-e^2)/(2f)$ and $b\sin\frac{\alpha}{2} = e(f^2-e^2)/(2f)$, whose difference is $e\cdot 2f^2/(2f) = ef = a$. Both are machine-checked (`lean/ApexRigidity.lean`: `apex_drop_eq`, `apex_edge_eq`). ☐

Theorem 114 (The first chord is covered exactly)**.** *Let T_1, T_2, T_3 be the three tiles at A, with T_1, T_3 the outer ones. Then:*

(a) *T_1's remaining apex edge is its b, ending at a point P lying* exactly *on C;*
(b) *T_1's third edge JP lies* along *C and has length exactly a;*
(c) *T_2 has exactly the fraction b/c of its area above C, and its trace on C is the segment PP' joining P to the mirror point P' of T_3, of length $e(f^2-e^2)/f$;*
(d) *consequently $a + |PP'| + a = |C|$ and the area above C is $2 + b/c = N/f^2$ — so T_1, T_3 and that fraction of T_2 fill the region above C* exactly*, and no other tile has any area above C.*

Proof. (a) The drop of J below A is $c\cos\frac{3\alpha}{2}$ and that of P is $b\cos\frac{\alpha}{2}$; they agree by Lemma 113. (b) The horizontal offsets of J and P from the axis of symmetry are $c\sin\frac{3\alpha}{2}$ and $b\sin\frac{\alpha}{2}$, differing by a.

(c) T_2 presents α at A with edges b and c there, the b along AP; let Q be its far vertex, at drop $c\cos\frac{\alpha}{2}$. The chord meets AQ at the point R with $|AR|/|AQ| = \cos\frac{3\alpha}{2}/\cos\frac{\alpha}{2} = b/c$ by Lemma 113, so $|AR| = b$ and $R = P'$; and T_2 is split by the line PR through its vertex P, so the area ratio is also b/c (`middle_fraction`). The length of PP' is $2b\sin\frac{\alpha}{2} = e(f^2-e^2)/f$.

(d) $2ef + e(f^2-e^2)/f = e(3f^2-e^2)/f = |C|$, and $2 + b/c = (3f^2-e^2)/f^2 = N/f^2$ (`area_above_chord`), which is the area above C. Since the three tiles already supply all of it, nothing else can. ☐

Corollary 115 (No T-junction at P)**.** *T_1 and T_2 each present γ at P, and $2\gamma = \pi + \alpha$, so the angle left at P is $\pi - \alpha < \pi$. Hence no tile has P interior to one of its edges: every tile meeting P has a vertex there, and the residual figure is $\{2\alpha, 2\beta\}$ or $\{\beta, \gamma\}$.*

Proof. T_1 has sides c, b, a at A, P, J as in Theorem 114, so its angle at P is opposite c, namely γ; likewise for T_2. A tile with P interior to an edge would contribute π, exceeding the residual

$\pi - \alpha$. The residual is $2\alpha + 2\beta$, and the vertex arithmetic of Lemma 32 at $k = 1$ leaves only the two figures named. $\square$

Theorem 116 (The second edge of an equal side is a c)**.** *Assume $e^2 + ef < f^2$, equivalently $a < b$. Then on every equal side the tile edge immediately below the first junction J is a c.*

Proof. Let Y be the tile below T_1's a-edge, adjacent to it at P. By Corollary 115, Y has a vertex at P and an edge running from P along C towards J. Since J lies on the boundary at distance a, that edge has length at most a; as $a < b < c$ it is exactly a, so Y's a-edge is JP, matching T_1's. In particular Y does not present α at P, whose flanks are b and c.

At J the figure is straight, hence $\{\alpha, \beta, \gamma\}$ or $\{3\alpha, 2\beta\}$. T_1 presents β there (its angle opposite b). The angles at the two ends of Y's a-edge are β and γ, so Y presents γ or β at J. In the first case the figure must be $\{\alpha, \beta, \gamma\}$ and the third tile presents α; in the second it must be $\{3\alpha, 2\beta\}$ and the remaining three tiles all present α. Either way, T_1 occupies the extreme position against the ascending side and Y the next, so the tile against the *descending* side presents α. Its flanks are b and c, and the side edge below J is that tile's edge; by Lemma 43 it is not a b, hence it is a c. $\square$

Corollary 117 (A bound on the side parameter)**.** *If $e^2 + ef < f^2$ then $n_c \geq 2$, hence*

$$pe + 2 \leq f.$$

In particular $p = 2$ is impossible whenever $2e + 2 \leq f$ fails, and on the tight subfamily $f = 2e + 1$ the bound reads $pe \leq 2e - 1$, so $p \leq 1$.

Proof. The edge at the apex and the edge below J are both c by Theorem 116, so $n_c \geq 2$; with $n_c + pe = f$ this is $pe + 2 \leq f$. For $f = 2e + 1$, $p \geq 2$ would give $2e \leq pe \leq 2e - 1$ (`side_p_bound`, `p_le_one_of_tight`). $\square$

Remark 118 (What this settles, and exactly where it stops)**.** Corollary 117 is a second and independent proof of Theorem 130 on the tight subfamily, by a route that uses no chord bound and no height identity — only the apex identities and the boundary at J. It also completes the first bullet of Remark 132: there the $a\,c$ ending was excluded only in the figure $\{\gamma, \alpha, \beta\}$ and only with the α adjacent to C, whereas Theorem 116 excludes it in *both* straight figures and for every p. Hence at $(3, 7)$ the side word of an equal side must end $c\,c$; the two cases $X = \beta$ and $X = \gamma$ of Remark 132 both persist, but now only inside that ending.

It does not reach $p = 1$ at any new member, and the reason is arithmetic rather than incidental. Killing $p = 1$ by this bound needs $e \geq f - 1$; with $e^2 + ef < f^2$ that forces $f = e + 1$ and then $e^2 < e + 1$, so $e = 1$ and $f = 2$ — the member already closed by Theorem 1 (`scope_limitation`). The hypothesis $e^2 + ef < f^2$ says f/e exceeds the golden ratio, and $e \geq f - 1$ says it is close to 1; the two cannot hold together beyond $(1, 2)$.

The obstruction to iterating is equally precise. Corollary 115 needed *two* tiles presenting γ at P, and the second, T_2, exists only because the apex figure is 3α. At the analogous point one level down — the far end of the next c-tile's a-edge, which does lie exactly on the next chord, by the same identities — only one γ is forced. Supplying a second is what an induction would require, and we do not have it.

Proposition 119 (The descent is self-similar)**.** *Suppose a tile Z lays a c-edge of an equal side between consecutive junctions J' (nearer the apex) and J, and presents α at J'. Let W be its third vertex. Then W lies* exactly *on the chord C through J; the segment JW is Z's a-edge, of length a, lying along C; and Z presents β at J and γ at W.*

Proof. The two edges of Z at J' are its flanks at α, namely b and c; the c runs down the side to J and the b into the interior, at angle α from the descending side, hence at $\frac{3\alpha}{2} - \alpha = \frac{\alpha}{2}$ from the downward vertical — the same inclination as the apex tile's b-edge in Theorem 114. So the drop from J' to W is $b\cos\frac{\alpha}{2}$ and that from J' to J is $c\cos\frac{3\alpha}{2}$, equal by Lemma 113; and the horizontal offsets differ by $c\sin\frac{3\alpha}{2} - b\sin\frac{\alpha}{2} = a$. The angles at J and W are those opposite b and c, namely β and γ. □

Remark 120 (Why the descent does not close, and it is not for want of trying)**.** Proposition 119 says the configuration at J is a verbatim copy of the configuration at the first junction, with (A, J_1, P) replaced by (J', J, W). If the argument of Theorem 116 could be run at J as it was at J_1, it would force the side edge below J to be a c as well, and induction down the side would give $n_a = 0$, that is $p = 0$ — exactly Hypothesis 28. It cannot, and the obstruction is sharp.

The argument at J_1 turns on Corollary 115: *two* tiles present γ at P, so the residual is $\pi - \alpha < \pi$ and no tile can have P interior to an edge. The second γ comes from T_2, which exists only because the apex figure is $\{3\alpha\}$. At W only Z's γ is forced, and no counting argument can supply the other:

Lemma 121 (A single γ never excludes a T-junction)**.** *Let x be an interior point at which one tile presents γ and one tile has x interior to an edge. Then the remaining tiles at x present angles totalling exactly $\alpha + \beta$, realised by one α and one β. The angle count is therefore always consistent.*

Proof. $2\pi = \gamma + \pi + R$ gives $R = \pi - \gamma = \alpha + \beta$, since $\alpha + \beta + \gamma = \pi$. □

So the obstruction is not a missing computation but the tile's own angle sum: a lone γ beside a straight edge balances at every member and at every point. Exclusion by angle count needs two γ's, and the apex is the only place that supplies two.

We also record a near miss, since it shows how narrow the gap is. At W the T-junction *can* be excluded when it would sit on the tile V angularly adjacent to Z at J': if V lays its c along Z's b-ray, the budget is $\gamma + \pi$ with residual $\pi - \gamma < \pi$, so no *second* straight angle fits, and the tile below Z's a-edge is forced to have a vertex at W and to lay exactly a — which is all Theorem 116 needs. What defeats this is that the T-junction need not sit on V at all: the tile *below* C may carry an edge running along C from J through W, of length b or c, with its vertex on the boundary at J. That is precisely the tile the argument was trying to pin, so it closes on itself. Settling whether that placement can occur is, as far as we can see, the whole of what remains at $p = 1$.

Definition 122 (The points $W(J)$)**.** Let J be a junction of an equal side whose edge *above* it is a c, laid by a tile presenting α at the junction above. By Proposition 119 that tile's third vertex lies on the chord through J at distance exactly a from J; call it $W(J)$.

Theorem 123 (An a-edge forces a T-junction)**.** *Assume $e^2 + ef < f^2$. Let J be as in Definition 122 and suppose the side edge* below *J is an a. Then $W(J)$ lies in the* interior *of an edge of some tile.*

Proof. Write Z for the c-tile above J; by Proposition 119 it presents β at J and its a-edge is JW with $W = W(J)$. Let M be the tile below sharing the chord ray with Z, and let X be the last tile in angular order, standing against the descending side; X lays the next side edge on one of its flanks, and that flank is not a b by Lemma 43.

The figure at J is $\{\alpha, \beta, \gamma\}$ or $\{3\alpha, 2\beta\}$, with Z first. Exhausting the orders with first entry β:

figure	order	M	side edge below J
$\{\alpha, \beta, \gamma\}$	(β, α, γ)	α	a
$\{\alpha, \beta, \gamma\}$	(β, γ, α)	γ	c
$\{3\alpha, 2\beta\}$	$(\beta, \alpha, \alpha, \alpha, \beta)$	α	a or c
$\{3\alpha, 2\beta\}$	$(\beta, \alpha, \alpha, \beta, \alpha)$	α	c
$\{3\alpha, 2\beta\}$	$(\beta, \alpha, \beta, \alpha, \alpha)$	α	c
$\{3\alpha, 2\beta\}$	$(\beta, \beta, \alpha, \alpha, \alpha)$	β	c

(the side edge is read off X's flanks with b removed: $\alpha \mapsto c$, $\beta \mapsto a$ or c, $\gamma \mapsto a$). In both rows permitting an a, M presents α. Its flanks are then b and c, so the edge M lays along the chord ray has length b or c, and $e^2 + ef < f^2$ gives $a < b < c$. Since $|JW| = a$, the point W lies strictly inside that edge. ☐

Corollary 124 (A sufficient condition for Hypothesis 28)**.** *Assume $e^2 + ef < f^2$. If for every junction J as in Definition 122 the point $W(J)$ is a* vertex *of every tile meeting it, then $p = 0$ on every equal side, i.e. Hypothesis 28 holds at that member.*

Proof. Suppose some equal side carries an a-edge, and take the first one counted from the apex. Every edge above it is a c, so the junction J immediately above it satisfies Definition 122, and Theorem 123 puts $W(J)$ interior to an edge — contrary to hypothesis. Hence $n_a = 0$, that is $p = 0$. ☐

Remark 125 (The crux, in its sharpest form)**.** Corollary 124 replaces "exclude the surviving configuration" by a single question about an explicitly located finite set of points. The exclusion succeeds at the first junction: there $W(J_1) = P$, and Corollary 115 rules out a T-junction because two tiles present γ at P. It is unavailable at every later junction, and not by accident: Lemma 121 shows a single γ beside a straight edge always balances, so no angle count can do it, and Proposition 119 shows the configuration one level down is a verbatim copy, so no new local information appears. What is needed is a second γ at $W(J)$ from geometry rather than from counting.

We note the numerical consequence is weak and should not be oversold: only a $c \to a$ transition forces a T-junction, and a side word such as $c\,c\,c\,c\,a^7$ has exactly one. The theorem gives one forced T-junction per equal side carrying an a, not one per a-edge, and in particular it does not revive the counting argument retracted in Remark 112.

5.9. The tight subfamily $f = 2e + 1$.

Corollary 109 leaves the two cases $p \in \{1, 2\}$. The case $p = 2$ is not uniformly hard: on one infinite subfamily it is completely rigid, and the rigidity is visible in a counting budget.

Lemma 126 (The γ-injection budget at $p = 2$)**.** *A side of parameter p carries $n_a = pf$, $n_b = 0$, $n_c = f - pe$, hence $k = pf + (f - pe)$ edges and $k - 1$ junctions, so the γ-injection bound $\#a + \#b \le k - 1$ has slack $f - pe - 1$. At $p = 1$ the slack is $f - e - 1 > 0$ for every member. At $p = 2$ it is $f - 2e - 1$, which vanishes exactly when $f = 2e + 1$.* (`gamma_slack`, `p_two_tight_iff`.)

The subfamily $f = 2e + 1$ is infinite and automatically coprime, with $N = 3f^2 - e^2 = 11e^2 + 12e + 3$: the members $(1, 3)$ at $N = 26$, $(2, 5)$ at 71, $(3, 7)$ at 138, $(4, 9)$ at 227, $(5, 11)$ at 338. The first is closed by Theorem 50 and the second by search, so $(3, 7)$ is the smallest tight member not already settled.

Proposition 127 (The $p = 2$ side is forced)**.** *Let $f = 2e + 1$ and suppose an equal side has $p = 2$. Then $n_c = f - 2e = 1$, and the side word is $a^{2f} c$, read from the base corner to the* *apex. Every one of its* $2f$ *junctions carries exactly one* γ *and is the figure* $\{\gamma,\alpha,\beta\}$*; each* a*-tile presents* β *at its left junction and* γ *at its right; the corner tile lays* a *on the side and* c *on the base.*

Proof. $n_c = f - 2e = 1$ and $2f \cdot ef + f^2 = f^2(2e+1) = f^3$, so the profile is consistent (`p_two_single_c`). Tightness makes the γ-injection a bijection, so every junction carries exactly one γ; a π-vertex carrying a γ is $\{\gamma,\alpha,\beta\}$ (`pi_vertex_with_gamma`). The apex figure is 3α and the sides carry no b-edge (Lemma 43), so the apex-end edge is a c; with only one c available it sits there, fixing the word. The corner figure is a single β with flanks $\{a,c\}$, and the base-corner end being an a-edge puts the c on the base. The alternation follows: the corner tile presents γ at J_1, and each junction having a unique γ forces the next a-tile to present β there. □

Remark 128 (The last junction, and what is still missing)**.** At J_{2f} the two boundary neighbours are the a-tile A_{2f}, presenting γ with a b-ray inward, and the apex c-tile, presenting β with an a-ray inward; the filler's rays are b and c. Since neither b nor c equals a, J_{2f} *cannot be edge-to-edge*, and each placement leaves a residual run: $c-a$, or $b-a$ together with $c-b=e^2$. All three are gaps of $\langle a,b,c\rangle$ — $b-a$ by `gap_b_sub_a`, e^2 by `gap_e_squared`, and $c-a$ by `gap_c_sub_a_tight`, which holds precisely for $e \geq 2$ since $c-a=2a$ at $(1,3)$. Moreover $e^2 < a$, so no tile edge fits inside that run at all. Hence the covering *must* overrun a filler vertex.

Each placement therefore forces the covering to overrun a filler vertex. That alone is not an exclusion, and the exclusion comes instead from a chord bound, which we now state.

Lemma 129 (The chord at the last junction)**.** *Let* $f = 2e+1$ *and let a side have* $p = 2$*. The apex* c*-tile presents* β *at* J_{2f} *with its* a*-ray pointing along the horizontal, and* J_{2f} *lies at height fraction* $2e/f$*, so the horizontal chord from* J_{2f} *to the opposite side has length exactly* $e(3f^2-e^2)(f-2e)/f = eN/f$*. At* $(3,7)$ *this is* $414/7$*.*

Moreover the tiles meeting that chord from either side meet it in whole *edges: a tile whose interior misses the chord line lies in a closed half-plane of it, so its contact is a face (*`tile_contact_face`*), and at a non-vertex point that face is an edge (*`contact_is_edge`*). Disjointness of tile interiors then makes the covering a chain starting at the end of the* c*-tile's* a*-edge, and convexity of the target keeps every endpoint within the chord.*

Theorem 130 ($p = 2$ is excluded on the tight subfamily)**.** *Let* $f = 2e+1$*. Then no equal side of the base-*β *target at* $m = 1$ *has* $p = 2$*.*

Proof. By Proposition 127 the side word is $a^{2f}c$ and the last junction carries $\{\gamma,\alpha,\beta\}$, so by Remark 128 a tile must sit above the chord of Lemma 129 with an edge beginning at chord position a. Enumerate those placements: the edge on the chord is a, b or c, each with two chiralities.

If the edge is c it leaves the target, since $(a+c)f - eN = f^3 - 2ef^2 + e^3 = (2e+1)^2 + e^3 > 0$.

If the edge is a, the height of the tile over its a-edge is $2\,\mathrm{Area}/a$, and the apex stands $H(f-2e)/f$ above the chord where H is the target height. Writing A for the target area, these are $2A/(Nef)$ and $2A(f-2e)/(eNf)$, equal precisely when $f-2e=1$. So on this subfamily the third vertex lies at exactly apex height, and the target meets that height in one point: the apex. The two chiralities then demand $N(f-2e-1) = 2f^2$ and $N(f-2e+1) = 4f^2$, i.e. $0 = 2f^2$ and $N = 2f^2$, the latter giving $e = f$. Both are impossible. (At $(3,7)$ the tile would be $(21,40,41)$.)

If the edge is b, one chirality overruns the chord when $-e^3+3e^2+4e+1 > 0$, i.e. for $e \leq 4$, and for $e \geq 5$ its third vertex falls outside on the left, since $(f^2-e^2)(2f-e) \geq e(3f^2-e^2)$ reduces to $e^3 \leq 3e^2+4e+1$, false there; the ranges are complementary. The other chirality places the third vertex at chord position $a + (2f^2-e^2)/2$, exceeding the chord by $(7e^2+8e+2)/(2f) > 0$ for every e.

No placement survives. □

Theorem 131 (The last-junction dichotomy)**.** *Let (e, f) be any member with $f = 2e + 1$, and consider the last junction J of an equal side — the junction where the final c-edge begins, at distance $f^3 - f^2$ from the base corner, hence at height fraction $(f - 1)/f$ for every value of p. Let X be the tile angularly adjacent to the apex c-tile C on the interior side of C's a-ray. Then X lays an a-edge along that ray, coinciding with C's. Equivalently, X* does not present *α at J.*

Proof. X lays one of its own flanks along C's a-ray. If it lays c, the run $c - a$ must be covered inside the chord of Lemma 129 and that window is empty. If it lays b, the window pins the covering to a single a-edge, and the height identity of Theorem 130 then places that tile's third vertex at the apex, where its remaining sides fail to be tile sides. So X lays a. The flanks of α are b and c, so X is not an α-vertex. (`alpha_cannot_lay_a`, `adjacent_angle_not_alpha`.) □

Remark 132 (What the dichotomy settles, and what it leaves)**.** Theorem 130 is the case $p = 2$: there the tightness of Proposition 127 forces X to be the α-filler, which Theorem 131 forbids. The dichotomy is strictly more general — it is stated for every p and every vertex figure — and at $p = 1$ it forces structure without closing the case:

in the figure $\{\gamma, \alpha, \beta\}$ it forces the angular order $C(\beta), \gamma, \alpha$, so the side word cannot end $a\,c$ with the α adjacent to C;

in the figure $\{3\alpha, 2\beta\}$ it forces X to be the second β. That β shares C's a-edge and is the reflection of C across it, so both present γ at the far end Q of that edge. Since $2\gamma = \pi + \alpha < 2\pi$, the point Q is interior, and its figure is $\{2\alpha, 2\beta, 2\gamma\}$ or $\{\beta, 3\gamma\}$.

At $(3, 7)$ the reflected tile is (J, Q, R) with $R = (207, 100\sqrt{187}/7)$, a genuine $(21, 49, 40)$ placement lying inside the target, and both figures at Q are consistent. So the two surviving configurations at $p = 1$ — the mirror partner and the γ — are a single phenomenon, and neither is excluded by the chord or the height identity. Closing $p = 1$, and with it Hypothesis 28 on the whole subfamily $f = 2e + 1$, requires a further mechanism — and §5.10 rules out the two most natural candidates, so that mechanism must be metric.

Remark 133 (Scope, and where $e \geq 2$ is easier)**.** Theorem 130 closes $p = 2$ on the whole subfamily $f = 2e+1$, hence at $(1, 3)$, $(2, 5)$, $(3, 7)$, $(4, 9)$, $(5, 11)$ and onward; verified independently by exact enumeration of the six placements for every member with $e \leq 400$. Combined with Corollary 109 it leaves $p = 1$ as the only obstruction to Hypothesis 28 on that subfamily.

It does not extend beyond $f = 2e + 1$: the height identity used above holds exactly there.

Two by-products. First, $c - a$ is a gap of $\langle a, b, c\rangle$ precisely for $e \geq 2$, failing at $(1, 3)$ where $c - a = 2a$; this is the first point in the problem where $e \geq 2$ is easier than $e = 1$. Second, at $(3, 7)$ the base walk $(0, e, 2e)$ carries no a-edge, so the corner tile lays c on the base and a on the side, forcing $p \geq 1$ on both equal sides; with $p = 2$ excluded, Hypothesis 28 at $(3, 7)$ therefore requires the base walk (f, e, e).

5.10. **Two mechanisms that provably cannot close $p = 1$.** Remark 132 ends by saying that closing $p = 1$ "requires a further mechanism". That is a hedge, and it can be replaced by theorems. This subsection records two natural mechanisms and proves that neither can work, and one arithmetic fact that survives and points the other way. The member is $(e, f) = (3, 7)$ throughout: tile $(a, b, c) = (21, 40, 49)$, target $(343, 343, 414)$, $N = 138$.

Fix an equal side with $p = 1$. Its word is a permutation of a^7c^4 ending in c ($7{\cdot}21{+}4{\cdot}49 = 343$), so it has 11 edges and 10 interior junctions. Each edge is laid by a tile, and the two angles of that tile at the ends of the edge are the two angles *not* opposite it: an a-edge is flanked by β and γ, a c-edge by α and β. Call the choice of which flank sits at the lower end the *orientation* of the edge.

Lemma 134 (Endpoint angles)**.** *The angle at the top of the last edge is α, and the angle at the bottom of the first edge is β.*

Proof. By Lemma 82 the apex fills uniquely as $\{3\alpha\}$ and each base corner uniquely as $\{\beta\}$. The last edge of the side is a c-edge, whose flanking angles are α and β; the tile at the apex presents α, so the top angle is α. The single tile at the base corner presents β, so the bottom angle of the first edge is β. □

Proposition 135 (The junction automaton is consistent, and hugely so)**.** *Regard the side as a transfer system: a state is the angle at the top of the current edge, and a transition is admissible when the pair (top angle of one edge, bottom angle of the next) extends to a straight figure at their junction. Then the only inadmissible pair is (γ,γ), which can occur only at a junction between two a-edges. Consequently, over all 120 words a^7c^4 ending in c and all 2^{11} orientations, exactly 10 560 configurations satisfy every junction constraint together with Lemma 134.*

Proof. A junction of an equal side carries a straight figure, so by Lemma 82 it is $\{\gamma,\alpha,\beta\}$ or $\{3\alpha,2\beta\}$. The figure $\{\gamma,\alpha,\beta\}$ has one angle of each kind, so it admits a pair of side angles precisely when the two are distinct. The figure $\{3\alpha,2\beta\}$ contains no γ, so it admits precisely the pairs drawn from $\{\alpha,\beta\}$, all four of which leave an admissible remainder (3α, $2\alpha+\beta$, $2\alpha+\beta$, $\alpha+2\beta$ respectively). A pair is therefore inadmissible exactly when it is equal and involves γ; and γ occurs only as a flank of an a-edge. The count is a finite check over $120\cdot 2^{11}$ cases. □

Proposition 136 (The census contributes one relation, and it is already spent)**.** *Let x be the number of junctions of the side carrying $\{\gamma,\alpha,\beta\}$. Then $x\geq 7$, and the α- and β-counts along the side are linearly dependent: they sum to the identity $13=13$. Both $x=7$ and $x=10$ occur in solutions, so the census does not determine x.*

Proof. The 7 a-edges contribute one β and one γ each, the 4 c-edges one α and one β each: 7γ, 11β, 4α in all. By Lemma 134 the apex absorbs one α and the base corner one β, leaving 7γ, 10β, 3α distributed over the 20 side-tile slots at the 10 junctions. A $\{3\alpha,2\beta\}$ junction contains no γ and a $\{\gamma,\alpha,\beta\}$ junction exactly one, so the 7 γ's occupy 7 distinct junctions of the latter type, whence $x\geq 7$.

Write u for the number of junctions whose γ lies on a side tile and whose other side slot is β, and (a_i,b_i) with $a_i+b_i=2$ for the side slots at the $10-x$ junctions of type $\{3\alpha,2\beta\}$. Counting α and β separately,

$$(7-u)+(x-7)+\textstyle\sum_i a_i \;=\; 3, \qquad u+(x-7)+\textstyle\sum_i b_i \;=\; 10.$$

Adding and using $\sum_i(a_i+b_i)=2(10-x)$ gives $7+2x-14+20-2x=13$, an identity. The two equations are therefore dependent. Taking $x=10$, $u=7$ satisfies both, as does $x=7$, $u=4+\sum_i a_i$ for any $\sum_i a_i\leq 3$. □

Remark 137 (What the two no-go results mean)**.** Proposition 135 shows that the local combinatorics of the side is almost unconstrained: the automaton forbids one adjacency and leaves 10 560 configurations. Proposition 136 shows that adjoining the census adds nothing, because the two counting equations are dependent. This is the third independent confirmation of the warning in Lemma 82 that corner counting cannot see p, and it is also the exact reason the retraction described in Remark 112 was possible: a counting argument at $(3,7)$ has one relation to spend and it is spent.

The conclusion is directional, not merely negative. Any mechanism that closes $p=1$ must use metric information — positions, not multiplicities. Proposition 138 is the first such fact.

Proposition 138 (Every junction chord is straddled)**.** *Let J be a junction of an equal side at distance d from the apex, and let C be the chord of the target through J parallel to the base. Then some tile of the tiling meets both open sides of C.*

Proof. Since d is a sum of edge lengths 21 and 49, we have $d = 21i + 49j$ and the height fraction is $\rho = d/343 = (3i + 7j)/49$. Write $n = 3i + 7j$, so $1 \le n \le 48$ for a junction strictly between the apex and the base. The chord cuts off a triangle similar to the target with ratio ρ, whose area in units of the tile area is

$$138\rho^2 \;=\; \frac{138\,n^2}{2401}.$$

Now $2401 = 7^4$ and $138 = 2 \cdot 3 \cdot 23$, so $\gcd(138, 2401) = 1$ and the quantity is an integer if and only if $7^4 \mid n^2$, i.e. $49 \mid n$ — impossible for $1 \le n \le 48$. Hence the area above C is not an integral number of tiles. If no tile met both open sides of C, every tile would lie weakly on one side and the area above C would be a sum of whole tile areas, a contradiction. □

Remark 139 (Why this is the right direction)**.** At the last junction, $n = 7$ and the area above the chord is $138/49 = 2.816\ldots$ tiles, while the chord has length $414/7 = 59.14\ldots$ and a tile meets a line in a segment of length at most $c = 49$; so at least two tiles meet that chord, and at least one straddles it. This is exactly the region below the chord that the enumeration of Theorem 130 does not reach — see the discussion of the $c\,c$ ending in Remark 132 — and it is the reason the surviving configurations survive: not that the chord bound fails there, but that it was never applied there. Whether the straddling constraint can be sharpened into a contradiction is open.

Proposition 140 (Chord decomposition at the last junction)**.** *Let C be the chord of the target through the last junction J of an equal side at $(3, 7)$, so $|C| = 414/7$. Let $\mathcal{U}$ be the set of tiles whose interiors meet the open half-plane above C. Each $T \in \mathcal{U}$ either* straddles C *(its interior meets both open sides) or is* flush from above *(it lies weakly above the line of C). Then:*

- (a) *a flush tile meets the line of C in a full tile edge, of length* 21, 40 *or* 49*;*
- (b) *the traces on C of the tiles of $\mathcal{U}$ have pairwise disjoint interiors and cover C, so their lengths sum to* $414/7$*;*
- (c) *at most two tiles of $\mathcal{U}$ are flush, and the multiset of flush lengths is one of* $\varnothing, \{21\}, \{40\}, \{49\}, \{21, 21\}$*;*
- (d) *at least one tile straddles, and if none is flush then at least two straddle.*

Proof. (a) For a tile lying weakly above the line of C, that line is a supporting line, so the intersection is a face of the tile — a vertex or an edge (`contact_is_edge`). A trace of positive length excludes the vertex case, and the edge lengths are $a = 21$, $b = 40$, $c = 49$.

(b) The closed region above C is covered by the tiles, and a point of C interior to the target is a limit of points of that region, hence lies in a tile of $\mathcal{U}$, which is closed; so the traces cover C. If two tiles of $\mathcal{U}$ had traces overlapping in a segment, both would occupy area immediately above that segment, contradicting disjointness of interiors. Additivity of length gives the sum.

(c) The flush lengths are integers drawn from $\{21, 40, 49\}$ and, by (b), sum to at most $|C| = 414/7 = 59.142\ldots$. The only multisets meeting that bound are the five listed: already $21 + 40 = 61 > |C|$, and $21 + 21 + 21 = 63 > |C|$.

(d) By the computation in Proposition 138 the area above C is $138/49$ tiles, not an integer, so a straddler exists. If no tile of $\mathcal{U}$ is flush, the straddling traces alone sum to $|C| = 59.142\ldots$; a segment contained in a tile has length at most the tile's diameter, which for a triangle is its longest side $c = 49$. Hence at least two straddlers. □

Remark 141 (What the decomposition gives, and the rung that remains)**.** Proposition 140 is the first statement that constrains the configuration *below* the chord at J, the region Remark 132 identifies as untouched. It reduces the possibilities there to a finite list: a flush total in $\{0, 21, 40, 42, 49\}$ together with a straddling remainder of $59.142\ldots$, $38.142\ldots$, $19.142\ldots$, $17.142\ldots$ or $10.142\ldots$, distributed over at least one and at most a bounded number of straddling tiles.

What it does not give is rigidity: the cross-section of a straddling tile varies continuously with its position, so no individual value is forced. The natural hope is that the γ-grading of §5.2 restores discreteness — a straddler's edges have directions that are integer multiples of γ modulo π, so one might expect its cross-section to be drawn from a countable set fixed by its label. *That hope is unfounded*, and we record why, since it is the first thing a reader will try.

The grading constrains directions, not positions. Tile vertices lie in the $\mathbb{Z}$-module generated by the admissible edge vectors, and for a horizontal chord what matters is the projection of that module to the vertical axis. Since $\gamma = (\pi + \alpha)/2$ with α/π irrational, that projection is generated by the values $\ell \sin(k\gamma)$ for $\ell \in \{21, 40, 49\}$ and k in the label window, and these are rationally independent: the subgroup they generate is dense in $\mathbb{R}$. So no cross-section is forced, and the chord decomposition cannot by itself close $p = 1$.

What the decomposition does establish is that the obstruction is located strictly *below* the chord. Above it there is only $138/49 = 2.816\ldots$ of a tile's area, so that region is enumerable up to boundedly many tiles — which is precisely the enumeration Theorem 130 performs. The surviving configurations of Remark 132 place the decisive tile below the chord, where no area bound confines it. Closing $p = 1$ therefore requires a constraint that reaches downward, and we do not have one.

Remark 142 (The chord programme is exhausted)**.** It is natural to hope that the family of chords does collectively what no single chord does: there are 37 admissible heights $\rho = n/49$ with $n = 3i + 7j$, each carrying its own non-integral area $138\rho^2$, and a tile lying below the chord at J lies above the chord at some lower junction. We record that this hope is also unfounded, so that the reader need not pursue it.

For a tile T and a chord C write $\varphi_T(C) \in [0, 1]$ for the fraction of T's area lying above C; thus T straddles C exactly when $\varphi_T(C) \in (0, 1)$. The area statement at height ρ is

$$\sum_T \varphi_T(C_\rho) \;=\; 138\rho^2,$$

and this is nothing but additivity of area over the tiling: the left-hand side *is* the area above C_ρ, decomposed tile by tile. It therefore holds identically, for every tiling of the target, and the 37 admissible heights supply 37 instantiations of one identity rather than 37 constraints.

The only information extractable from the family is the integrality bit — where $138\rho^2 \notin \mathbb{Z}$, some $\varphi_T(C_\rho)$ lies strictly between 0 and 1 — and that is precisely Proposition 138. The nesting adds nothing further: the set of heights straddled by a given tile is an interval, and 138 tiles cover 37 heights with room to spare.

This is the same degeneracy as Proposition 136, in a metric rather than a combinatorial register: a single global identity, instantiated many times, cannot separate tilings. Taken together with Remark 141 it closes the chord programme. What the chord method yields at $(3, 7)$ is exactly Propositions 138 and 140, and closing $p = 1$ requires a mechanism that is neither counting nor chord-area. We do not have one, and we prefer to say so.

Remark 143 (Close pairs: what the unconditional filters leave)**.** At a close pair ($f^2 \le 2ef + e^2$) Lemma 88 no longer applies and the base admits extra columns, all with $y = e + kf$ for some $k \ge 1$. Three unconditional filters apply: the γ-trap removes $z = 0$; Theorem 92 removes $x = 0$; and the corner parallelogram, which keeps b-edges out of the first two and last two positions, removes any column with $x + z \le 3$ in a word of at least four edges (`cornerpara_filter`). Over the range $e \le 9$, $f \le 15$ these leave 54 of the 62 extra columns standing, the first being $(3, 7, 2)$ at $(3, 4)$; every survivor has $x < f$. Closing them requires an argument bounding the number of base b-edges, and no such bound follows from size: writing $y = e + kf$, the size constraint $yb \le e(3f^2 - e^2)$ reads $k \le 2ef/(f^2 - e^2)$, whose vanishing is precisely the separation condition

$f^2 > 2ef + e^2$. The extra columns at close pairs are permitted by size by construction, so a bound on y there must be geometric.

Close pairs are not a finite exception: the condition $f \leq (1+\sqrt{2})e$ admits infinitely many coprime pairs, and among the smallest are $(5,6)$ and $(4,7)$, that is $N = 83$ and $N = 131$. Member by member certification therefore cannot settle this family.

Remark 144 (The extra base columns, in closed form)**.** The extra columns at a close pair all have $y = e + kf$ with $k \geq 1$ (Remark 143), and the obstruction is that no bound on k follows from size. Substituting $y = e + kf$ into the base equation collapses it: the difference is f times a relation in x and z alone,

$$x(ef) + (e+kf)(f^2-e^2) + zf^2 - e(3f^2-e^2) \;=\; f\big(xe + zf - (2ef - kb)\big), \qquad b = f^2 - e^2,$$

an identity (`base_column_reduction`). So for each k the surviving columns are the solutions of the two-variable linear Diophantine equation

$$xe + zf = 2ef - kb,$$

not of a three-variable search. Verified against brute enumeration on every close pair with $e \leq 25$, $f \leq 59$: 655 surviving columns, no exception.

Two consequences. At $k = 1$ the column is pinned, $(x,y,z) = (e,\, e+f,\, 2e-f)$, so it occurs exactly when $f \leq 2e-1$ (`base_column_k_one`). And the equation bounds k. Using only $x, z \geq 1$ gives $kb \leq 2ef - e - f$ (`base_column_k_bound`). The corner parallelogram also forces $x + z \geq 4$, and since $e < f$ the minimum of $xe + zf$ subject to $x, z \geq 1$, $x + z \geq 4$ sits at $(x,z) = (3,1)$, so

$$kb \;\leq\; 2ef - 3e - f \qquad (\texttt{k\_bound\_sharp}),$$

better by $2e$. Checked against brute enumeration on every close pair with $e \leq 25$, $f \leq 59$: no violation, and the bound is *attained* — equal to the largest surviving k — at 186 of the 198 members carrying survivors. There is exactly one surviving column for each k in range. For fixed k the solutions form a chain $x = x_0 + tf$, $z = z_0 - te$, so there are $\lfloor (z_0-1)/e \rfloor + 1$ with $z \geq 1$, and exactly one when $z_0 \leq e$; writing $ke = qf + r$ this is $e(1+q) \leq kf$, and since $e < f$ we have $ke/f < k$, so $q \leq k-1$ and $e(1+q) \leq ek < fk$ (`one_column_per_k`; neither coprimality nor the close-pair hypothesis is used). Hence the surviving set at a member is described completely by

$$k = 1, \ldots, \left\lfloor \frac{2ef - 3e - f}{b} \right\rfloor.$$

The description can be made exact. Reducing $xe + zf = 2ef - kb$ modulo f gives $e(x - ke) = f(2e - kf - z)$, so $f \mid e(x-ke)$ and, by coprimality, $f \mid x - ke$ (`column_x_congruence`). For each k the admissible x therefore form a single residue class modulo f, and the surviving columns of that k are the members of that class with $z = (2ef - kb - xe)/f \geq 1$ and $x + z \geq 4$. Over every close pair with $e \leq 25$, $f \leq 59$ this criterion reproduces the surviving k-set at all 198 members carrying survivors, with no exception. The 12 members at which the bound $kb \leq 2ef - 3e - f$ is not attained are exactly those where $xe + zf = 2ef - kb$ has no solution with $x, z \geq 1$ — a Frobenius condition on $\langle e, f \rangle$, not a further geometric filter; at $(e,f) = (15,23)$ and $k = 2$, for instance, the residue 44 is a gap of $\langle 15, 23 \rangle$.

All of this is a description of the surviving set, not the required $k = 0$. What it buys is that the residual question is one linear congruence per k, with k in an explicit finite range, in place of an unstructured three-variable family.

Proposition 145 (The extra base columns at a close pair)**.** *Let* $\gcd(e,f) = 1$ *and* $1 \leq e < f$, *let* $b = f^2 - e^2$, *and let* (x,y,z) *be a base column with* $y = e + kf$, $k \geq 1$, *satisfying* $x, z \geq 1$ *and* $x + z \geq 4$. *Then*

(i) *the base equation is equivalent to* $xe + zf = 2ef - kb$;

(ii) *$f \mid x - ke$, so x lies in a single residue class modulo f;*
(iii) *$kb \le 2ef - 3e - f$;*
(iv) *for each k the pair (x, z) is unique.*

Consequently the extra columns at a close pair are exactly one for each $k = 1, \dots, \lfloor(2ef - 3e - f)/b\rfloor$ that admits a solution, with x determined modulo f and z determined by x. (`close_pair_column`, `close_pair_column_unique`, `one_column_per_k`.)

The clause "that admits a solution" in Proposition 145 is not decided by (iii): the bound there ignores the residue class of x, and over-permits. At $(e, f) = (8, 17)$ it allows $k = 1$, while $8x + 17z = 47$ with $x \equiv 8 \pmod{17}$ has no solution with $x, z \ge 1$, so no column exists; the same happens at 551 of the 1558 close pairs with $e \le 60$. Restoring the residue class makes the test exact.

Proposition 146 (Sharp criterion for an extra base column)**.** *Let $\gcd(e, f) = 1$, $1 \le e < f$, $b = f^2 - e^2$. For $k \ge 1$ let x_k be the least positive integer with $x_k \equiv ke \pmod f$, and write $x_k = ke - m_k f$, so that $1 \le x_k \le f$. Put*

$$g(k) = (m_k + 2)e - 1 - kf, \qquad z_k = \frac{2ef - kb - x_k e}{f}.$$

Then:

(i) *an extra base column with parameter k exists if and only if $g(k) \ge 0$ and $x_k + z_k \ge 4$, and it is then $(x_k,\ e + kf,\ z_k)$;*
(ii) *$g(k+1) \le g(k) - (f - e)$, so g is strictly decreasing;*
(iii) *$g(1) = 2e - 1 - f$. Hence no member with $f \ge 2e$ carries an extra column at any k, and in general the admissible k form an initial segment $1, \dots, K$ with $K \le 1 + \lfloor(2e - 1 - f)/(f - e)\rfloor$.*

Machine-checked in `lean/Frontier.lean` as `column_criterion_identity`, `column_criterion`, `k_one_forces_f_le_two_e_sub_one`, `residue_step` and `g_strict_drop`, axiom-free beyond `propext`, `Classical.choice` and `Quot.sound`.

Proof. (i) By Proposition 145(i),(ii) a column satisfies $xe + zf = 2ef - kb$ and $x \equiv ke \pmod f$; conversely, for any x in that class f divides $2ef - kb - xe$, because $xe \equiv ke^2$ and $2ef - kb \equiv ke^2 \pmod f$. So the column exists precisely when some $x \ge 1$ in the class has $z = (2ef - kb - xe)/f \ge 1$, that is $xe + f + kb \le 2ef$. The left side increases with x, so it suffices to test $x = x_k$, and then $x = x_k$ by Proposition 145(iv). Writing $x_k = ke - m_k f$, the identity

$$(ke - mf)e + f + k(f^2 - e^2) - 2ef \;=\; f\bigl(kf - (m+2)e + 1\bigr)$$

turns that inequality, after dividing by $f > 0$, into $kf \le (m_k + 2)e - 1$, i.e. $g(k) \ge 0$.

(ii) From $x_k = ke - m_k f$ and $x_{k+1} = (k+1)e - m_{k+1} f$ one gets $x_{k+1} - x_k = e + (m_k - m_{k+1})f$. Both x_k, x_{k+1} lie in $[1, f]$, so the left side lies in $[1 - f, f - 1]$; with $0 < e < f$ this forces $m_k \le m_{k+1} \le m_k + 1$. Hence $g(k+1) - g(k) = (m_{k+1} - m_k)e - f \le e - f < 0$.

(iii) Since $1 \le e \le f - 1$, the least positive member of the class of e is e itself, so $m_1 = 0$ and $g(1) = 2e - 1 - f$. If $f \ge 2e$ then $g(1) < 0$, and g is decreasing, so $g(k) < 0$ for every $k \ge 1$. In general $g(k) \le g(1) - (k - 1)(f - e)$, and $g(K) \ge 0$ gives the stated bound on K. ∎

Remark 147 (What the criterion settles, and what it leaves)**.** The whole $k \ge 1$ question is now confined to $e < f < 2e$, a strict sub-range of the close-pair range $f \le (1 + \sqrt{2})e$. Of the 17296 close pairs with $e \le 200$, 5065 have $f \ge 2e$ and so carry no extra column at all: for those the base column is forced to $k = 0$ outright. At $e = 1$ the condition $f \le 2e - 1 = 1$ is incompatible with $f > e$, so no $(1, f)$ member has an extra column, which is the arithmetic content of Theorem 20 recovered independently.

The criterion is exact, not merely necessary: it was checked against brute enumeration of the base equation on all 17296 close pairs with $e \leq 200$, covering 67218 columns, with no disagreement, and the initial-segment shape and the bound on K hold in every case. The corner-parallelogram condition $x_k + z_k \geq 4$ is not redundant — it rejects an otherwise admissible k in 119 of the pairs with $e \leq 120$ — so it is carried explicitly in (i).

What this does *not* do is exclude the surviving columns: 21837 of them exist as solutions of the base equation, and no arithmetic condition can remove them, since they are genuine solutions. The arithmetic layer of the close-pair problem is now exactly described; the exclusion must come from the interior, as Remark 143 already recorded.

Remark 148 (The wedge-filler hypothesis, and where it fails)**.** Several steps of the corner chain fill a wedge of angle exactly α with a single α-tile, justified in Lemma 67 by "α is the minimal angle". That is not the load-bearing fact. What the step needs is that no two smaller angles fit, that is $2\beta > \alpha$, which with $3\alpha + 2\beta = \pi$ reads

$$\alpha < \tfrac{\pi}{4}.$$

The two conditions are different, and both directions of the difference occur. At $(e, f) = (3, 4)$ one has $b < a$, so β is the minimal angle, yet $\alpha < \pi/4$ still holds and the step survives.

The criterion is exact and integral. Since $b^2 + c^2 - a^2 = (f^2 - e^2)(2f^2 - e^2)$ and $2bc = 2f^2(f^2 - e^2)$,

$$\cos\alpha = \frac{2f^2 - e^2}{2f^2} = 1 - \frac{e^2}{2f^2},$$

which at $e = 1$ is the familiar $(2f^2 - 1)/(2f^2)$. So $\alpha < \pi/4$ iff $2f^2 - e^2 > \sqrt{2}\, f^2$, and squaring (both sides positive, as $f > e$) gives

$$e^4 - 4e^2 f^2 + 2f^4 > 0.$$

The separation condition $f^2 > 2ef + e^2$ implies it (`separated_wedge_ok`): from $f^2 - e^2 > 2ef$ one gets $f^4 + e^4 > 6e^2 f^2$ on squaring, and the claim follows with room to spare. So the step is safe at every separated member, and every failure is a close pair. Among $e \leq 11$, $f \leq 19$ the failures are

$$(4,5),\ (5,6),\ (6,7),\ (7,8),\ (7,9),\ (8,9),\ (9,10),\ (9,11),\ (10,11),\ (10,13),\ (11,12),\ (11,13),$$

including $(5, 6)$, which is $N = 83$. No claim in either paper is affected: Lemma 67 is invoked only at $e = 1$, where $a = f < f^2 - 1 = b$ and $\alpha < \pi/4$ both hold, and Proposition 150 lists four ingredients not including it. What the criterion supplies is a precise location for the obstruction: close pairs differ from separated members geometrically, not merely in having weaker arithmetic filters, and an argument aimed at them must either avoid the wedge-filler step or treat $e^4 - 4e^2 f^2 + 2f^4 \leq 0$ separately.

Two arithmetic strengthenings do *not* help there, which is why the bound must be interior. Reducing the base equation modulo e gives $y + z \equiv 0 \pmod{e}$ by coprimality; and the γ-injection is exactly a bipartite matching of the a- and b-edges to the junctions, whose sharp criterion is Hall's condition rather than the counting bound. Neither removes any of the 54 columns that survive the three unconditional filters over $e \leq 9$, $f \leq 15$: the first is implied by the base equation, and the second collapses to the γ-trap $z \geq 1$, since with $z \geq 1$ a matching always exists.

Lemma 149 (A c-corner carries a side a-edge)**.** *The base corner angle is β, with flanks $\{a, c\}$. If a corner tile lays c on the base it lays a on the side, so that side's a-count P' is positive; by Lemma 87 $P' = fp$, hence $P' \geq f$, and with $pe + R' = f$ and the γ-trap $R' \geq 1$ one gets $1 \leq p \leq (f - 1)/e$. So the side condition of Hypothesis 28, namely $p = 0$, fails at any c-corner:*

under that hypothesis both base ends are a-edges, exactly as Theorem 20 concludes at $e = 1$. What remains for $e \geq 2$ is to exclude $1 \leq p \leq (f-1)/e$, which is the analogue of the entire $e = 1$ programme rather than a single lemma; the boundary alone does not do it, since a and c are both available at the first base position once the walls column is in force.

Proposition 150 (The chain machinery is member-independent)**.** *Every ingredient of the corner chain holds at all members (e, f), not only at $e = 1$:*

- (i) *the branch relations $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta$ define the family, so the unique fills of β, α and $\alpha+\beta$ (`fill_beta`, `fill_alpha`, `fill_alpha_beta`) and the straight and full vertex classifications are unchanged;*
- (ii) $\cos\gamma = -e/2f < 0$*, so $2\gamma > \pi$ and every reflected partner dies at a straight junction, at every member;*
- (iii) $b \sin\gamma = c \sin\beta$*, so an a-up tile spans its strip, at every member;*
- (iv) *for $1 \leq j < f$ the only decomposition of $j\,b$ is j copies of b. Writing $j = y + z + w$, the equation gives $e^2(z+w) + x\,ef = wf^2$, so $f \mid e^2(z+w)$ and hence $f \mid z+w$ by coprimality; since $z + w \leq j < f$ this forces $z = w = 0$, so $y = j$ and $x = 0$. Both reductions are derived rather than assumed (`partition_jb_gen`). Verified for all coprime (e, f) with $e < 12$, $f < 20$.*

The single parameter that changes is the overshoot $c - b = e^2$, which is 1 only at $e = 1$. The $e = 1$ arguments that used $c - b = 1$ as a unit therefore carry over with e^2 in its place, while those that used its being the smallest positive integer do not.

Lemma 151 (The α-vertex gap, general)**.** *A cover piece of a horizontal a-edge has its foot at $(k + 1)a$ from the row origin if it is direct, and at $(k + 3)a - e^3/f$ if it is mirrored, since $2\,cu_x - 3a = -e^3/f$. The mirrored foot is a lattice point only if $f^2 \mid e^2$, and at row 1, where feet land on the base and every junction there is at an integer abscissa, it is a junction only if $f \mid e^3$; coprimality forces $f = 1$ in both cases (`alpha_vertex_gap_gen`). So the mirrored placement is excluded at row 1 at every member with $f \geq 2$, with e^3 playing the role that 1 plays at $e = 1$.*

Remark 152 (The two corners are not equally rigid)**.** At a c-corner the side junction carries γ and the remaining $\alpha + \beta$ has the unique fill $\{\alpha, \beta\}$, so the partner is forced immediately; this is what makes Theorem 92 short. At an a-corner the side junction carries α and the remaining $2\alpha + 2\beta$ admits both $\{2\alpha, 2\beta\}$ and $\{\beta, \gamma\}$, so the side branches at once. The base end behaves the other way: there the a-corner leaves $\alpha + \beta$, whose unique fill again forces the partner and confines the next base edge to $\{a, c\}$. The chain therefore runs along the base from either corner, and what it cannot do is pass a base b-edge; this is the mechanism behind Theorem 60 at $e = 1$ and it is available verbatim at every member.

Lemma 153 (The corner step, unconditional at every member)**.** *Let a base corner tile lay a on the base. It presents γ at the far end of that a-edge, where the junction is straight, so the remainder is $\alpha + \beta$ with the unique fill $\{\alpha, \beta\}$. The corner tile's b-edge joins a base vertex to a side vertex, so both of its ends lie on the boundary and the tile across it lays a whole b; that tile presents α at the junction, since a straight junction carries at most one γ. Hence the α-tile is adjacent to the corner tile and the β-tile is adjacent to the base, and the second base edge, lying among the β-tile's flanks, is an a or a c.*

Remark 154 (Where the chain stops being free)**.** Lemma 153 iterates only as far as the same two ingredients survive. The fill of $\alpha + \beta$ is unique at every member, so the two tiles are always an α and a β; what decides whether the next base edge is confined to $\{a, c\}$ is their angular order, and that is settled by whether the α-tile is the partner across the previous tile's whole b-edge. At the corner this is unconditional, because the b-chord runs from a base vertex to a side vertex. At the k-th step the chord's upper end is interior, and pinning it is precisely the crossing question of Remark 14. Both remaining cases, the residual configurations at $e = 1$ and the exclusion of $1 \le p \le (f-1)/e$ at $e \ge 2$, reduce to that single question.

That question is not answered by the hypothesis $m = 1$ alone. Instrumenting the certified exhaustions to record every partial configuration they build shows the overshoot occurring freely: at $(1, 2)$ it appears in 31 of the 52 recorded frontiers and at $(1, 3)$ in 1128, the first already at depth 3 with three tiles placed, a c-edge from $(2, 0)$ to $\left(\frac{11}{2}, \frac{\sqrt{15}}{2}\right)$ covering a b-edge from $(2, 0)$ and overshooting it by $c - b = 1$. Every such configuration is eventually refuted, but not locally: the partner pinning cannot be established by excluding the overshoot near the chord.

Remark 155 (The thick program)**.** The three verified facts above change the character of the $e \ge 2$ case. The chain machinery ports with strictly stronger tools: the wall climb and the first-run chirality are member-independent (their proofs use only the angle identities and the boundary overshoot $c - b = e^2 > 0$), and Lemma 86 removes the c-chord fork entirely, so the thick chain branches strictly less than the thin one. What remains is the finite bookkeeping: the base letters run through a trichotomy rather than a free word, the side violations are quantized to $p \in f\mathbb{N}$, and the reach analysis of the thin case applies verbatim where the blocks are short. The $(2, 3)$, $(2, 5)$ and $(3, 4)$ members are certified by full exhaustions (19,677, 553,417 and 825,724 nodes; the second is the prime $N = 71 \equiv 11 \pmod{12}$, the third a close pair). The close pair refutes deeper ($8.3f$ against $5.7f$ and $5.2f$), consistent with its additional base decompositions.

Remark 156 (First thick-member certificates)**.** The full $m = 1$ search of the $(2, 3)$ target ($N = 23$, prime, $23 \equiv 11 \pmod{12}$) is EXHAUSTED at 19,677 nodes: no tiling exists, and Hypothesis 28 holds at $(2, 3)$. The refutation profile differs from the thin members: the semigroup-run prune carries 57% of the kills (against 8% at $(1, 5)$), consistent with Lemma 86 strengthening the run arithmetic, and the maximal depth is $5.7f$ against $4.8f$.

Lemma 157 (The crossing kill, and solitude of branches)**.** *A transversal edge crossing a ladder's line within parameter $b + c$ of its start is fatal: the covers cannot cross it, every T-junction forces the next cover edge, and Lemma 76 forbids termination below $b + c$. A branch's own ceilings never trigger this — the k-th strip crossing is endpoint-tangent, sitting exactly $(k-1)c$ east of the corresponding ceiling's end (`crossing_tangency`) — but a* different *same-row branch's ceilings do whenever it sits 1 to $2f - 1$ columns west: the offset Df lands the crossing inside a ceiling for every $-2f < D < 0$, at parameter b or $2b < b + c$. Hence any strip junction of an A-branch with another A-branch 1 to $2f - 1$ columns to its west admits no straddler, and its fills are mates: at most one A-branch per $2f$-column window per row-band carries any straddler at all.*

Lemma 158 (The mirrored piece is charged)**.** *A cover piece of a horizontal c-edge is direct, with foot at the row lattice, or mirrored, with foot displaced by $1/f$; the mirrored piece presents β at its left junction, as does its left neighbour, so that junction carries $n_\beta \ge 2$ and its figure is $\{3\alpha, 2\beta\}$ (`mirrored_left_junction`). By the census (Lemma 82) each such junction forces an additional $\{\beta, 3\gamma\}$ vertex: a single mirrored piece already gives $v_1 \ge 2$ (`escape_charge`). The off-lattice escape available at rows $j \ge 3$ is therefore paid for globally, in climber vertices, of which the kernel-checked tilings carry between one and four.*

Lemma 159 (Climbers detect deviation)**.** *At an interior lattice point of the brick/mate structure exactly six tiles meet, contributing β, γ, α from the row above and α, γ, β from the row below; the counts are $(2, 2, 2)$ and the figure is $\{2\alpha, 2\beta, 2\gamma\}$. Hence a $\{\beta, 3\gamma\}$ vertex never occurs where the six incident tiles are the brick/mate six: by Lemma 82 every tiling contains such a vertex, and by Lemma 158 each mirrored cover piece forces a further one, so each escape requires two points* *at which the tiling deviates from the forced pattern. This is a detector, not an exclusion: the kernel-checked tilings do carry climbers at lattice points, at places where their local structure is not brick/mate.*

Remark 160 (What reach buys, and what it does not)**.** The four kills at reach R empty the window exactly for $f \le R+2$: reach 3 gave $f \le 5$, reach 4 gives $f \le 6$, and each further row would give one further member. Theorem 77 raises the reach by one and no more, because the clearance bound it uses is the fixed constant $T/b > 2$. A proof for all f therefore requires killing straddlers at *every* row, not at the first three. The natural route is that a ladder descends into strips the chain has already forced, where every line of its direction is a shared b-edge chain broken at the strip junctions on both sides at once — unstaggered — while a ladder is staggered by $c-b=1$; a ladder in forced territory would have to run along such a line or cross tile interiors, and both are impossible. What is missing is the bookkeeping: that the region a ladder descends through, roughly one half-column east per strip, lies inside the forced staircase at the moment the fork is being decided.

Remark 161 (The residue at $f \ge 6$)**.** For $f \ge 6$ the surviving words form an explicit thin set, symmetric under reversal; at $f=6$ it is a single class: b at position 5, c at position 7 ($a\,a\,a\,a\,b\,a\,c\,a$), read from a corner whose block is at least 3. The blocking structure is always the same: the chain's column-3 line needs a row-3 brick, the row-3 fork's A-branch lays its south cover on the row-2 floor, and the floor is edged only as far as the staircase has reached — which the b itself interrupts. Closing the row-3 fork is the entire remaining content of Conjecture 57 at $e=1$.

Remark 162. The residual class for $f \ge 5$ is precisely: some side's first a-run behind an initial c-block of length at least 2. At $(1,4)$ the single such side word dies by the line-total kill of Remark 52, whose three gap facts $b-a$, $b-e^2$, $2b-a \notin \langle a,b,c\rangle$ hold for every $f \ge 3$; what does not yet generalize is the case analysis behind them when the initial block exceeds two or the run splits. Closing this class for all f is the last step of Conjecture 57 at $e=1$.

5.11. **Assembly.** Fix a hypothetical tiling. Its base walk is one of the three; each funnels as above to a configuration killed by the T_{mid} corner, the pentagon lemma, or the walk arithmetic (A1-type shielding, the a-prefix forcing with P6-type end constraints, the β-wedge). No case survives; no tiling exists. $\square$

5.12. **Scope, certificates, verification.** Every arithmetic component is kernel-checked (`lean/`: BAdjacency, Rigidity, W2Core, MidTriangle, SurplusLattice, GeneralPillars, MasterLemmas, Pentagon; core toolchain, axiom-clean). For Lemma 39 the file `Pentagon.lean` proves, general in (e,f) and axiom-free, both facts the wedge analysis consumes: that $(0,\min(a,b))$ is a gap of the numerical semigroup $\langle a,b,c\rangle$, and that the stub $e^2 \bmod b$ lies in that gap *immediately* — splitting on $e^2 < b$: below, the stub is e^2, smaller than $a = ef$ since $e < f$ and smaller than b by hypothesis; above, $b \le e^2 \le ef$ bounds the remainder by both. The second initial stub $|a-b|$ can exceed $\min(a,b)$ (e.g. $5 > 3$ at $(1,3)$), so it escapes that gap; it is settled instead by Lemma 40 below, which removes the iteration from the argument entirely. Both stubs are therefore machine-checked, general in (e,f) and axiom-free. Certified seeded refutations: $(1,3)$, $(1,4)$ at $j=1,2$, $(2,3)$ side-break, $(2,3)$ pentagon. Engine exhaustions independently confirm $N = 11, 23, 26, 47, 59, 66, 71, 107$. The exposition above is the assembled argument; an extended version with expanded strip/recursion diagrams is in preparation and does not affect the claims.

6. Realizability: the collar induction and the first complete member

Theorem 29 is the exclusion side of the branch: no member has an instance at $m=1$. This section supplies the construction side. For the member $(e,f)=(1,2)$ — tile $(2,3,4)$, unit target

Δ_1 with base $Y_1 = 11$, sides $X_1 = 8$, $N_1 = 11$ — it settles *every* remaining instance, making $(1, 2)$ the first member of any incommensurable-angles branch whose instance spectrum is completely determined.

Write Δ_m for the instance target at scale m (base mY_1, sides mX_1, count $N = m^2 N_1$), and subdivide Δ_m by the standard m-fold edge subdivision into m^2 cells, each a translate of Δ_1 ("up-cells") or of its half-turn ("down-cells"). A cell is *not* individually tileable — that is exactly the $m = 1$ exclusion — so all constructions below use two larger blocks whose interfaces are full straight segments.

Lemma 163 (Unit parallelogram)**.** *The parallelogram P_1 with vertices $(0, 0)$, $(Y_1, 0)$, $(Y_1 + X_1 \cos\beta,\ X_1 \sin\beta)$, $(X_1 \cos\beta,\ X_1 \sin\beta)$ — one up-cell and one down-cell, glued along a full cell edge — admits a tiling by $2N_1 = 22$ congruent copies of the tile. The tiling was produced by the exhaustive engine (125 nodes) and is kernel-verified in* `lean/PgramTiling22.lean`*: exact $\mathbb{Z}[\sqrt{15}]$ coordinates scaled by 4, with congruence of all 22 tiles, containment in the closed parallelogram (4 half-planes), an explicit separating edge-line for each of the 231 tile pairs, and the exact area identity $\sum 2A_i = 528\sqrt{15}$ — no axioms, no imports.*

Lemma 164 (Collar decomposition)**.** *For $m \geq 3$, let $\Delta_{m-2}^{\text{apex}} \subset \Delta_m$ be the scale-$(m-2)$ copy anchored at the apex. The closure of $\Delta_m \setminus \Delta_{m-2}^{\text{apex}}$ — the* collar*, the bottom two cell rows, $4(m-1)$ cells — decomposes into*

$$(m-2)\ \textit{parallelogram columns} \ +\ \textit{one scale-2 corner triangle},$$

where the j-th column ($0 \leq j \leq m-3$) is the sheared strip with base $[jY_1, (j+1)Y_1]$ and top $[(j+1)Y_1, (j+2)Y_1]$ at height $2X_1 \sin\beta$ — the union of two translates of P_1 stacked flush — and the corner triangle is the scale-2 up-triangle on the base segment $[(m-2)Y_1,\ mY_1]$. Every interface (column-to-column, column-to-corner, collar-to-$\Delta_{m-2}^{\text{apex}}$) is a full straight segment, so the pieces tile independently. In tiles: $44(m-2)+44 = 44(m-1)$, and $N_1 m^2 = N_1(m-2)^2+44(m-1)$ exactly.

Proof. The subdivision arithmetic is immediate (and is kernel-checked, shape-free, as `collar_cells` in `lean/Collar.lean`); the only geometric content is flushness. The j-th column's lateral sides run parallel to the left leg (direction $(X_1 \cos\beta, X_1 \sin\beta)$, slope $\tan\beta$); its right side, from $((j+1)Y_1, 0)$ to $((j+2)Y_1,\ 2X_1 \sin\beta)$, coincides with the left side of column $j+1$, and for $j = m-3$ with the left edge of the corner triangle. The corner triangle's right edge lies on the right leg of Δ_m; the collar's top edge, from $(Y_1,\ 2X_1 \sin\beta)$ to $((m-1)Y_1,\ 2X_1 \sin\beta)$, is exactly the base of $\Delta_{m-2}^{\text{apex}}$ ($X_1 \cos\beta = Y_1/2$ by the isosceles altitude). Each column splits into two translates of P_1 along the full segment at height $X_1 \sin\beta$. □

Theorem 165 (Realizability, member $(1, 2)$)**.** *$N = 11m^2$ is realizable for every $m \geq 2$. With Theorem 29 the instance spectrum of the member $(1, 2)$ is completely determined:*

$$\Delta_m \textit{ is tileable} \iff m \neq 1.$$

Proof. Bases: $m = 2$ and $m = 3$ are the kernel-verified certificates `Tiling44.lean` and `Tiling99.lean`. Step: if Δ_m is tileable ($m \geq 2$) then so is Δ_{m+2}, by Lemma 164 applied at scale $m+2$: tile Δ_m^{apex} by hypothesis, each of the m columns by two copies of Lemma 163, the corner by the $m = 2$ certificate. The two-step induction from bases $2, 3$ reaching all $m \geq 2$ is `two_step` in `lean/Collar.lean` (kernel-checked); the exclusion at $m = 1$ is Theorem 29. □

Remark 166 (Relation to Zhang's constructions)**.** Zhang's construction paper [6] builds tiling families for the six sporadic targets whose tile has a $2\pi/3$ angle (with a $\pi/3$ variant), using ideal trapezoids as macro-tiles; the families there carry lower bounds and residue classes in m (e.g. $m \geq 9$, or $m \in 5\mathbb{N}_0 + 7\mathbb{N}_0$), leaving the small members undecided. The branch treated here, $3\alpha+2\beta=\pi$, is disjoint from those six cases ($\gamma=2\pi/3$ forces $\beta=0$ in this branch), and no construction family for it appears in the literature known to us; the 44-, 99- and 22-tilings above were found by our engine. The collar plays the role of Zhang's trapezoids — a macro-tile calculus— but the induction it drives has no lower bound and no residue class: every admissible m is realized. Member $(1,2)$ is thus, to our knowledge, the first incommensurable-angles (target, tile) pair with no undecided instance.

Theorem 167 (The family is not uniform)**.** *For the member* $(e,f)=(1,3)$ *— tile* $(3,8,9)$*,* $N_1=26$ *— the scale-2 target admits no tiling:* $N=104$ *is not realized.* (Exhaustive search, 43 541 916 nodes.) *Hence the conclusion of Theorem 165 does not extend along the family:* Δ_2 *exists for* $(1,2)$ *and does not exist for* $(1,3)$*.*

This is the reason every attempt to build the seeds at $f=3$ fails, and it is worth listing them because each is now explained rather than merely unsuccessful: the cevian piece C_2 (68 tiles), the symmetric middle M_2 (32 tiles), the maximal interior of the $a\times c$ lattice (34 tiles left to place) and of the b-glued Q_c lattice (34 likewise) all return NO TILING, and a seeding by unit parallelograms is impossible by counting, since Δ_M has $M(M+1)/2$ up-cells against $M(M-1)/2$ down-cells and each parallelogram consumes one of each, leaving M untileable cells. There was nothing to construct.

Two consequences. First, Remark 33 acquires content: $(1,2)$ is the unique member of its family with $\beta<\pi/3$, and it is also the only one known to realize its scale-2 target. Second, the collar of Theorem 165 has no base at $m=2$ for $(1,3)$; its bases must come from $m\geq 3$. Since Δ_4 would require $\Delta_1+\Delta_3$ and Δ_5 would require $\Delta_2+\Delta_3$, both excluded, the set reachable from Δ_3 alone is exactly the multiples of 3. So if Δ_3 tiles, the spectrum of $(1,3)$ is plausibly $N=26m^2$ with $3\mid m$, a shape quite unlike $\{m\geq 2\}$; and if Δ_3 does not tile, the member contributes nothing. We state this as a question rather than a conjecture, and the search is running.

Remark 168 (The program for the other members)**.** Lemma 164 is stated for $(1,2)$ but its geometry is member-free: the subdivision, the column shear, and the flushness argument use only the isosceles altitude identity, valid for every (e,f). For any member, realizability of *all* $m\geq 2$ therefore reduces to three finite seeds: a tiling of Δ_2, of Δ_3, and of the unit parallelogram P_1. The m-direction is closed once and for all; what remains per member is the three seed searches (or a uniform seed construction in (e,f), which we have not yet found). The seeds grow quickly — for $(1,3)$ they are 104, 234 and 52 tiles — so this is engine territory; the collar lemma converts each trio of successes into a complete member. The parallelogram seed of $(1,3)$ is already in hand: the tile $(3,8,9)$ tiles its unit parallelogram into 52 copies, found by the engine and kernel-verified in `lean/PgramTiling52.lean` (exact $\mathbb{Z}[\sqrt{35}]$, 1326 separating lines, zero axioms).

For the family $e=1$ the seed is not needed at all: the parallelogram can be written down.

Theorem 169 (The unit parallelogram, $e=1$)**.** *Let* $e=1$ *and* $f\geq 2$*, so* $(a,b,c)=(f,\ f^2{-}1,\ f^2)$ *and* $N_1=3f^2-1$*. Then* P_1 *admits an explicit tiling by* $2N_1$ *copies of the tile.*

Proof. Work in the affine basis $u=(1,0)$, $w=(\cos\beta,\sin\beta)$, in which P_1 is the box $[0,Y_1]\times[0,X_1]$ with $Y_1=3f^2-1$ and $X_1=f^3=f\,c$; recall $X_1\cos\beta=Y_1/2$. The $a\times c$ cell of this basis is the parallelogram obtained by gluing two tiles along their b-edge (sides a,c, angle β), so every whole cell carries two tiles and a cell cut along its own b-diagonal carries one. Split P_1 by the two parallel lines through $(fa,0)$ and $(fa+b,0)$ with direction $X_1w-fa\,u$:

- Λ the triangle $(0,0),(fa,0),(0,X_1)$. Its hypotenuse drops exactly a per row of height c, hence passes through cell corners; cells with $i+j\leq f-2$ are whole and those with $i+j=f-1$ are cut along their b-diagonal, giving $2\binom{f}{2}+f=f^2$ tiles.

Σ the strip between the lines: the parallelogram spanned by $b\,u$ and $X_1w - f\,a\,u$. Put $\sigma = (X_1w - f\,a\,u)/b$. Then

$$|\sigma|^2 = \frac{b^2 + 4f^6 - N_1^2}{4b^2} = f^2, \qquad \cos\angle(\sigma, u) = \frac{1}{2f} = \cos(\alpha + \beta),$$

so σ has length a and makes angle $\alpha + \beta$ with u. Consequently the parallelogram spanned by $b\,u$ and σ has sides b, a and angles $\alpha + \beta, \gamma$: it is the parallelogram obtained by gluing two tiles along their c-edge, and its long diagonal has length c because $c^2 = a^2 + b^2 + 2ab\cos(\alpha + \beta)$. Stacking b of them along σ closes exactly on $(0, X_1)$ and gives $2b$ tiles.

P the remainder, which is the Λ-staircase in the lattice translated by b, run to $u = Y_1$; this is legitimate since $Y_1 - b = 2f^2 = 2f\,a$. It gives $2\big(f^2 + \binom{f}{2}\big) + f = 3f^2$ tiles.

The three pieces are disjoint with union P_1, and $f^2 + 2b + 3f^2 = 4f^2 + 2(f^2 - 1) = 2N_1$. □

The construction has been generated and verified exactly in $\mathbb{Q}(\sqrt{D})$ — congruence, containment, pairwise separation, area — for $f = 2, \ldots, 12$, i.e. 22, 52, 94, 148, 214, 292, 382, 484, 598, 724 and 862 tiles; only $f = 2, 3$ had previously been obtained by search.

Remark 170 (The obstruction at $e \geq 2$)**.** The mechanism is specific to $e = 1$. For general (e, f) the strip is always the parallelogram spanned by $eb\,u$ and $X_1w - f\,a\,u$, whose slant has length exactly fb and whose angle satisfies $\cos\theta = e/(2f) = \sin(\alpha/2)$, i.e. $\theta = \alpha + \beta$; but in the sheared basis it is combinatorially an $s \times fb$ rectangle to be filled by $a \times b$ bricks, and the de Bruijn–Klarner criterion requires $a \mid s$ or $a \mid fb$. The latter says $e \mid f^2$, i.e. $e = 1$; the former contradicts the compatibility condition $s \equiv -e^3 \pmod{ef}$ unless $f \mid e^2$. So no admissible width admits a brick interior when $e \geq 2$, and exhaustive search confirms that no interior of any kind exists in the cases tested: on $(2, 3)$ the widths $s = 4, 10, 16, 22$ all return NO TILING (1, 30, 181, 855 nodes), and on $(3, 4)$ the width $s = 21$ returns NO TILING (72 nodes), while the $e = 1$ control is found in 10 nodes.

Theorem 171 (The dichotomy)**.** *The unit parallelogram of the member $(e, f) = (2, 3)$ — sides $Y_1 = 46$ and $X_1 = 27$ with angle β, tile $(6, 5, 9)$, 46 tiles — admits no tiling.* (Exhaustive search, 4 334 789 nodes.) *Together with Theorem 169 this separates the branch: at $e = 1$ the unit parallelogram exists for every f, and at $e = 2$ it does not exist at all.*

The obstruction of Remark 170 is therefore not a limitation of the two-lattice architecture: at $e \geq 2$ there is nothing for any architecture to find. At $(2, 3)$ both the unit cell (Theorem 29) and the unit parallelogram are untileable.

Remark 172 (What this does to the induction)**.** The collar of Theorem 165 runs on unit columns, so it is unavailable at $e \geq 2$; only the wide columns of Proposition 173 remain, and they advance m in steps of f. Consequently the clean law "tileable iff $m \neq 1$" proved for $(1, 2)$ need not persist: the spectrum may genuinely depend on e. For $(2, 3)$, where $f = 3$, the subdivision closure and the wide-column law together would give every $m \geq 4$ and every even m from a single seed Δ_2, leaving exactly $m = 3$ unreached from below; so the decisive experiment for the whole $e \geq 2$ story is the realizability of Δ_2 at $(2, 3)$, i.e. $N = 92$. That search is running; we record the reduction rather than guess its outcome.

Proposition 173 (Wide columns, all (e, f))**.** *A parallelogram with sides $p\,Y_1$ and $q\,X_1$ is a pure $a \times c$ lattice — an $(pN_1/f) \times (qf)$ grid of cells, hence tileable by $2pqN_1$ tiles — if and only if $f \mid p$. Consequently, for every member, Δ_M is tileable whenever Δ_t is tileable and $f \mid M - t$: run the collar of t rows with columns f cells wide. Together with the subdivision closure (Δ_m tileable $\Rightarrow \Delta_{mk}$ tileable) this is a residue-class realizability law valid in every member; at $e = 1$ the unit column exists and the law upgrades to all $m \geq 2$.*

7. What is known to be dead

The following approaches to this branch are closed, and are recorded so that they are not retried.

(1) **The whole class of linear direction invariants.** Weighting a directed edge by a function of its direction alone, antisymmetric under $\theta \mapsto \theta+\pi$, yields a one-parameter family of identities; every member is a specialisation of the master identity of [3, Rem. 4.9]. The associated ideal-membership condition is satisfied identically for every root of unity and every (e, f), and the optimal ℓ^1 bound the family can produce is $m(3f-2e)(f+e)/f$, which is smaller than N by a factor at least $11/6$ and growing like mf. So no choice of weight, kernel, moment ladder or duality can exclude a candidate. Both statements are machine-checked in `lean/LambdaFactor.lean`.

(2) **Closing $e = 1$ by proving $R \geq 2$ on every side.** False at the base: the walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$. On the base, "$R \geq 2$" restates the remaining problem rather than reducing it. It does hold on the two equal sides, and is proved there.

(3) **Boundary tile counts.** The number of tiles meeting ∂ABC grows like the number of boundary edges, and on the genuine certificates it is 28 of 44 and 44 of 99 — a falling fraction. Comparing it against N therefore never yields a contradiction.

(4) **The far side of the mismatch ray**, under the hypothesis that the ray is a complete wall: that side is the tile scaled by f, and is tileable. No obstruction can come from there.

(5) **Local combinatorics of an equal side, with or without the census**, for $p = 1$ at $(3, 7)$. The junction automaton forbids a single adjacency and leaves 10 560 configurations (Proposition 135); adjoining the corner census adds nothing, its two counting equations being dependent (Proposition 136). Together with Lemma 82 this is the third confirmation that corner counting cannot see p.

(6) **The chord/area programme**, beyond Propositions 138 and 140. Two continuations both fail: the γ-grading does not discretize straddle positions, because it constrains directions and the vertical projection of the vertex module is dense (Remark 141); and the family of 37 admissible chord heights supplies 37 instantiations of one identity — additivity of area — rather than 37 constraints (Remark 142). Closing $p = 1$ needs a mechanism that is neither counting nor chord-area.

(7) **Boundary-word invariants (Conway–Lagarias tiling groups)** [1]. These are the natural non-abelian strengthening of the direction invariants of item 1, and are known to be strictly stronger than colouring and weighting arguments — which is exactly the class item 1 kills here. They nevertheless do not apply, for two independent reasons. First, the method is formulated for regions assembled from cells of a regular lattice, and the vertices of a tiling of our target lie in no lattice: the edge vectors have lengths in $\{21, 40, 49\}$ and directions $k\gamma$ with $\gamma = (\pi + \alpha)/2$ and α/π irrational, so the vertical projection of the $\mathbb{Z}$-module they generate is dense in $\mathbb{R}$. Second, tilings here need not be edge-to-edge, so the boundary word of a region is not determined by its combinatorics.

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